

MAGNETOM

Troubleshooting Guide System

Magnet System

Avanto
Espree

Document Version

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1.1 General

This chapter describes the test strategy for the **Magnet, Refrigerator system, Alarm Box** and **Magnet Supervisory Unit (MSUP)**.

The Alarm Box and Magnet Supervisory Unit are **Field Replaceable Units**. If an internal error develops, the complete unit must be replaced.

Prerequisites

n.a.

Tools

Handheld MSUP-Display Unit

Additionally required documents

n.a.

1.2 Magnet

Symptom	Area of Trouble (component or function)	Probable Cause	Service action
01. Liquid helium loss observed	Refrigerator system	Interrupted cold head operation	Maintain
		Compressor pressure too low	Leak check the helium compressor. Top off compressor pressure.
		Reduced efficiency of the cold head	Check the cold head temperature.
			Replace the cold head.
		High shield temperature	Check 50K shield link and bore temperature.
	Pressure heater	Pressure heater is permanently on	Check the status of the pressure heater. LED on the MSUP. Hand-held MSUP-Display Unit Check magnet system pressure.
		Pressure heater power > 0 W	Leak check of the venting system including quench valve. Check magnet system pressure. Ensure EIS is off.
	Venting system	16.0 psi (A) valve of the venting system is leaking	Check the magnet pressure. Perform a leak test.
		Bypass valve is open	Close the bypass valve.
	Helium fill port	Leakage on helium baffle	Check the baffle. Replace the gasket.

Symptom	Area of Trouble (component or function)	Probable Cause	Service action
02. High 50K shield temperature High cold head temp (CHT)	Water cooling	Problems with the water cooling circuit	Check the SEP. Check the Cooler.
	Cold head	Low efficiency of the cold head	Replace the cold head. Contact Siemens Magnet HSC.
	The compressor unit	Low gas pressure of the compressor unit	Refill the compressor unit.
03. Quench	ERDU	ERDU button pressed. Persistent quench	Check the ERDU buttons, release the button. Check the tamper-proof seal. Check the event log file and quench log in the MSUP. Refer to: Post Quench Procedure.
04. Magnet pressure < 15.3 psi	Pressure heater	Pressure heater is defective or off.	Check the status of the pressure heater with the Hand-held MSUP-Display Unit Ensure pressure heater LED is on. Check fuse F7 on the MSUP. Check MSUP connections.
	Venting system	Leakage in the venting system.	Leak check of the venting system including quench valve.
05. Average heater power < 0W System pressure > 15.3 psi (A)	EIS heater	EIS heater is on	Ensure that EIS heater is off, or times out.
	Helium syphon	Helium syphon is fitted	Remove syphon.
	OVC vacuum	Suspected OVC vacuum	Contact Siemens Magnet HSC.
	Refrigerator system	Long-term increase in shield temperature	Check history of shield temperature by SeSo. Contact Siemens Magnet HSC.

1.3 Refrigerator

Symptoms of the Refrigerator systems are listed in the following table.

1.3.1 Troubleshooting

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action
01. System shutdown LCD reads Helium Temp - ERR.	Compressor	Low oil level or blocked oil circulation.	Try five times to restart the compressor. If it does not start, contact MR Magnet Technology HSC .
		Faulty gas thermistor or PCB.	Contact MR Magnet Technology HSC .
	Water Supply	Water flow rate is inadequate.	Check the water flow rate.
		Water flow is reversed.	Check the water lines.
		Supply water temperature is too high	Scroll LCD display. If Water Temp-ERR is indicated, supply water temperature is too high. Provide 5° C to 25° C cooling water
02. System shutdown LCD reads Motor Temp - ERR.	Compressor	Compressor motor winding high temperature switch opens. Compressor motor windings have overheated.	Check the water flow rate. Allow 30 minutes for the windings to cool enough for the switch to reset. Contact MR Magnet Technology HSC if the problem persists.
		Reversed phase or loss of phase.	Check mains power. Fuse(s) opened or poor power quality.
		Fuse(s) opened or poor power quality.	Check fuses
03. System shutdown LCD reads Phase Seq-ERR.	Compressor	Fuse(s) opened or poor power quality.	Check fuses.
04. System shutdown LCD Return Press-ERR.	Compressor	Compressor has lost helium charge.	Refer to Helium Gas Charging F70 If no pressure, replace compressor.
05. System shutdown LCD reads FAIL instead of ERR.	Compressor	Sensor is disconnected.	Contact MR Magnet Technology HSC .

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action
06. Compressor and cold head motor do not start when the start switch is pressed.	Electrical Supply	No electrical power.	Check that the power source is on and connected.
		Wrong voltage.	Check mains voltage.
		Main power phase sequence is wrong.	Interchange any two incoming power leads (except ground).
	Compressor	Defective component in the power circuit.	Replace compressor.
		Tripped motor circuit protector on the front panel.	Re-set the protector - turn knob OFF and ON. Contact MR Magnet Technology HSC if problem persists.
	Magnet	Helium vessel pressure is very low.	Check for gross leaks on the magnet venting.
		Pressure switch faulty.	Check pressure manometer.
		Sensor cable broken or disconnected.	Check cable and connections at compressor and manometer.
07. Compressor starts but shuts down later.	Compressor	Insufficient coolant for the compressor.	Check the coolant flow and temperature. Refer to specifications.
08. System starts but gas pressure abnormally high or low.	Compressor	Wrong equalization pressure.	Refer to Service -Troubleshooting Guide - Gas Charging procedure.
		Gas line couplings are not fully engaged.	Check the gas line connections.
		Gas lines are wrongly connected.	Check the gas line connections. Re-connect.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action
09. Cold head motor does not start when the compressor starts.	Cold head	Cold head cable is not connected.	Stop the compressor. Connect the cable. Check connections at the cold head and at the compressor.
		Open circuit in the cold head cable.	Disconnect the cable. Check each conductor for continuity. Replace the cable if necessary.
		FU-4, FU-5 or FU-6 fuse is blown.	Replace the fuse.
10. Cold head motor hums but does not start.	Cold Head	Open circuit in the cold head cable.	Disconnect the cable. Check each conductor for continuity. Replace the cable if necessary.
		Valve disc stalled on the valve stem.	Check the system equalization pressure. Contact MR Magnet Technology HSC .
		FU-4, FU-5 or FU-6 fuse is blown.	Replace the fuse.
11. Cold head motor runs but there is no cooldown.	Cold head	Gas line couplings are not fully engaged.	Check gas line connections.
		Gas lines are wrongly connected.	Check gas line connections.
12. Intermittent operation.	Compressor	Compressor is cycling on and off.	Check input power, coolant flow and temperature.
13. Loss of refrigeration.	Compressor	Compressor malfunction	Check input power, coolant flow and temperature, and equalization pressure.
	Cold head	Worn displacer.	Check cold head run hours.
		Contaminated helium gas.	Contact MR Magnet Technology HSC .
14. Compressor shuts down, LCD display is blank.	Compressor	FU-1, FU-2 or FU-3 fuse is blown.	Replace the fuse.
	Compressor	Malfunctioning control boards.	Contact MR Magnet Technology HSC .

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action
15.Compressor is unresponsive to remote (DB-25 diagnostic interface) control.	Compressor	Configuration selector switch is in the wrong position.	Turn off power and move the switch to the correct position (down for 4K systems, up for 10K systems).
16.Remote status signals (DB-25 diagnostic interface) do not match actual status.	Compressor	Configuration selector switch is in the wrong position, or was in wrong position when power was supplied to the controller.	Turn off power and move the switch to the correct position (down for 4K systems, up for 10K systems).
17.LCD display shows configuration error	Compressor	Configuration selector switch is in the wrong position or was in the wrong position when power was supplied to the controller.	Turn off power and move the switch to the correct position.
18.Abnormal noise from the cold head.	Cold head	Contaminated helium gas.	Contact MR Magnet Technology HSC .
Remarks: Check the Event log of the MR system for errors and warnings generated by the refrigerator system or the water cooling system.			

1.3.2 Compressor F70

1.3.2.1 Compressor Restart time

Timeout time of the compressor:

- 20 min after power failure.
- 60 min after static pressure test.



The compressor timeout time can be 10% - 15% longer, depending on the load the MSUP controller sees.

1.3.2.2 Fuse Location F70



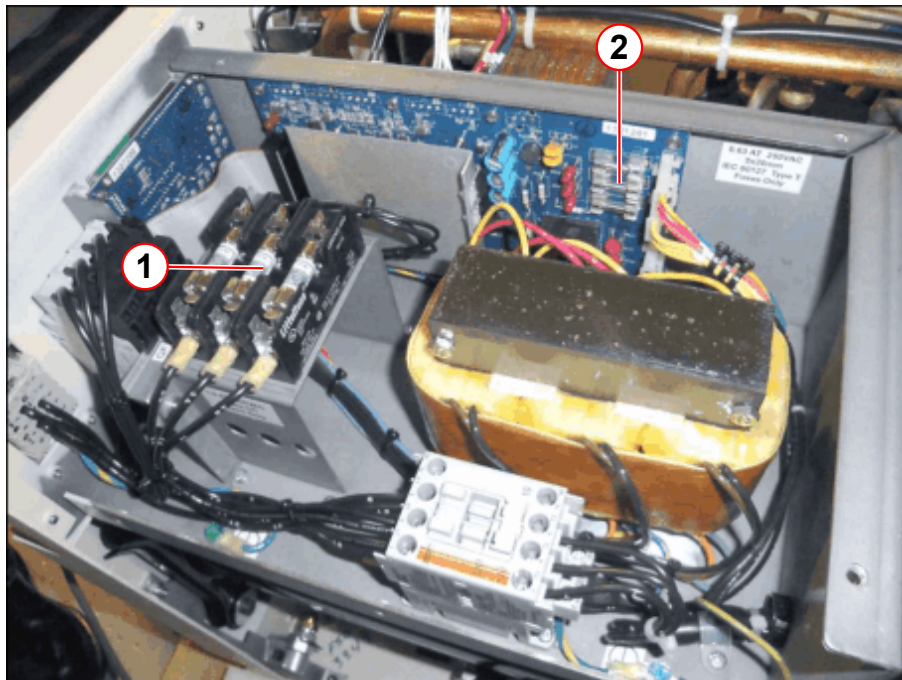
WARNING

AVOID ELECTRIC SHOCK

Failure to observe this warning can result in serious injury.

- » Disconnect the power to the compressor before troubleshooting the electrical components.
- » Permit only qualified electrical technicians to open electrical enclosures.

Fig. 1: F70 - Electrical Chassis



- (1) Fuses, FU-1, FU-2, FU-3
(2) PCB 1 with Fuses, FU-4, FU-5, FU-6

1.4 Magnet Supervision Unit (MSUP)

Error Messages

- Refer to (→ Healthcare Services Knowledge Base (SKB), <https://skb.healthcare.siemens.com/default.aspx>).

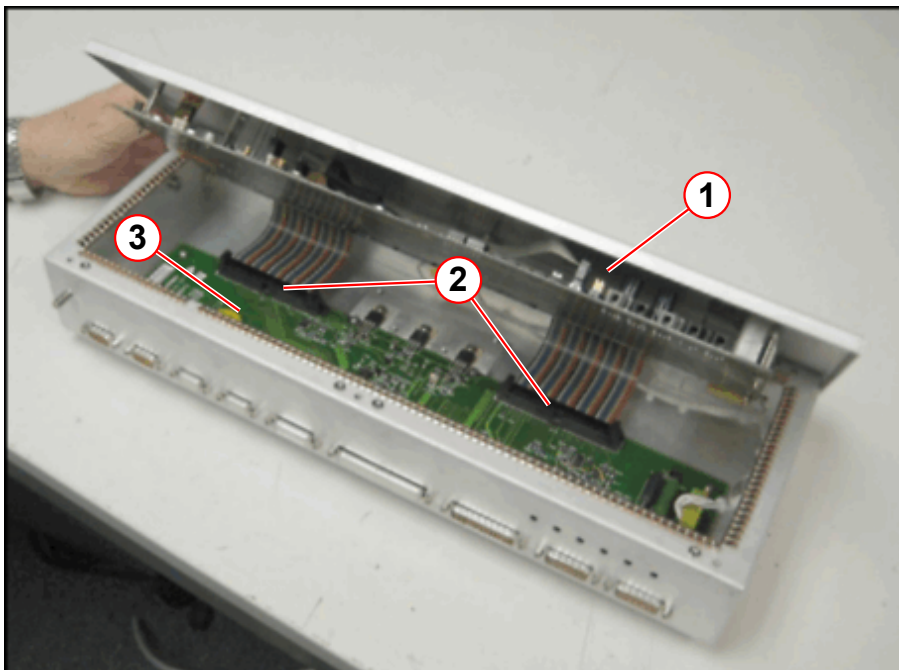
1.4.1 MSUP 2306

1.4.1.1 General

The **MSUP 2306** is a **Field Replaceable Unit**. It contains 2 assemblies:

- The **MSUP 2306 controller assembly** is attached to the underside of the lid and is a **replaceable** spare part. (→ 1/Fig. 2 Page 14)
- The **ERDU Test Load assembly** is mounted in the base of the MSUP and is **not separately replaceable**. If faulty, the **complete MSUP 2306 unit** must be replaced. (→ 3/Fig. 2 Page 14)

Fig. 2: MSUP 2306 complete unit



- (1) MSUP controller assembly (replaceable part)
- (2) Ribbon cables
- (3) ERDU assembly (not replaceable)

1.4.2 MSUP 2657

This chapter deals with troubleshooting the Magnet Supervision Unit (MSUP).

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
No response from MSUP	MSUP	36 V power supply incorrect.	Check fuse F2 on MSUP board. Check circuit breaker F2 in the ACC. Check fuse F3 in the ACC.	##
MSUP not serviceable with the display unit	MSUP	Defective MSU	Perform Com-test with Test Tool Replace MSUP	
Display unit suspected	Display unit	+5V is missing	Check +5 V at output of MSUP X9	
MSUP reports short circuit (S/C) after measuring helium level	Helium level probe	Incorrect level probe parameters	Check cabling Check fuse F3 on MSUP board. Alternately change pulse length and pulse height. » Change pulse length at 5 ms intervals, up to maximum 200 ms. » Change pulse height at 50 mA intervals, up to maximum 800 mA.	Speed info MR_45/04
Remarks: Check the Event log of the MR system for errors and warnings generated by the Magnet Supervision Unit.				

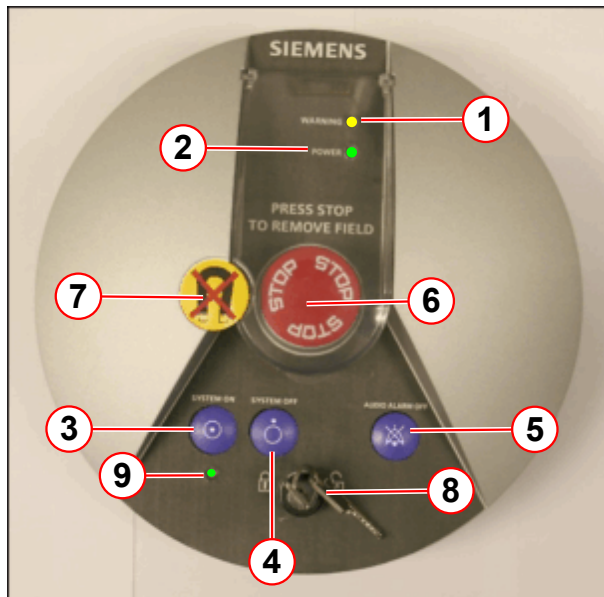
1.5 Alarm Box

The Alarm Box is an FRU. If an internal fault develops, the complete unit must be replaced.

1.5.1 Alarmbox (Mat. number 10523563) with MSUP 2306

1.5.1.1 Alarm box (Mat. number 10523563)

Fig. 3: Alarm box - LEDs and buttons



The Alarm Box is a Field Replaceable Unit (FRU). If an internal fault develops, the complete unit must be replaced.

- (1) LED WARNING = yellow
- (2) LED POWER = green
- (3) Button - SYSTEM ON
- (4) Button - SYSTEM OFF
- (5) Button - AUDIO ALARM OFF
- (6) Button - MAGNET STOP (ERDU)
- (7) Tamper-proof seal
- (8) Power switch locking key
- (9) SYSTEM ON LED = green

The alarm box provides two indicators concerning the health of the magnet system. When part of the system fails, the indicator will be illuminated to warn of the malfunction.

To silence the audible buzzer, press the AUDIO-ALARM OFF button.

Indicator - Title	Color	Function (when lit)	Service Action
System ON	Green	Indicates power has been applied to system.	n.a.
Warning	Yellow	Warns of fault in Magnet Supervisory system	Check at console for identification of problem.

1.5.1.2 LEDs

Tab. 1 Alarm box LEDs

LED	Description
LINE POWER	Power is supplied to the alarm box
WARNING	Helium level alarm or warning condition, or helium alarm Quench / switch heater error or ERDU button Magnet pressure too high / low Compressor stopped (pressure switch operated) ERDU battery voltage alarm or warning condition EIS drive error

1.5.2 Alarmbox (Mat. number 10096765) with MSUP 2657

The alarm box provides nine indicators concerning the health of the magnet system. When part of the system fails, the indicator will be illuminated to show the malfunction.

To silence the audible sounder, press the Acknowledge button.

Indicator - Title	Color	Function (when lit)	Service Action
System ON	Green	Indicates power has been applied to system.	n.a.
Line power	Green	Indicates power has been applied to alarm box	n.a.
Helium level	Red	Shows helium level in magnet is low.	Check the helium level. Check the refrigerator system.
Mag Stop	Red	ERDU button has been pressed Indicates an ERDU-related fault.	Check heater resistances via the hand held unit on the 2657 MSUP. Check the cable connections of the ERDU. Check fuses F3, F5, F6 and F7 on MSUP board Perform safety-related test. Download MSUP log. Email Magnet HSC
Pressure	Red	Shows when system pressure is too high/low.	Check the magnet pressure with hand-held unit. Ensure pressure heater circuit works correctly. Check the Event log for errors and warnings about the water cooling system. Switch on the compressor again.
Compressor	Green/Red	Red indicates that the compressor has stopped running.	Check the Event log for errors and warnings about the water cooling system. If no audible alarm sounds when the compressor stops, wait 30mins to see if the compressor restarts automatically.

Indicator - Title	Color	Function (when lit)	Service Action
Battery low	Red	Shows a fault in the emergency rundown back-up battery system.	Check fuse F4 on MSUP board. Check cable connection between battery and MSUP. Check hand held unit on the 2657 MSUP for the battery voltage. Replace the battery.
EIS	Yellow	Indicates a fault in external influence shield operation.	Check fuse F1 on the MSUP board Check hand-held unit on the 2657 MSUP for EIS condition.
Comms. Fault	Red	Indicates a break in the communications between the Magnet Supervision Unit and the alarm box.	Check cabling between Magnet Supervision Unit and alarm box. Check condition of 2657 MSUP.

1.6 Superconducting Shim (Espreo only)

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
Term 2.0 is out of specification	Superconducting Shim (Z-2 Shim)	Z-2 shim current	- Applying Z-2 shim current without using an Array Shim Device. - Perform a shim plot for calculating the Z-2 shim current	Online Help "Shim / Superconducting Shim"
Term 2.0 is out of specification	Superconducting Shim (Z-2 Shim)	Magnet was ramped	Applying the Z-2 shim current without using an Array Shim Device.	Online Help "Shim / Superconducting Shim"
Remarks: Check the Event log of the MR system for errors and warnings generated by the Magnet Supervision Unit.				

1.7 Shim power supply

Error Messages

- Refer to (→ Healthcare Services Knowledge Base (SKB), <https://skb.healthcare.siemens.com/default.aspx>).

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
SPSU suspect	Input power supply	Missing input power	Is power LED lit? Is fan running? Check input power at connector on front of SPSU	n.a.
Remarks: Check the Event log of the MR system for errors and warnings generated by the Magnet Supervision Unit.				

1.8 Magnet Power Supply

Error Messages

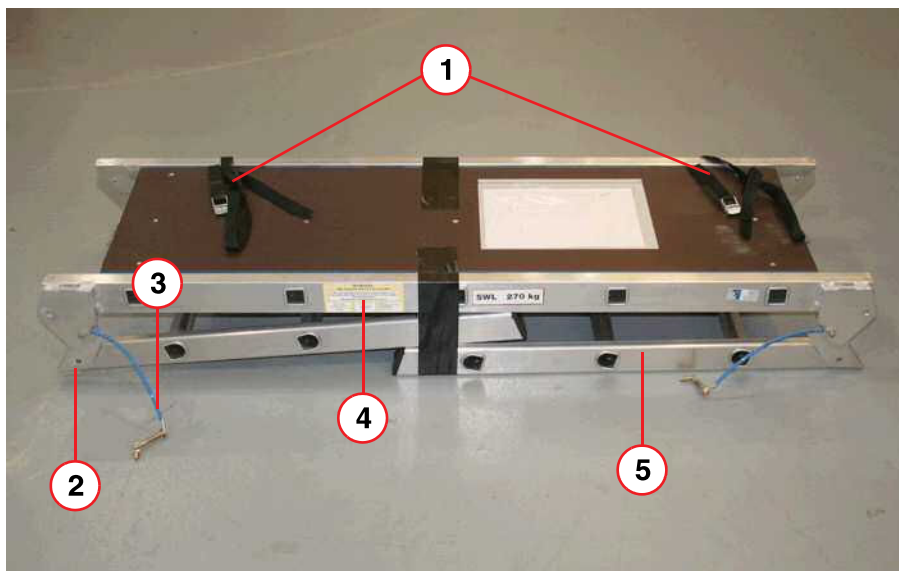
- Refer to (→ Healthcare Services Knowledge Base (SKB), <https://skb.healthcare.siemens.com/default.aspx>).

Symptom	Probable Cause	Service Action	Reference
Fault reported by HOST	n.a.	Check if fault is an MPS or magnet problem	n.a.
Temp error	Air vents Air temperature in the surrounding area	Clean air vents Check temperature of air in the surrounding area	n.a.
Power error	400 V is missing	Check 400 V on the power supply cable	n.a.
Overcurrent error	Defect on the MPS	Replace MPS	n.a.
MPS fails to obey HOST	Connection between HOST and MPS	Check RS232 connection between MPS and HOST	n.a.

2.1 Magnet

2.1.1 Service platform installation

Fig. 4: Service platform - as delivered



- (1) Safety straps (2 off)
- (2) Hinged locking pin location (each end)
- (3) Retaining wire and hinged locking pin (each end)
- (4) Warning label
- (5) Step legs (each end)

2.1.1.1 Service platform safety

Before using the service platform **read the following safety cautions** and perform the **required safety checks**.



CAUTION

Risk of personal injury

Failure to observe the following precautions may result in personal injury.

- » Always check the platform and hinge locking pins for damage.
- » Always check the hinge locking pins are secured.
- » Always secure the platform to the magnet frame or with the two long straps through the magnet bore.
- » Always ascend and descend (Walk UP or DOWN) the steps facing the platform.
- » Where possible use the frame or covers for additional hand support.
- » Never climb onto the platform from the side.
- » Never jump from the steps or top of the platform.

Fig. 5: Correct method - Up to platform



Fig. 6: Incorrect method - Down from platform

**⚠ CAUTION****Risk of personal injury**

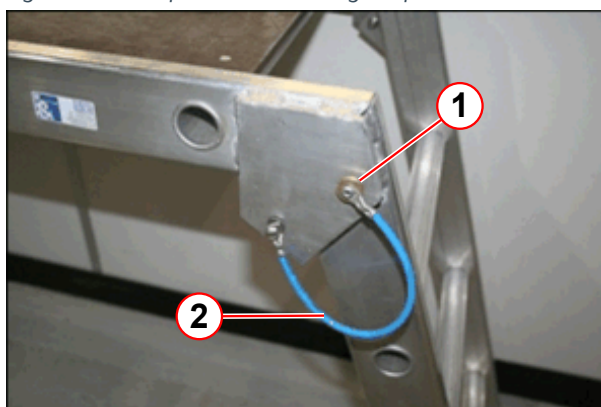
Carrying heavy components or tooling up or down the platform steps can lead to injury.

- » DO NOT carry heavy components up or down the platform steps.
- » Place heavy objects (i.e. cold head) onto the platform first.
- » Only lift the component when standing safely on the platform.

2.1.1.2 Platform assembly

- Remove any packaging.
- Unfold both step legs.

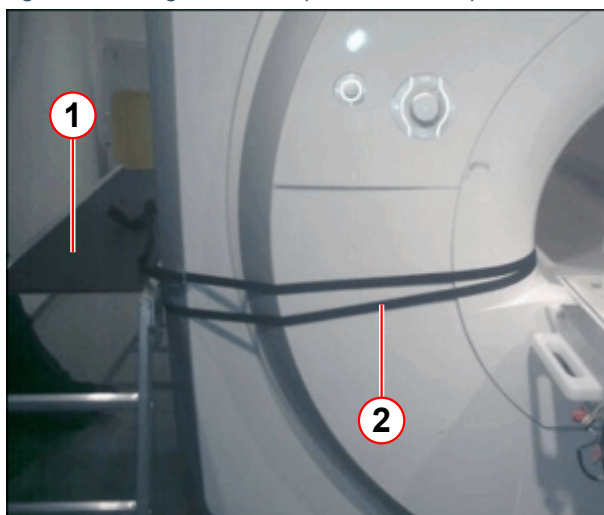
Fig. 7: Service platform - lock legs in position



- (1) Hinged locking pin
(2) Retaining wire

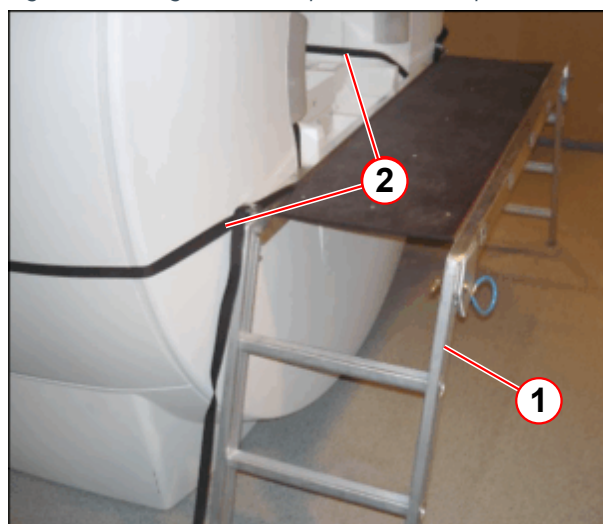
- Align the hole at the top of the leg with the **pin location hole** (pos 1).
- Push the **hinged locking pin** through the hole (pos 1) and fold back the hinge to secure.
- Repeat with the other leg.

Fig. 8: Securing the service platform (example)



- (1) Service platform
(2) Long straps (fitted through the magnet bore)

Fig. 9: Securing the service platform (example)



- (1) Service platform
(2) Long straps (fitted through the magnet bore)



Do not exceed the maximum load - the combined weight of 2 people must not exceed 270 kg.



The fiber optic connections in the top of RFIS are very fragile, so exercise due caution.

2.1.2 Depressurization of the helium vessel



CAUTION

Risk of asphyxiation and hypothermia if helium gas is vented into the magnet room

Failure to observe the following safety warning may result in dizziness or loss of consciousness for personnel.

- » Quench valve and venting system must be connected to the quench pipe.



CAUTION

Risk of injury

Danger of cold burns. Suitable protective clothing must be worn.

- » Do not undo joints on the HIGH pressure side until the system is fully depressurized.



Before opening up any of the turret ports, there must be a positive pressure of at least 7 mbarg (0.1 psig) and no more than 14 mbarg (0.2 psig) in the helium vessel, as displayed on the pressure gauge assembly.

Depressurization is performed via the **bypass valve assembly**, which bypasses the **quench valve**.

Depressurization of the helium vessel is required when:

- Helium filling - refer to the filling procedure described in the **System Startup**.
- Replacing parts on the venting system - **refer to Service - Replacement of Parts**.
- Replacing the cold head.

2.1.3 Helium filling procedure

Performing the procedure

Refer to the filling procedure described in the system start-up.

2.1.4 Venting leak check

2.1.4.1 General

For all recondensing systems, it is essential that the magnet system is leak tight to maintain a zero-loss cryogenic performance.

Individual leaks must be 1×10^{-4} mbar l/s or less.

Leak checking must be performed with an electronic leak detector capable of detecting leaks, at least as small as **1×10^{-4} mbar l/s**.

The magnet must be checked after any **service operations** have been performed, such as:

- During installation.
- After helium filling.
- After a cold head exchange.
- After any exchange or disturbance of a part on the high pressure side of the magnet.



CAUTION

Risk of icing

Fine leaks may not show any change in pressure. Over time, they will contribute to air ingress and icing.

» Leak check with a suitable leak check device.

2.1.4.2 Venting system

The document describes the procedure for **Mk1** and **Mk2** venting systems.

Fig. 10: Mk1 OR105 Venting System



Fig. 11: OR122 Mk1 Venting System

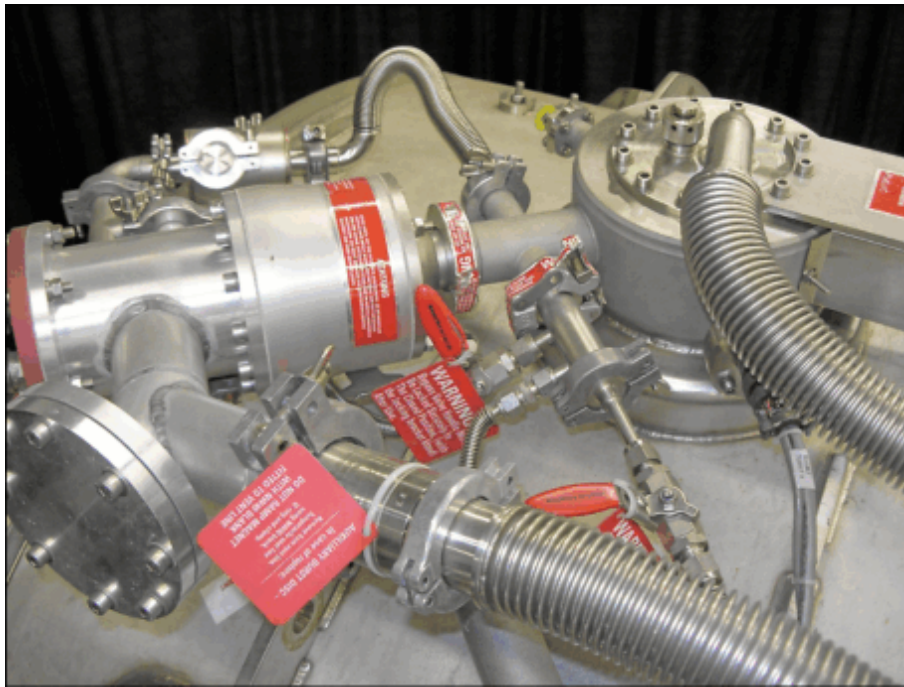


Fig. 12: Mk2 Venting System



2.1.4.3 Safety

! CAUTION

Pressurized helium vessel.

Cold helium gas.

Failure to observe the following safety information may result in physical injury.

- » Fittings on the high-pressure side must not be disconnected without first depressurizing the helium vessel.

! WARNING

Magnet at field.

Personal injury or damage to equipment may occur.

- » Do not use magnetic tools for repairing or tightening seals when the magnet is at field.
- » Only non-magnetic equipment may be taken into the magnet room.



It may also be necessary to disable the pressurization heater.

2.1.4.4 Equipment required

- Heat gun.
- MSUP-Display Unit.
- Paper towelling.
- Accepted leak detection equipment:
 - Helium mass spectrometer

Fig. 13: Edwards Leak detector



- Use the Edwards hand-held leak detector (part no **10614850**).

This is the **recommended equipment** for leaks **greater than 1×10^{-4} mbar l/s**.

NOTE: The hand-held device is calibrated in **ml/sec**:-

1×10^{-4} mbar l/s = 1×10^{-4} ml/sec at normal pressure and temperature.



The functionality of the Edwards leak detector can be compromised by magnetic fields.



The mass spectrometer is used for the detection of ultra fine leaks less than 1×10^{-5} mbar l/s. This equipment must only be used when trouble-shooting is required.

- Check that the equipment is detecting helium.
 - Check the equipment using helium from a known gas supply (i.e. regulated gas bottle).
 - Refer to the equipment manufacturer's operating instructions.
 - The calibration certificate must be valid.

2.1.4.5 Pre-requisites for leak checking

- A suitable electronic leak detector is available.
- A suitably long remote sensor must be used when magnets are at field.
- The helium vessel must at least be at **normal operating pressure (15.3 psi (A))** prior to leak checking.
 - Activate the pressure heater (if required) to raise pressure.
- All joints are dry and at room temperature (use a heat gun).

2.1.4.6 Leak checks at installation/service

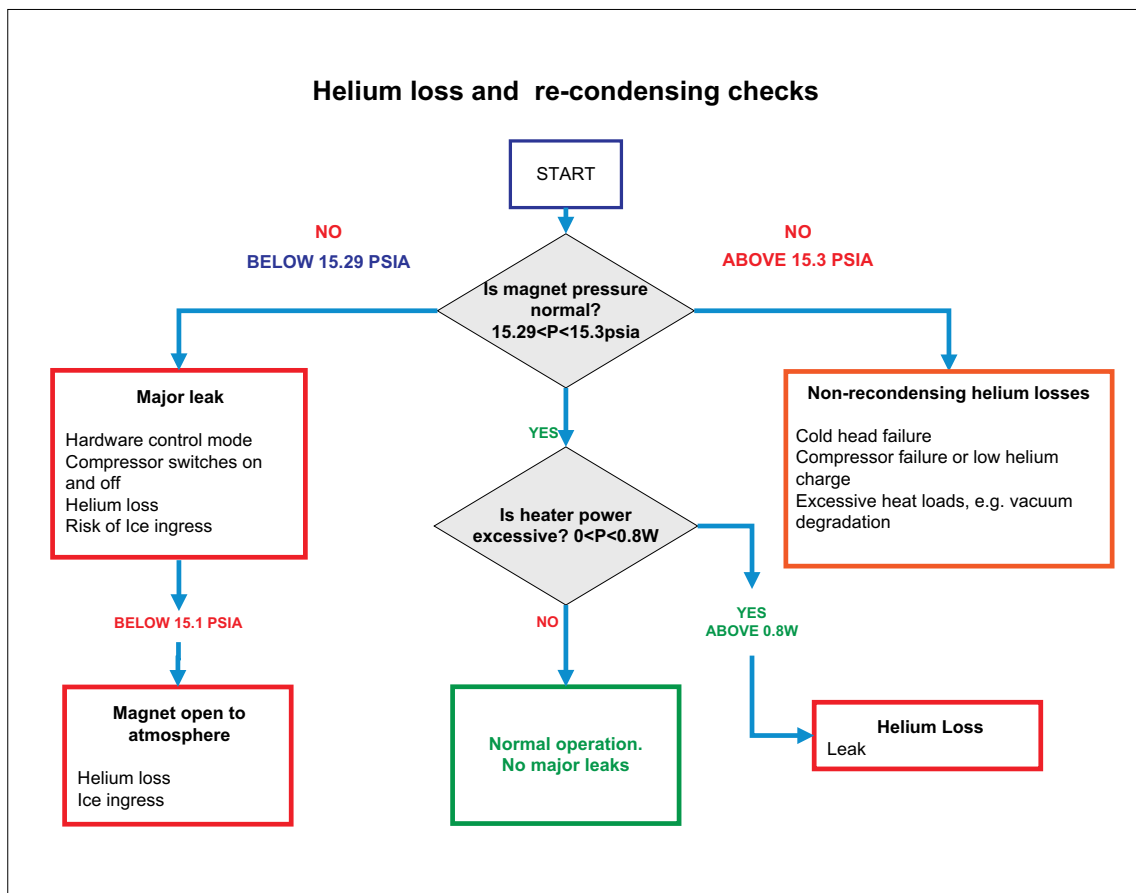
Installation

- Check the **system pressure** on arrival at the installation. If the system has **low**, or **zero** pressure then a **gross leak** could have occurred during shipment.
 - All seals, burst disks and fittings on the **high pressure side** of the vent valve must be checked **before** the cold head is turned on and the magnet run to field.
 - Any **joints** which are broken and re-made during installation must be checked.
 - Leaks can occur through the **internal venting system** (valves) as well as to atmosphere.

Service

- Leak check all **seals** that have been subjected to **prolonged low temperature** and the subsequent build up of frost. For example, those which have been cooled to low temperature by helium transfers, helium vessel de-pressurization, or magnet ramping.
 - Any **joints** which are broken and re-made during servicing must be leak checked.
 - The **turret, 16 psi (A) venting valve assembly** and quench valve must be **warmed up to room temperature** after any service procedures have been performed, prior to leak check.

Fig. 14: 4K Helium loss and re-condensing checks



The illustration above shows the condition in which the magnet should return, after a number of hours from use, and has not been affected by external heat loads.

2.1.4.7 Procedure



The leak detector must be set for maximum sensitivity.

- Switch on the leak detector and allow it to warm up.
- Check the equipment is working correctly (detecting helium). This should be done both before and after use.



Ensure that the atmospheric environment is not saturated with helium gas (i.e. do not leak check soon after a helium fill, or depressurization).

- Determine the background level in the magnet room.



The background must be at least one order of magnitude lower than the smallest leak expected, (i.e. leaks of 10^{-4} mbar l/s, require background of order 10^{-5} mbar l/s). This level will be undetectable by the hand-held device.

- Note response time of the detector.



During leak checking, move the instrument slowly over the leak point to give the detector time to respond.

Start at the **lowest** point:

- Hold the detector at the suspected leak point until the reading has stabilized.
- Move the detector away and check that the indication disappears.
- Repeat the process to check that a leak is still indicated.

The following points must **always** be observed:-

- Joints on the '**HIGH**' pressure side of the venting system will generally be fitted with **metal seals**. Before dismantling a joint, ensure a replacement seal is available.
- **DO NOT** use rubber seals as a substitute for metal seals.
- **DO NOT** re-use metal seals, always replace with a new seal.
- **DO NOT** use vacuum grease.



Do not use any form of vacuum grease on O-rings and seals. It is recommended that these be wiped clean and checked for damage prior to fitting. Vacuum grease dries over time and can cause a seal to leak.

- After repair, leak check the entire venting system, not just the joints which have been repaired.

2.1.4.8 External & internal leak checks

External : The venting and components on the **high pressure** side of the magnet system require leak checking to ensure that the re-condensing performance is achieved. This includes:

- Cold head.
- Turret top plate, including helium syphon port and venting.
- 16 psi (A) Vent valve assembly.
- Pressure gauge assembly.
- Connecting pipe couplings.



If no external leak is found but the system pressure is low, then it is possible that one of the internal valves is leaking. This can be caused by a cracked or broken burst disk, or a broken O-ring.

Internal:

- Auxiliary 1.8bar metal bursting disk.
- Quench valve.
- 16 psi (A) valve.



The auxiliary and quench valve burst disks (including the quench valve O-ring) must be checked after a magnet quench.



The magnet must be at ZERO FIELD before leak checking inside the quench valve or auxiliary vent.

If leaks are found at vent line connections, first tighten the clamp, or securing screws and re-check. **DO NOT OVERTIGHTEN.**

If this does not cure the leak, replace the **metal seal**.

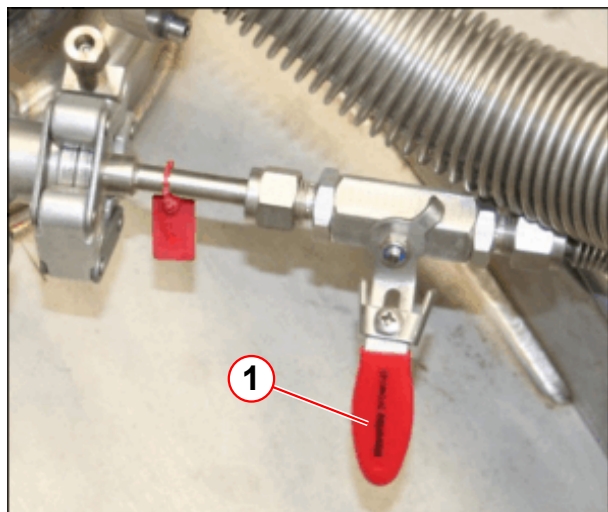
2.1.4.9 Cold head external checks

2.1.4.10 Mk1 and Mk2 venting



The cold head vent line is to remain connected with the hand-valve CLOSED during normal system operation.

Fig. 15: Mk1 Venting - Cold head vent line valve



(1) Valve (shown closed)

Fig. 16: Cold head vent line valve is closed



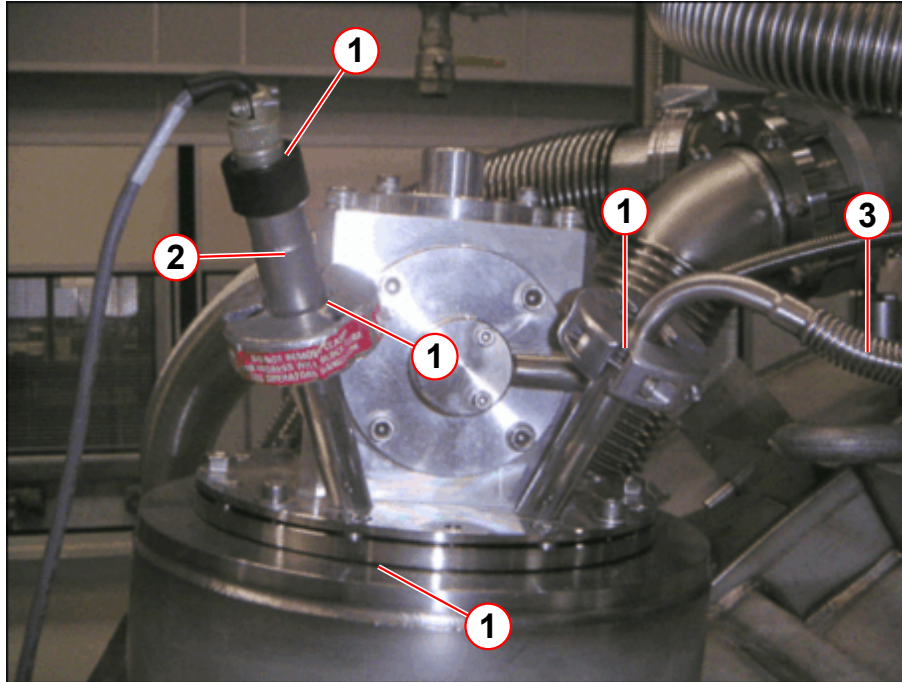
(1) Cold head vent line valve is closed

- Check the **cold head vent line seals** at both ends. (→ 3/Fig. 17 Page 37),
(→ 3/Fig. 18 Page 38)



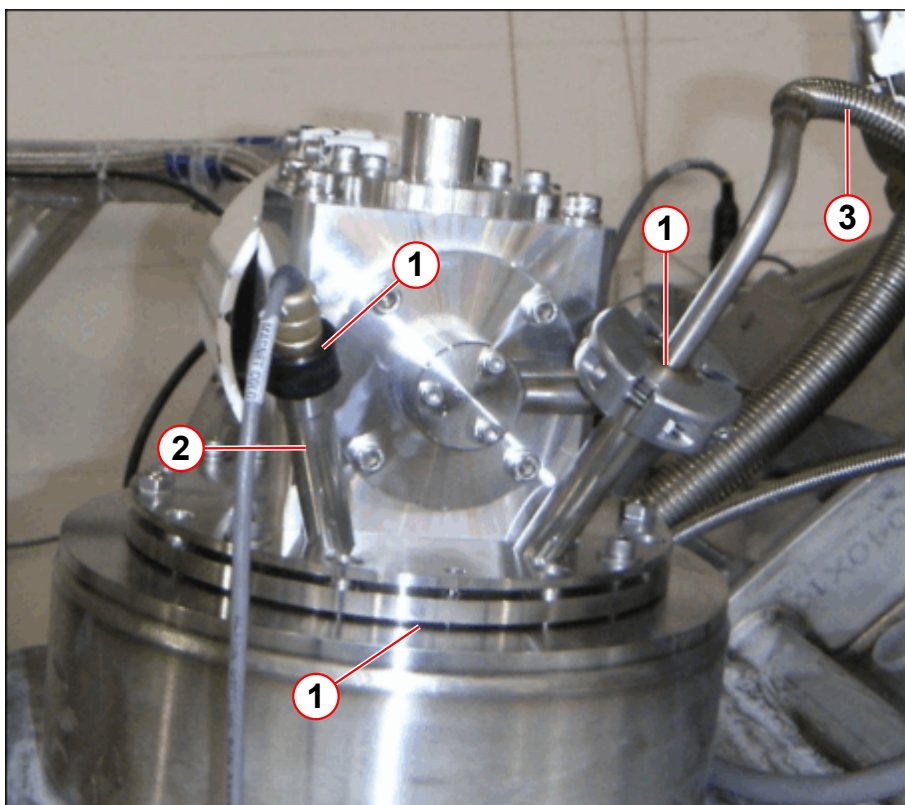
Earlier cold heads have an NW16 clamp fitted to the temperature sensor feed-through.

Fig. 17: Leak checking the cold head seals (early coldhead)



- (1) Cold head seals leak checkpoints
- (2) Temperature sensor feed-through
- (3) Cold head vent line

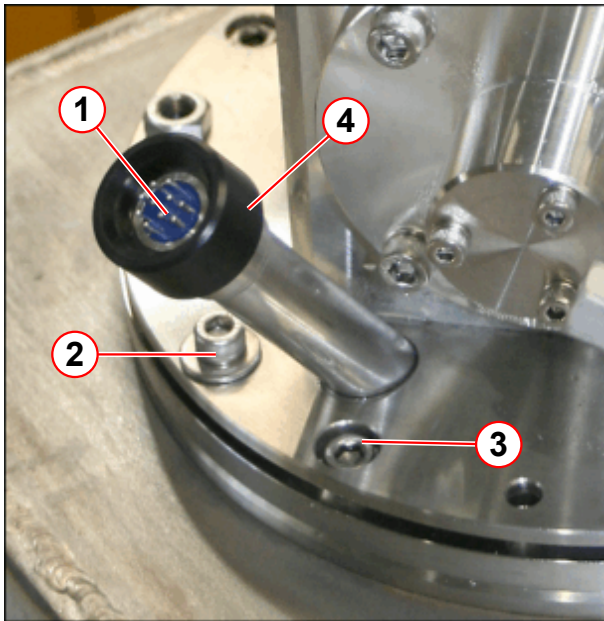
Fig. 18: Leak checking the cold head seals (later coldhead)



- (1) Cold head seals leak check points
- (2) Temperature sensor feedthrough
- (3) Cold head vent line

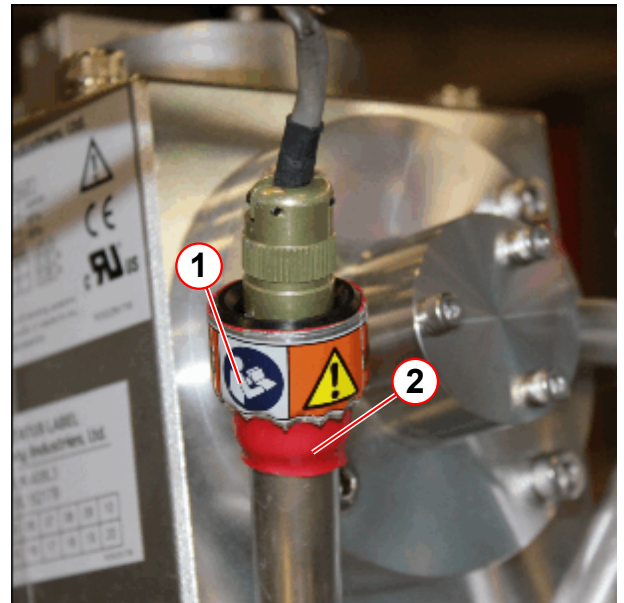
- Check the **hand-valve seals** on the cold head vent line. (→ Fig. 15 Page 36), (→ Fig. 16 Page 36)
- The **cold head** is sealed in position by an **O-ring**. (→ 1/Fig. 17 Page 37), (→ 1/Fig. 18 Page 38)

Fig. 19: Cold head 10 pin seal



- (1) Leak check point
- (2) 4 off M6 x 25 mm cold head flange screws, torque to 4Nm
- (3) 8 off M6 x 16 mm skt hd screws O-ring clamp plate, torque to 8Nm
- (4) Black sealing cap

Fig. 20: Protective heat shrink & warning label



- (1) Warning label
- (2) Heat shrink



New cold heads are now supplied (from July 2015) with protective heat shrink and a warning label fitted around the temperature sensor connector.

The cold head is fitted with **temperature sensors**:

- The **feed-through** for the wires is sealed with an **O-ring**.
- Check the **feed-through** and **cold head seals**.
- **Check the torque** of the cold head **screws**.



If a leak is found with the magnet at field, it is permissible to re-torque the 4 off cold head flange screws to a higher setting of 5 Nm.
If the leak persists, send a report to your local Uptime Service Center for repair by a specialist.

2.1.5 Repairing leaks on the turret

2.1.5.1 General

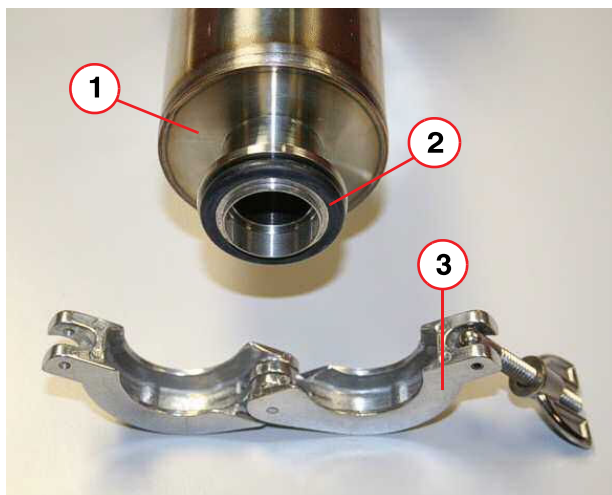
The following points must be observed at all times:

- **DO NOT** re-use metal seals, always replace with a new seal.
- Before dismantling a joint, ensure that a replacement seal is available.
- **DO NOT** use vacuum grease on metal seals.

- Having dismantled the joint, ensure that the sealing faces are clean and dry, and that there are no scratches across the sealing faces.

2.1.5.2 Low Pressure side

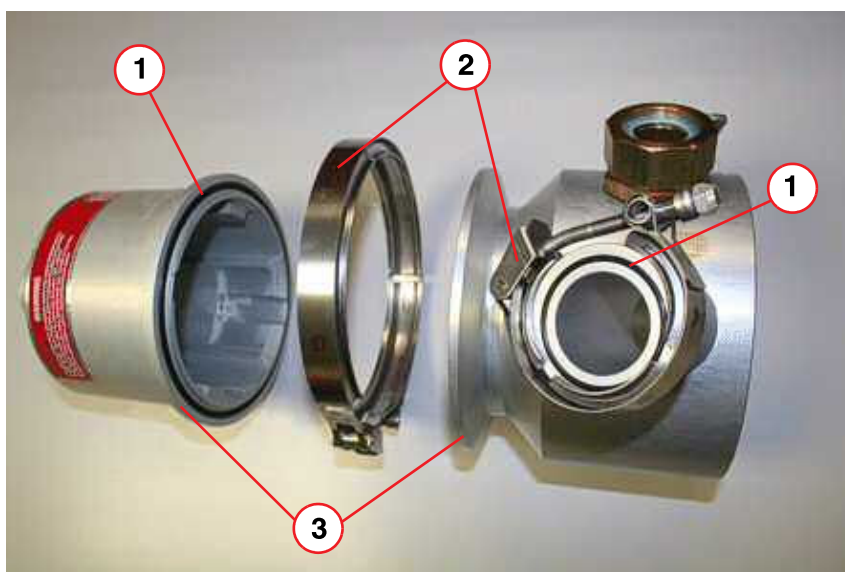
Fig. 21: Low pressure joint



- (1) Component
- (2) Rubber O-ring and carrier
- (3) 2 part hinged clamp

- Joints on the **low pressure** side of the venting system will always be fitted with **rubber** seals (pos 2 **10100529**).
- Always check clamps are tight (pos 3 **07040710**).
- Rubber 'O'- rings may be re-usable. Check for any damage prior to assembly.

Fig. 22: Low Pressure - Quench Valve joints



- (1) Recessed rubber 'O'-rings
- (2) Strap clamps
- (3) 'V' Flanges

Part Numbers - Quench Valve joints

Tab. 2 Quench Valve Low Pressure joints

	Rubber 'O'- ring	'V'Clamp
Large	10100147	10100047
Small	10100151	10100046

2.1.5.3 High Pressure side**CAUTION**

Pressurized helium vessel, cold helium gas.

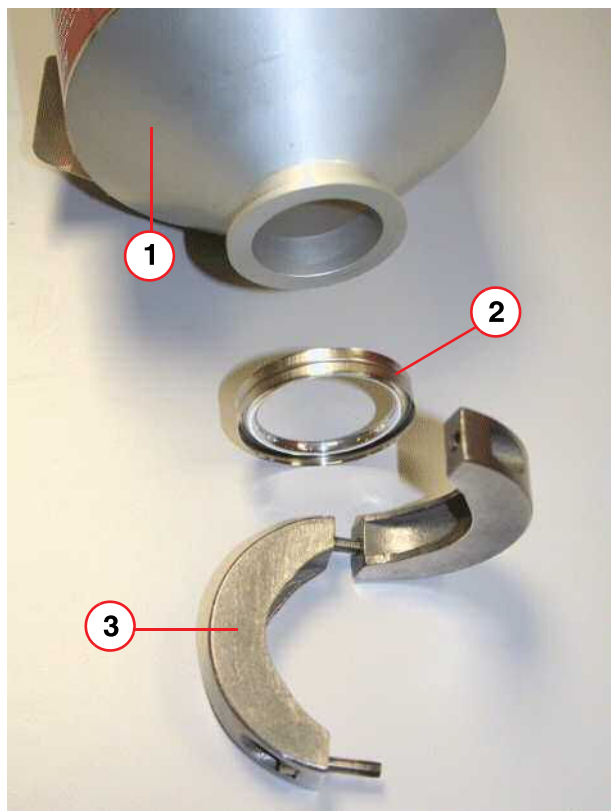
Personal injury may occur.

- » The magnet must be ramped to zero field.
- » Protective clothing, gloves, and eye protection have to be worn.
- » The magnet must be de-pressurized before undoing joints on the high-pressure side. Refer to (→ Depressurization of the helium vessel / Page 29)

For leaks found on the **high** pressure side, first try tightening the clamp or the securing screws. **Beware of overtightening.**

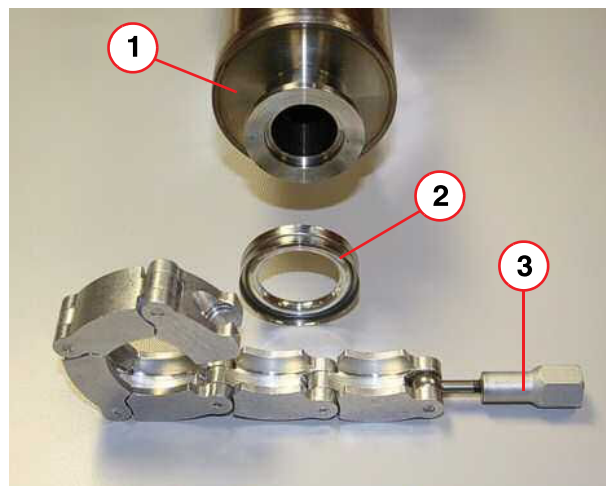
- In most instances, high pressure joints are fitted with a metal seal. **Do not replace metal seals with rubber or silicon type seals.**

Fig. 23: High Pressure - Quench valve joint



- (1) Quench valve
- (2) Metal seal
- (3) 2 part clamp

Fig. 24: High Pressure joint



- (1) Component
- (2) Metal seal
- (3) Chain clamp

Part Numbers - High pressure joints

Tab. 3 High Pressure joints

	Seal	Clamp
NW40	10100171	10100038
NW25	01564025	01645439

- If tightening does not solve the leak, disassemble the joint.



There is considerable danger of allowing air into the system if the joint is on the high-pressure side. As soon as the joint is undone, plug the open tube with a wad of paper to prevent air from flowing in.

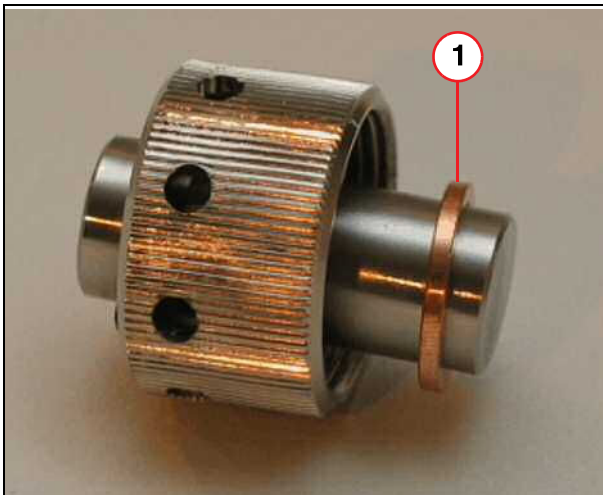
Do not have two joints open at the same time.

Do not leave the system open for longer than is necessary to complete the repair.

- Re-assemble the joint using a **new seal**. Ensure the seal is undamaged and clean, and is seated correctly.
- After repair, leak test the **entire venting system**, not just the joints that have been repaired.

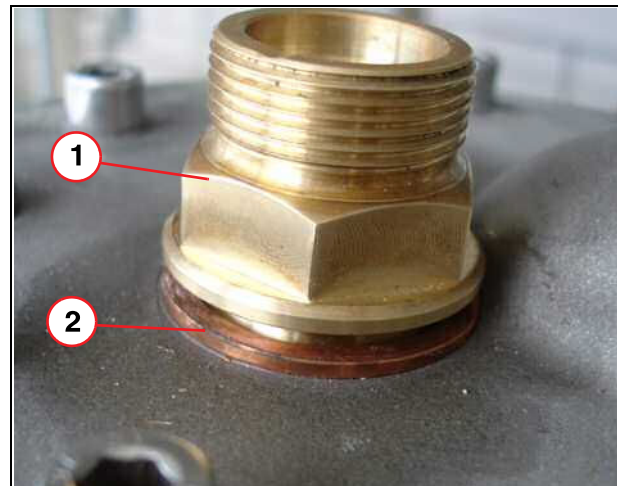
2.1.5.4 Syphon port

Fig. 25: Mk1 syphon blanking plug



(1) Copper seal

Fig. 26: Mk2 Fill adaptor



(1) Fill adaptor
(2) Copper seal

- Fit a new, clean undamaged **copper seal** -
 - Mk1 - **03108094**
 - Mk2 - **10103250**
- Ensure that the port surface is clean and dry. Tighten the ring nut firmly.



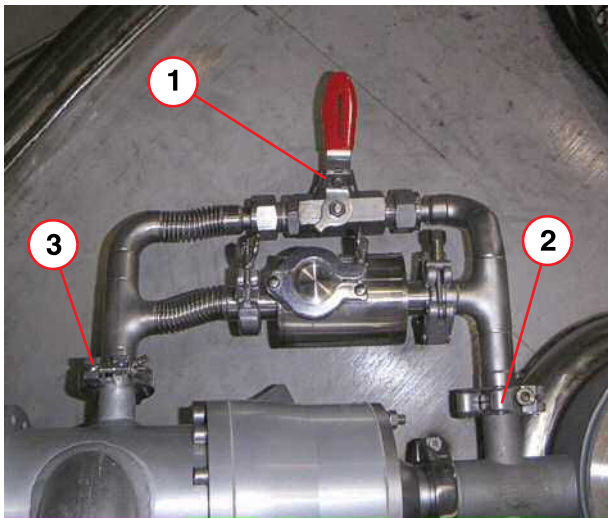
Do not tighten the syphon port blanking plug when the threads are cold and frosty. Always warm up first.

2.1.5.5 Quench valve

If the quench valve is leaking refer to "Replacing the Quench Valve Burst Disc" (→ Magnet System / M6-020.841.17)

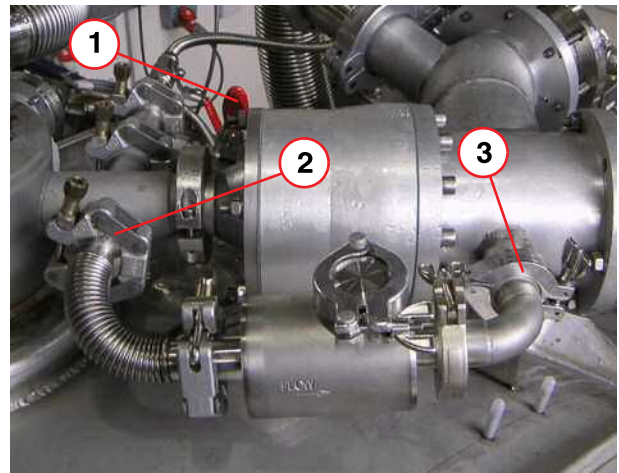
2.1.5.6 Bypass valve

Fig. 27: OR105 Mk1 Vent valve assembly



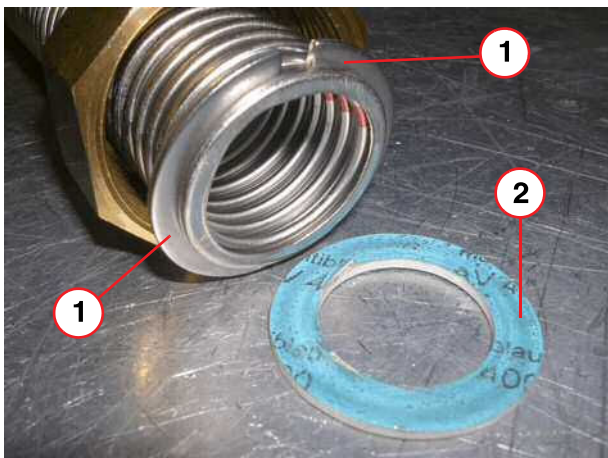
- (1) Bypass valve
- (2) NW25 High pressure clamp (turret)
- (3) NW25 Low pressure clamp (manifold)

Fig. 28: OR122 Mk1 Vent valve assembly



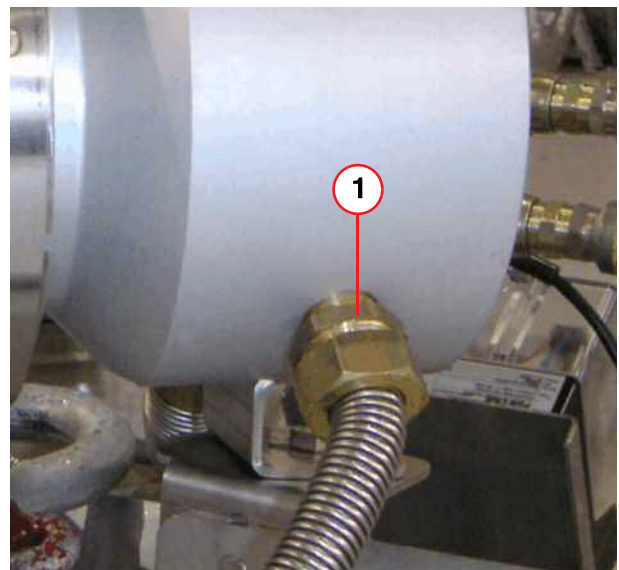
- (1) Bypass valve
- (2) NW25 High pressure clamp (turret)
- (3) NW25 Low pressure clamp (manifold)

Fig. 29: Mk2 16 psi (A) vent manifold connection



- (1) Split rings
- (2) Gasket

Fig. 30: Mk2 Manifold connection

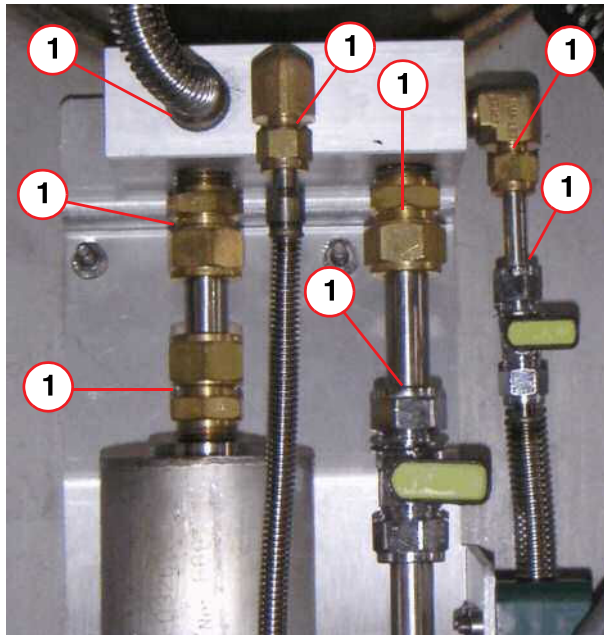


- (1) Manifold connection

Part Numbers - 16psia valve

- MK2 16psia gasket
 - Large - **10113279**
 - Small - **10113277**

Fig. 31: Mk2 Venting 16 psi (A) Valve Assembly



(1) Leak test points

If any bypass valve compression seals continue to leak after tightening, the complete 16psia bypass valve assembly must be replaced..

Refer to Replacing the Vent Assembly

If the 16psia valve leaks internally due to a failed **valve** the complete 16psia valve assembly must be replaced.

To determine if a 16psia valve has a internal leak:

- Remove the flexible line from the quench manifold (→ Fig. 30 Page 44).
- Fit a surgical glove over the end of the open flexible line and seal.
 - If the glove expands with the magnet pressure **lower** than 16psia (i.e.15.1psi) then the valve is not seating and the complete assembly will need to be replaced.

2.1.5.7 Cold head bypass line

If the valve body, the seals at the ends of the line, or the seals at the valve are leaking, and the seals leaks cannot be closed by carefully tightening, contact the **Siemens Magnet HSC**. (→ Fig. 32 Page 46)

2.1.5.8 Refrigerator temperature sensors

The cold head is fitted with temperature sensors. The feed-through for the wires is sealed with an "O" ring. (→ 2/ Fig. 32 Page 46)

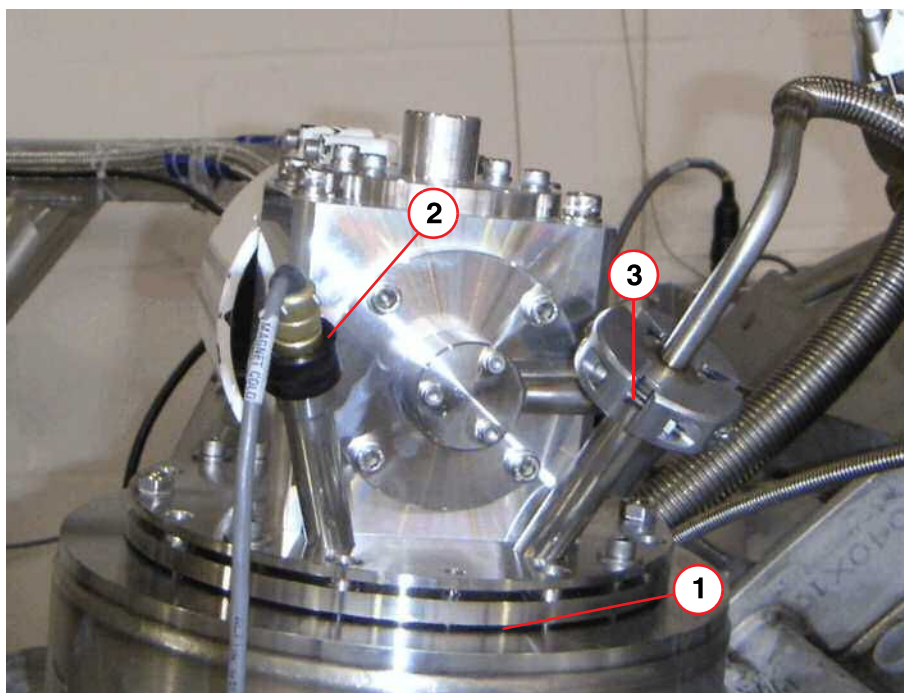
If the feed-through is damaged or leaking, contact **Siemens Magnet HSC**.

2.1.5.9 Cold head interface

- Check the torque of the cold head flange bolts. (8Nm and 4Nm)

Replacement of the cold head 'O'-ring interface seal requires a specialist. (→ 1/ Fig. 32 Page 46). If a leak is suspected, contact the **Siemens Magnet HSC**.

Fig. 32: Mk2 Cold head seals leak checks



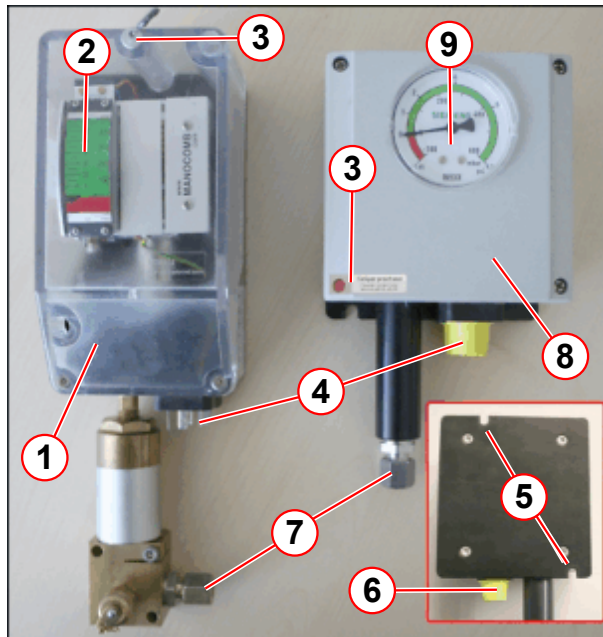
- (1) Cold head 'O'-ring
- (2) Cold head wiring plug and seal
- (3) Cold head vent line seal

Part Numbers - MK2 Cold head

- NW10/16 (pos 3)
 - Clamp - **03420783**
 - Seal - **08396496**

2.1.5.10 Pressure Gauge

Fig. 33: Pressure switch transducers Manocomb and WIKA



- (1) Manocomb Transducer assembly
- (2) Linear gauge
- (3) Tamperproof seal
- (4) Front/rear electrical connections
- (5) Mounting slots (rear view)
- (6) Protection caps
- (7) Pressure line connection
- (8) WIKA Transducer assy (part no. 10126452)
- (9) Dial gauge

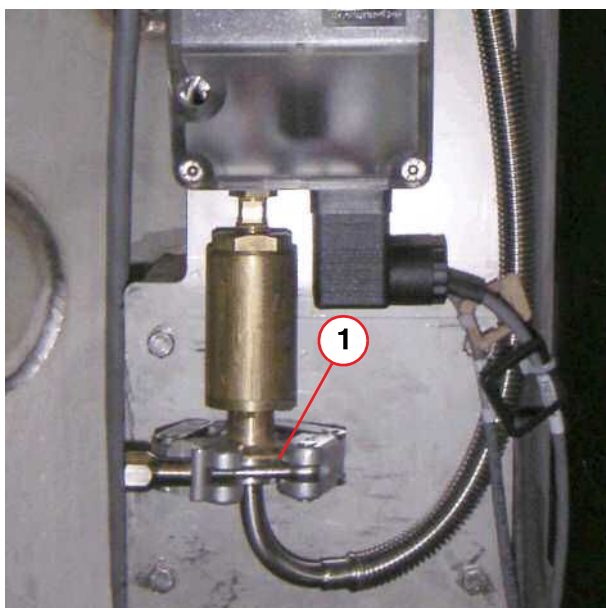
The **WIKA** Pressure switch transducer assembly will **replace** the **Manocomb** pressure switch when stocks are exhausted.

The functionality of the WIKA unit is identical, and the same fixings are used for installation.



The WIKA Pressure switch transducer is NOT directly interchangeable with the MK1 Manacomb switch. Contact Magnet HSC for assistance.

Fig. 34: Mk1 Pressure Gauge



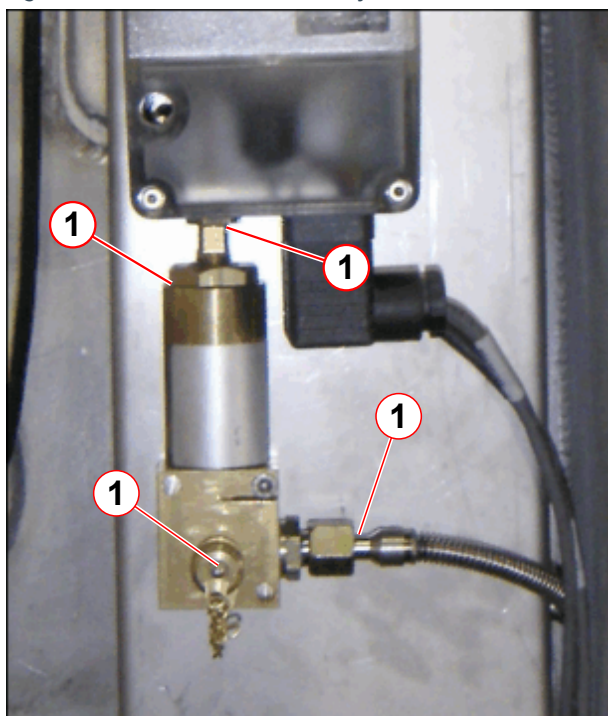
(1) Leak test point

- **Mk1** - If the leak cannot be closed by carefully tightening the **clamp** replace the **NW25 metal seal**.

- **Mk2** - If the leak cannot be closed by carefully tightening the **compression nut**, replace the **pressure line**, **compression nut** and **ferrule**.
- **Do not overtighten** the compression nut.

Manocomb leak check points

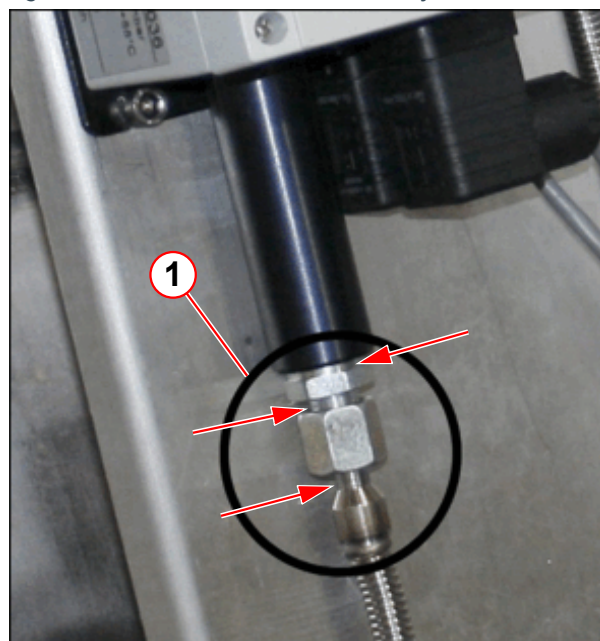
Fig. 35: Pressure switch assembly - leak checks



(1) Leak check points

WIKA leak check points

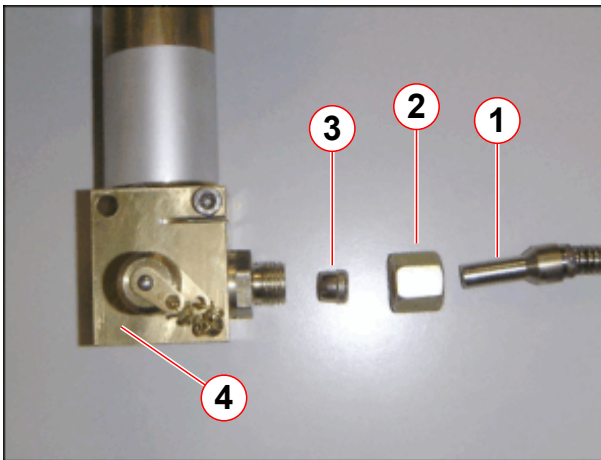
Fig. 36: WIKA Pressure switch assembly - leak checks



(1) Leak check points

Manacomb compression joint

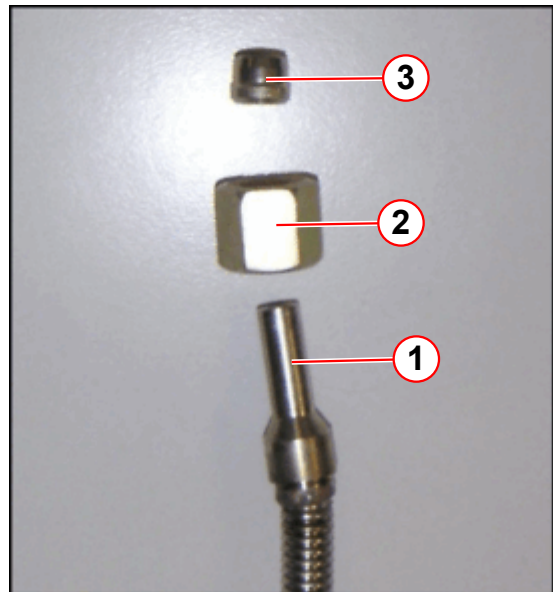
Fig. 37: Pressure switch compression joint



- (1) Pressure line
- (2) Compression nut
- (3) Compression ferrule
- (4) Transducer body

WIKA compression joint

Fig. 38: WIKA Pressure switch compression joint



- (1) Compression line
- (2) Compression nut
- (3) Ferrule

Part Numbers - Mk2 Compression Joint

- Pressure Line - **10111085**
- Compression nut - **10113284**
- Compression ferrule - **10113289**

Refer to Replacing the Pressure Switch Transducer.

2.1.6 What to do after a Quench (Avanto, Avanto Dot Upgrade, Espree with MSUP K2256)

2.1.6.1 General

What to do after a Quench (Avanto, Avanto Dot Upgrade, Espree with MSUP K2256).

Should the magnet be quenched deliberately (via the ERDU) or accidentally, (i.e. persistent mode quench), a large volume of helium gas will be generated within the helium reservoir. The quench valve will open, and stay open until the flow subsides. The valve will then re-seat, re-sealing the venting and thus preventing any air (caused by cryo-pumping) from being sucked back into the helium vessel. There is a possibility that the graphite burst disks might have ruptured.

Mobile systems are fitted with a **High Altitude Burst disk**, which will rupture when the magnet quenches.

The MSUP should automatically maintain the system pressure above atmospheric. **If this is not the case, contact the Magnet HSC immediately.** (→ Magnet HSC, <mailto:magnet.wscshsc.healthcare@siemens.com>)



When a quench is detected, Magnet HSC must be notified.

Fill out the **Quench Report database** (→ Product Information MR, <https://bps-healthcare.siemens.com/productinformation/ProductInformation.aspx?uuid=7abf6bb6-5daf-440c-9920-fab2d4640084>) - Links - MR - Quality Database (and Quench Report) and send to Magnet HSC.



CAUTION

If the graphite burst disk breaks.

Failure to observe the following will result in ICE FORMING within the magnet.

- » Replace the graphite or auxiliary vent bursting disk IMMEDIATELY.
- » DO NOT RAMP THE MAGNET.
- » Inform Magnet HSC. (→ Magnet HSC, <mailto:magnet.wscshsc.healthcare@siemens.com>)

NOTICE

The magnet helium level can be recovered if the magnet temperature is between LN2 and He temperature.

The CCR (CARBON RES 1) reading must be in the range of between 1200 and 3400 ohms. Compare to the values on the Magnet Acceptance Document, or with HSC.

The magnet must be re-filled immediately with helium.

To re-fill the magnet from zero helium, or warm carbon resistor temperatures, perform a BOTTOM FILL procedure.

If proceeding with a helium fill, immediately place an order for 2000 liters of liquid helium.

- » Contact Magnet HSC IMMEDIATELY before proceeding.



CAUTION

IF the magnet is allowed to warm up above 77K, a full nitrogen pre-cool will be required.

Helium filling will not cool the magnet sufficiently to collect liquid helium.

A full nitrogen pre-cool takes about 10 days.

- » Contact Magnet HSC IMMEDIATELY before proceeding.

Fig. 39: OR105/122 Magnet Acceptance Document (CR)

SIEMENS		Magnet Acceptance Documentation Installation Data		Siemens Magnet Technology Quality Regulations	
Cryostat Number	21199812	SMT Reference	21201632 02.09		
System Description 1.494T Actively Shielded Superconducting Twin Turret Magnet System OR105					
Measured Magnet Centre (Offset)		-1 (S) mm			
If magnetic Offset is outside the spec (+_5mm) the magnet will be marked (+_5mm) and Siemens Magnet Technology Ltd will be required to Shim the Magnet					
Operating Parameters					
Description	Specification value	Measured value			
Operating Current @ 1.494T	< 600 AMPS	509.95		AMPS	
Field/Current Ratio	N/A	12.48		kHz/O.1 Amp	
Carbon Resistor Values					
POSITION	Room Temperature	Helium Temperature			
Top	162.4	1140			
One	134.4	903			
Two	132.3	994			
TDC	140.4	943			
Helium Level Probe Resistance (When using a 250 mA supply)					
		Probe 1	Probe 2		
@ 0 Resistance	Ohms	85.36	85.36		
0 % Calibration		774	774		
100 % Resistance	Ohms	0.22	0.22		
100 % Calibration		2	2		
Diode Table No		-1			
Pressure Heater Res	Ohms	19			
Final Temperature Sensor Recon Values					
50kK	50kK Lin	Coldhead	Recon Margin		
48	45	42	K 467 Millivolts		
Checked by	K Suraina		Signature		
Date	02/02/2009				



Ceramic Carbon Resistors.

Carbon resistors have been replaced with Ceramic Carbon resistors (CCRs), which have a different resistance range.

Refer to the Magnet Acceptance Document.

Fig. 40: OR105/122 Magnet Acceptance Document

SIEMENS		Magnet Acceptance Documentation Installation Data	
MR Magnet Technology Quality Regulations			
Cryostat Number	21243234	SMT Reference	21245901 06.10
System Description		OR105	
1.494T Actively Shielded Superconducting Twin Turret Magnet System			
Measured Magnet Centre (Offset)		-2 (S) mm	
If magnetic Offset is outside the spec (+_5mm) the magnet will be marked (+_5mm) and Siemens Magnet Technology Ltd will be required to Shim the Magnet			
Operating Parameters			
Description	Specification value	Measured value	
Operating Current @ 1.494T	> 600 AMPS	510.05	AMPS
Field/Current Ratio	N/A	12.48	kHz/O.1 Amp
Carbon Resistor Values			
POSITION	Room Temperature	Nitrogen(77K)	Helium Temperature
TDC d - e	878		2567
Top a - e	926		2938
Mid b - e	985		2715
Bottom c - e	877		2613
TDC Calibration Values:	850.4	1122.9	2731.7
Helium Level Probe Resistance (When using a 250 mA supply)			
		Probe 1	Probe 2
@ 0 Resistance	Ohms	85.36	85.36
0 % Calibration		774	774
100 % Resistance	Ohms	0.22	0.22
100 % Calibration		2	2
Diode Table No		-1	
Pressure Heater Res	Ohms	19	
Final Temperature Sensor Recon Values			
50kK	50kK Lin	Coldhead	Recon Margin
53	47	44	530 mW
Checked by		Signature	
K Suraina			
Date			
02/06/2010			
Page 1 of 1		25-AD EPS 000 01	

Tab. 4 Expected Carbon Resistor Values

41 way plug Measured in Ohms	Expected Values Room Temperature	Expected Values 77K Temperature	Expected Values 4.2K Temperature
Carbon One a - e (Bottom)	140 - 180	170 - 200	700 - 1500
Carbon Two b - e	120 - 160	130 - 180	700 - 1500
Carbon Three c - e	120 - 160	130 - 180	700 - 1500
Carbon Four d - e (Top)	120 - 160	130 - 180	700 - 1500
All Ceramic Carbon Resistors (CCRs)	850 - 1050	950 - 1350	2300 - 3200



The above table is a guide only. Always compare the readings with those on the Magnet Acceptance Document.

Safety



Only Siemens CSEs trained in ramping magnets are allowed to perform this work in the event of a quench.



Switch off and isolate the magnet power supply from the mains before further action.



WARNING

Risk of asphyxiation.

Failure to observe the following safety warning may result in dizziness and loss of consciousness in personnel working with cryogenics.

- » Do not vent helium gas into the magnet room during helium transfer. As an exception, a small quantity of helium gas can be vented to cool the syphon.
- » Ensure the room is ventilated.
- » Ensure that the turret venting kit is connected to the quench pipe.
- » O₂ monitors must be used in the imaging suite.



CAUTION

RISK OF ICING!!

Air ingress will cause INTERNAL ICING OF THE TURRET.

- » Replace broken or damaged burst disk/O-rings IMMEDIATELY.
- » Check that there is a free flow of gas through the magnet bypass valve. If there is no gas flow, there may be an ice blockage.
- » Report the icing to Magnet HSC IMMEDIATELY.
- » Do not fill or ramp the magnet until advised to do so by Magnet HSC.
- » DO NOT OPEN OR REMOVE THE TURRET TOP PLATE OR VENTING.

Required tools

- Helium remote display unit, (material number **07758522**).
- Heat gun.
- Standard tool kit.
- Pressure service tool (material number **03861221**)
- Laptop and RS232 cable
- O₂ monitor

Fig. 41: Edwards Leak detector



- Use the Edwards hand-held leak detector (part no **10614850**).

This is the **recommended equipment** for leaks **greater than 1×10^{-4} mbar l/s**.

NOTE: The hand-held device is calibrated in **ml/sec**:-

1×10^{-4} mbar l/s = 1×10^{-4} ml/sec at normal pressure and temperature.

Post Quench Steps

Follow the work steps in the table below for the different types of quench.

	Type of quench		
	Quench during ramping	Quench during He filling	Persistent quench or ERDU button pressed
Step 1	- Isolate the MPSU from the magnet. - Download the 3600 MPSU ramp log files.	Stop filling.	Check that no ERDU buttons are pressed. Re-set ERDU button.
Step 2	Check the liquid helium level using the MSUP-Display Unit.		
Step 3	Order about 1000 liters liquid helium ¹		
Step 4	Warm up and check the burst disk are intact: Auxiliary burst disk:- (→ Auxiliary vent burst disk / Page 62) Quench Valve burst disk:- (→ Burst disk Inspection / Page 68)		
Step 5	After 2 hours : Download the Mag Supervisory log files. Helium fill. Refer to Start up .		
Step 6	Leak check the quench valve and venting. Refer to Turret leak check .		
Step 7	Fill out the Quench report. Refer to Quench report .		
Step 8	Ramp the magnet. Refer to Start up		
Step 8	- Perform the Shim check » If the shim check exceeds the specifications, a re-shim is necessary.		
Step 9	Contact Magnet HSC. » Email Mag Power Supply and Mag Supervisory logs to: Magnet HSC with magnet serial no. and all log data. Refer to Startup		

1. The quantity of the order depends on the helium volume already on site.

2.1.6.2 Procedure

Unattended quench

A magnet may quench with, or without a trained CSE in attendance. For **unattended quenches** indicators are:

- If the **ERDU** has been activated, an alarm LED will show on the alarm box.

- If the **helium level** is low, an alarm LED will show on the alarm box.
- If the **pressure** is very low, the cold head will have switched off.
- Imaging will not be possible if there is no magnetic field.

Log file capture

Magnet Supervisory

- The **quench data (VQ) will be lost**, if:
 - a) The system is **re-booted** within 2 hours of a quench
 - b) The system is **turned off** within 2 hours of a quench
 - c) The data is downloaded within 2 hours of a quench
- Download the MSUP log files.

If the magnet quenches during ramping:-

- Download the **Magnet Power Supply** log file (see Installation - Startup - Downloading MPS Log files).
- **Switch off and disconnect** the MPSU from the magnet.



The MPSU log files MUST be downloaded BEFORE the next ramp is performed, otherwise previous data will be overwritten!

If the magnet quenches during helium fill

- **Stop** the transfer.



Do not remove or disconnect the syphon as the magnet will be at high pressure.

- **Reactivate** the pressure control.



For all types of quench, allow up to 1 hour for the quench pressure to subside.

Use the Helium remote hand held unit to ensure that the MSUP pressure control system is working. If a display is not available, use the SESO interface on the Host service screens.

If the pressure control system is NOT working, the pressure inside the cryostat will fall to atmospheric levels, at which point the **compressor switches off**.

Pressure heater control

The **Pressure Service Tool** is used to **deactivate** the pressure heater control circuit. The **electrical connectors** are labeled **pressure sensor** (rear), and **pressure switch** (front).

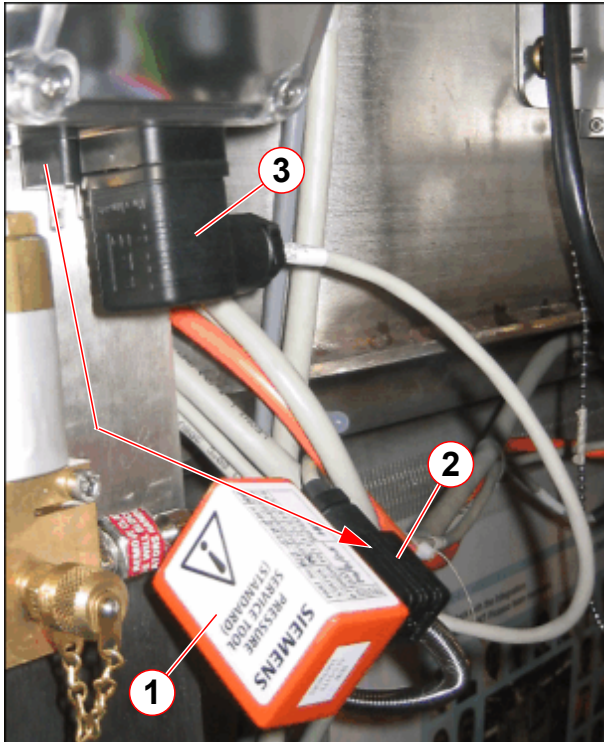
- **Fit the pressure service tool:**
 - **Disconnect** the rear electrical connector on the **pressure switch transducer**.
 - Fit the **pressure service tool** to the connector lead. (→ 2/ Fig. 43 Page 58)

- The **Alarm Box** will sound **PRESSURE ERROR**.
- Press the **ACKNOWLEDGE** button to cancel the alarm.
- The **pressure** reading will be set to **~18 psi (A)**, which will **disable the pressure heater**.

To **reactivate** the pressure heater control circuit, re-connect the cable to the rear connector on the pressure switch transducer. Cancel any alarms.

Mk2 Manocomb Pressure switch

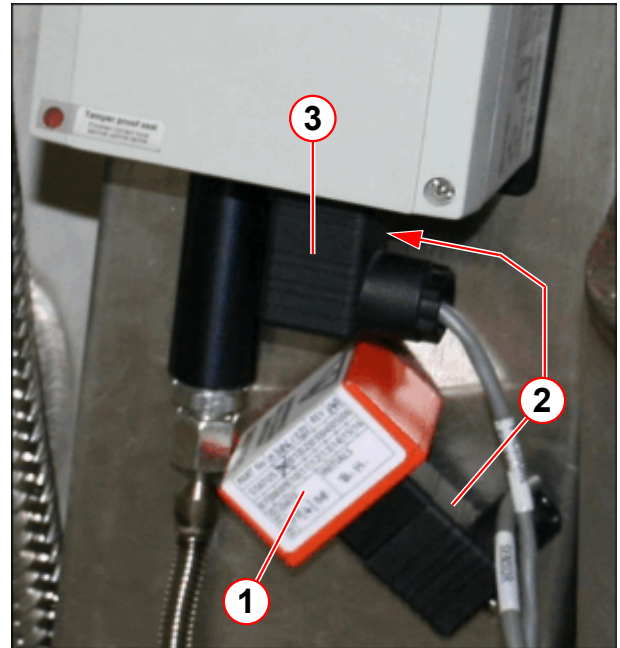
Fig. 42: Fit the pressure service tool



- (1) Pressure service tool
- (2) Rear pressure sensor
- (3) Front pressure switch

WIKA Pressure switch

Fig. 43: Fit the pressure service tool

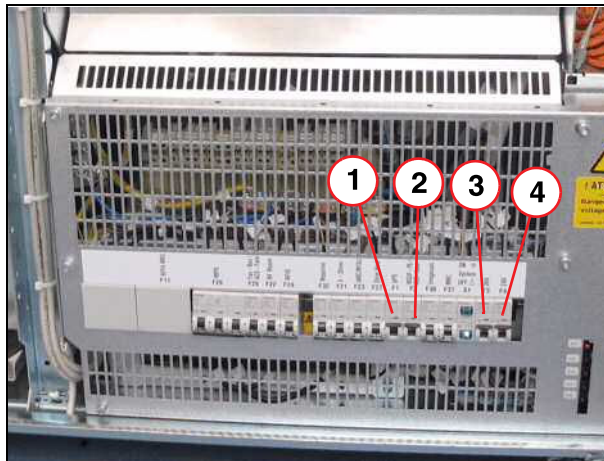


- (1) Pressure service tool
- (2) Rear pressure sensor
- (3) Front pressure switch



Do not leave the system unattended with the pressure service tool attached.

Fig. 44: ACC fuse (inside)



- (1) Fuse F1
- (2) Fuse F2
- (3) Fuse F3
- (4) Fuse F4

If the service **pressure tool** is **not available**,

- **Switch OFF** the MSUP at circuit breaker **F3** (fuse) in the **ACC**.
 - » **Pressure heater is powered down.**
- The **ERDU circuit** is still functional.
- **Helium level** and **system temperatures** can still be obtained via the MSUP-Display Unit (via battery power only).

NOTE: On battery power, **only 2 minutes** of power is available.

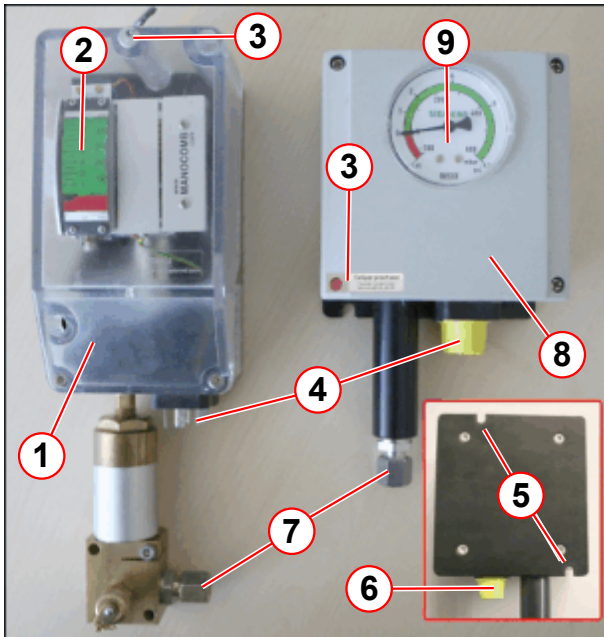
Checking the helium vessel pressure

Pressure switch transducer assembly

- The **WIKA** Pressure switch transducer assembly , **replaces** the **Manocomb** pressure switch, when stocks are exhausted.
- The **new** WIKA switch (part number **10615712**) will **replace** the **current unit** (part number **10126452**), when stocks are exhausted.

- The **functionality** of the new switch is **identical**, and the same fixings are used for installation.

Fig. 45: Pressure switch transducers Manocomb and WIKA

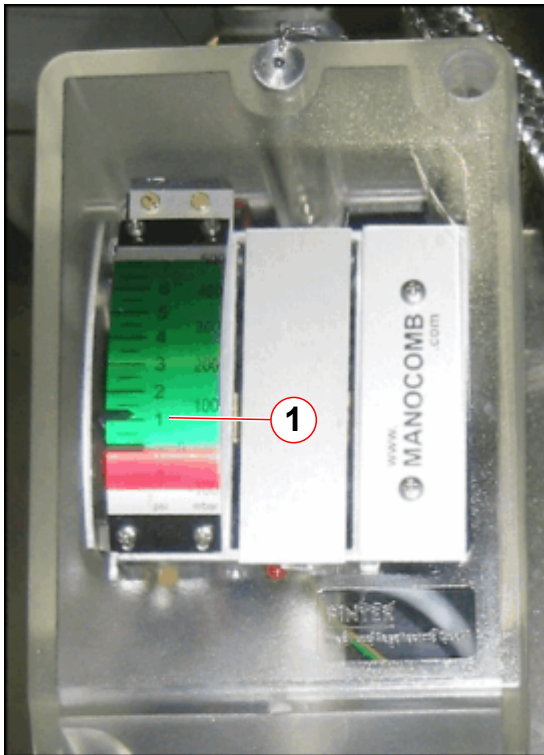


- (1) Manocomb Transducer assembly
- (2) Linear gauge
- (3) Tamperproof seal
- (4) Front/rear electrical connections
- (5) Mounting slots (rear view)
- (6) Protection caps
- (7) Pressure line connection
- (8) WIKA Transducer assy (part no. 10126452)
- (9) Dial gauge

Fig. 46: WIKA Pressure switch transducer (part no 10615712)

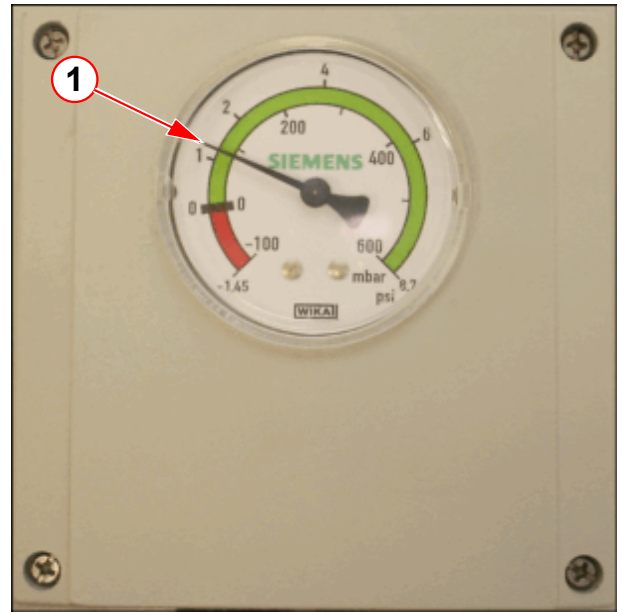


Fig. 47: Pressure gauge (expected pressure)



(1) Expected pressure 1.3 psi (G)

Fig. 48: WIKA pressure switch - expected pressure



(1) Expected pressure 1.3 psi (G)

Monitor the helium vessel pressure:

- Magnet pressure should read **15.3 psi (A)** on the helium remote display unit..
- Check for **positive pressure** on the pressure gauge.
- If the **pressure gauge** is in the **RED zone**, the magnet is **sub-atmospheric**.
 - » **DO NOT** open the magnet **bypass valve**, or disturb any high pressure fitting.
 - » If the magnet does not stabilize via the pressure heater control circuit, manually induce pressure into the cryostat.

Check the quench valve and auxiliary burst disks are intact

- Check the **auxiliary burst disk** first, as it is the most accessible.
- **Mk1** only - Remove the OVC burst disk for easier access the quench valve.
- Under normal quenching circumstances, the OVC burst disk should remain intact.
- **Disable** the pressure heater by fitting **the pressure service tool** .
- **Slowly** open the **magnet bypass valve** to depressurize the helium vessel.
- When the magnet has reached **zero pressure**, warm up the quench valve and venting to room temperature, using a heat gun.

⚠ CAUTION

RISK OF ICING!!

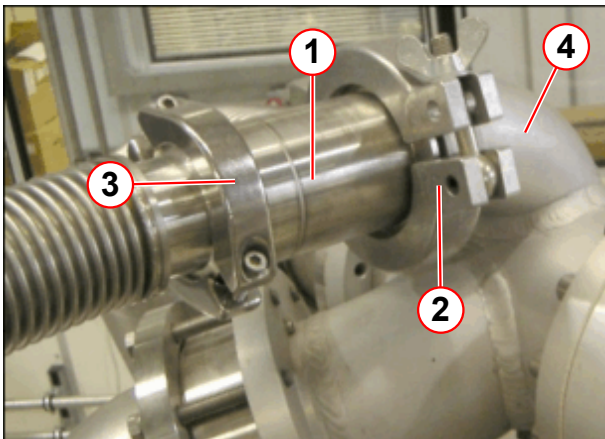
Air ingress will cause INTERNAL ICING OF THE TURRET.

- » Replace broken or damaged burst disk/O-rings IMMEDIATELY.
- » Check that there is a free flow of gas through the magnet bypass valve. If there is no gas flow, there may be an ice blockage.
- » Report the icing to Magnet HSC IMMEDIATELY.
- » Do not fill or ramp the magnet until advised to do so by Magnet HSC.
- » DO NOT OPEN OR REMOVE THE TURRET TOP PLATE OR VENTING.

Auxiliary vent burst disk

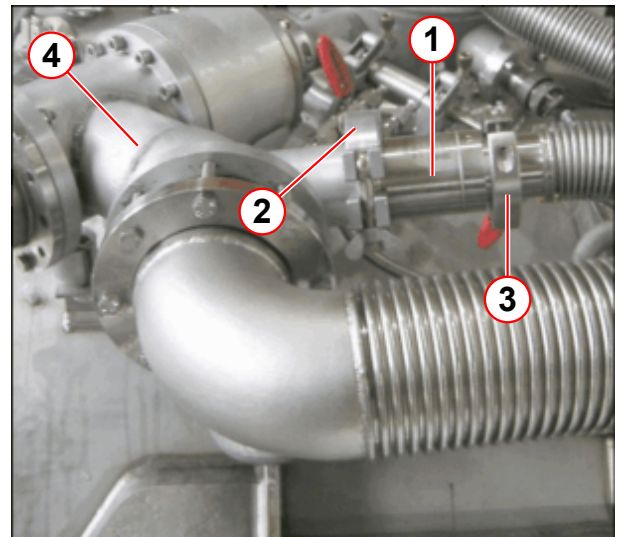
- **Mk1** - Remove the **NW50 clamp** from the manifold on the **low pressure** side of the auxiliary burst disk. (→ 2/ Fig. 49 Page 62)

Fig. 49: OR105 Mk1 Auxiliary burst disk assembly



- (1) Aux burst disk assy
- (2) NW50 low load clamp (low pressure)
- (3) NW40 high load clamp (high pressure)
- (4) Low pressure manifold

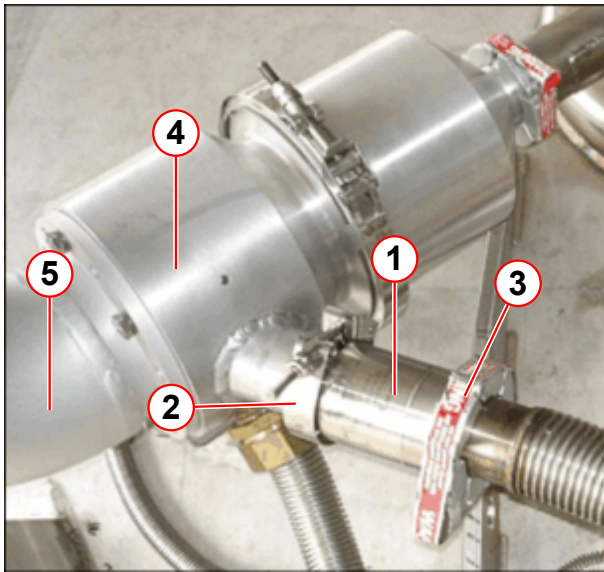
Fig. 50: OR122 Mk1 Auxiliary burst disk assembly



- (1) Aux burst disk
- (2) NW50 low load clamp (low pressure)
- (3) NW40 high load clamp (high pressure)
- (4) Low pressure manifold

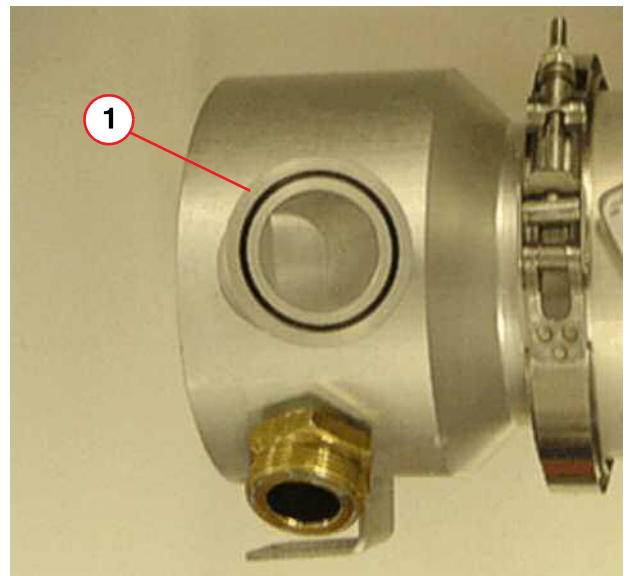
- **Mk2** - Undo and remove the hinged **V-clamp** from the low pressure side of the auxiliary burst disk

Fig. 51: Mk2 Auxiliary burst disk assembly



- (1) Aux burst disk
- (2) V-clamp (low pressure)
- (3) NW40 clamp (high pressure)
- (4) Manifold
- (5) Quench elbow

Fig. 52: Mk2 Auxiliary vent interface



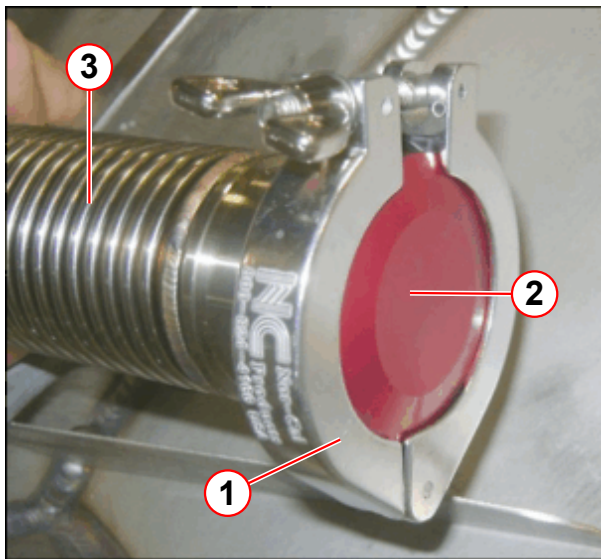
- (1) O-ring seal for aux vent

- Visually inspect the **internal membrane** of the burst disk for damage.
- Visually inspect the **O-ring** on the low pressure side.



The membrane is very fragile - DO NOT TOUCH THE SURFACE.

Fig. 53: Aux vent - fitting the NW40 blanking plate



- (1) NW40 high-load clamp
- (2) NW40 red anodized blanking plate
- (3) Aux vent line

If damaged in any way:

- Undo and remove the **NW40 high pressure clamp**
- **Remove** the broken burst disk and metal seal.
- Fit the **red NW40 blanking plate** in place of the burst disk.
- **Refit** the NW40 high pressure clamp.
 - » Contact Magnet HSC immediately
 - » order a replacement auxiliary bursting disk
- Place the blanked **auxiliary vent line** to one side and continue with checking the **quench valve**.



WARNING

Safety device removed!!!!

The magnet must not be ramped with the auxiliary burst disk removed or broken.

- » **Replace the auxiliary vent burst disk before ramping.**

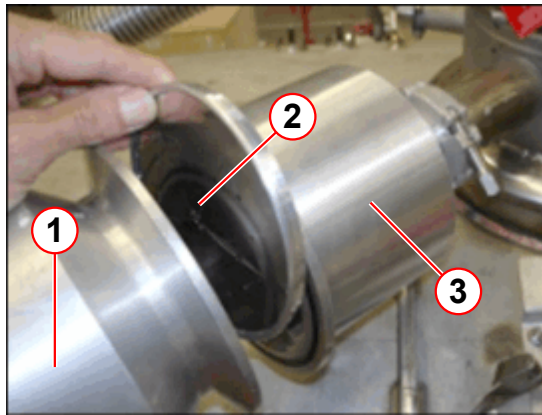
Mobile Systems - High Altitude Quench Valve Burst Disk



Mobile systems are fitted with a high altitude burst disk, in addition to the quench valve burst disk.

If the magnet quenches, the high altitude disk will rupture.

Fig. 54: OR105/122 Mobile - High altitude burst disk (Mk2 Venting)



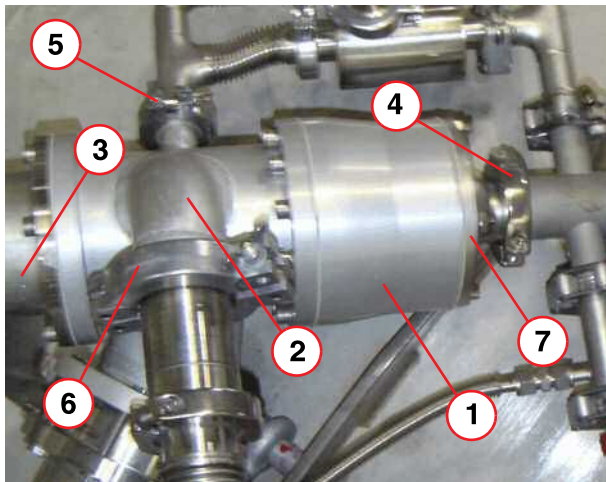
- (1) Manifold
- (2) High altitude burst disk
- (3) Quench valve

Quench valve burst disk and O-ring seal

Mk1

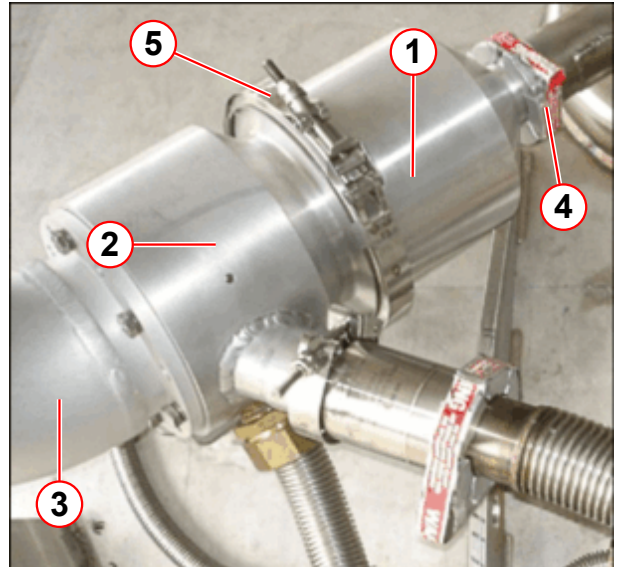
- To access the quench valve, first the OVC burst disk must be removed.

Fig. 55: OR105 Mk1 Manifold/Quench Valve



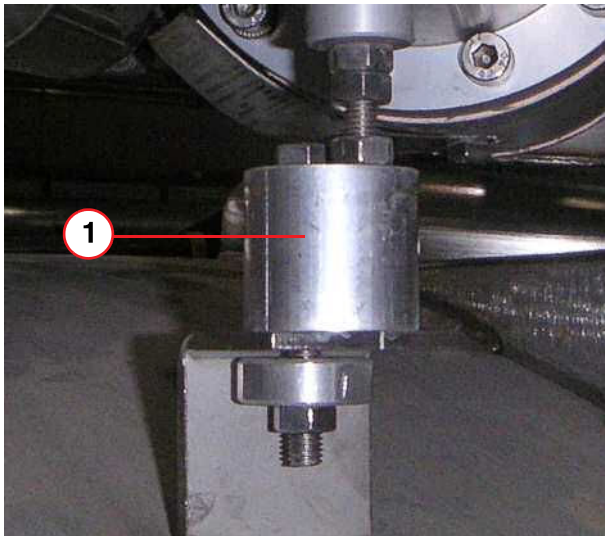
- (1) Quench valve
- (2) Mainfold
- (3) Quench elbow
- (4) NW40 high pressure clamp
- (5) NW25 low pressure clamp
- (6) NW50 low pressure clamp
- (7) Turret flange

Fig. 56: Mk2 Manifold/Quench Valve



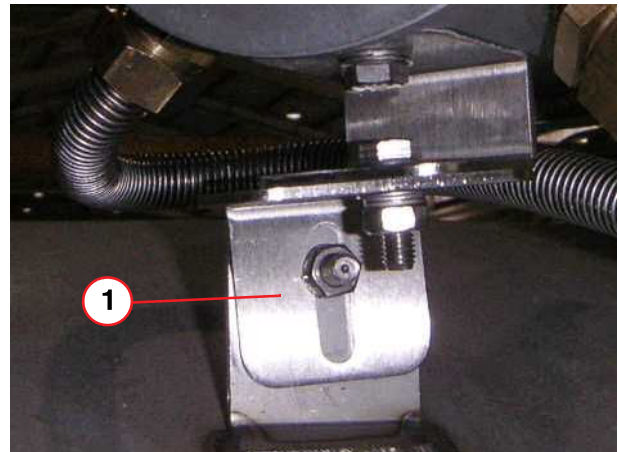
- (1) Quench valve
- (2) Manifold
- (3) Quench elbow
- (4) NW40 clamp (high pressure)
- (5) V-clamp

Fig. 57: Mk1 Support Bracket



(1) Support Bracket

Fig. 58: Mk2 Support bracket



(1) Support bracket

- **Remove the quench elbow**, or straight coupling if fitted. (→ 3/Fig. 55 Page 65)
(→ 3/Fig. 58 Page 66)
- **Slacken the support bracket** underneath the **quench valve manifold**.
(→ Fig. 57 Page 66), (→ Fig. 58 Page 66)
- **Mk1** -
 - **Remove** the NW25 (low pressure) **clamp** to release the **16 psi (A) valve assembly**.
(→ 5/Fig. 55 Page 65)
 - **Remove** the NW50 (low pressure) **clamp** to release the **auxiliary vent line**.
(→ 6/Fig. 55 Page 65)
 - **Remove** the (12 off) M8 cap **screws** securing the **quench valve manifold** to the quench valve. Place the manifold to one side, allowing access to the quench valve.
 - **CLOSE** the **magnet bypass valve**.
 - **Mark the orientation** of the quench valve, then **remove** the 4 off M8 nuts and Nordlock washers that secure the **quench valve** to the **turret flange**.
(→ 7/Fig. 55 Page 65)



Remove the 4 off M8 nuts and nordlock washers in a diametric sequence. Undo the nuts slowly and evenly - the valve is spring loaded.

- **Mk2**
 - **Remove** the **V-clamp** securing the **quench valve manifold** to the **quench valve**. Place the manifold to one side, allowing access to the quench valve.
(→ 5/Fig. 56 Page 65)
 - **CLOSE** the **magnet bypass valve**.
 - **Mark the orientation** of the quench valve, then **remove** the **NW40 (high pressure) clamp and seal**, securing the **quench valve** to the **turret**. (→ 4/Fig. 56 Page 65)
- **Re-open** the **magnet bypass valve** (to ensure zero pressure).

- **Lift the quench valve** from the turret flange.

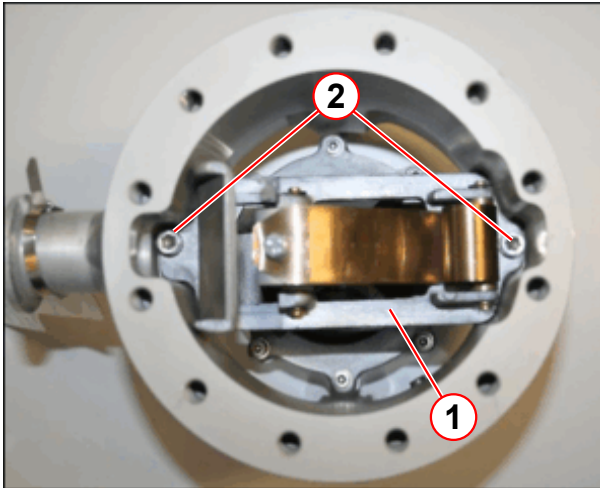


To minimize air ingress, immediately cover the open turret port temporarily, using the NW40 blank.

Use well compacted paper towelling if the blank is not available.

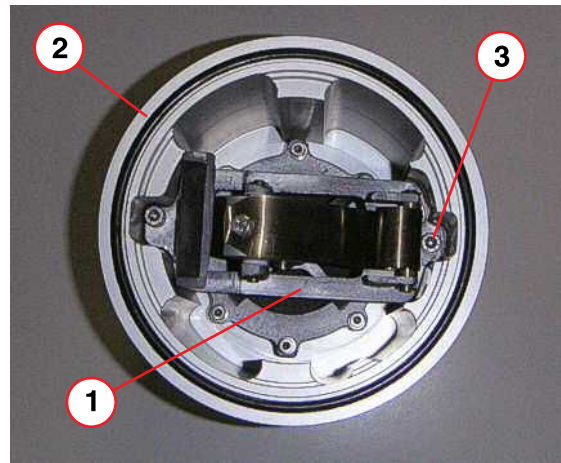
Quench Valve Inspection

Fig. 59: Mk1 Quench valve assembly (low pressure side)



- (1) Burst disk assembly
- (2) M5 cap head screws (2 off)

Fig. 60: Mk2 Quench valve



- (1) Burst disk assembly
- (2) O-ring
- (3) M5 Cap head screws (2 off)

- Evenly unscrew and remove the (2 off) M5 cap head screws and lift out the **burst disk assembly** from the quench valve body.

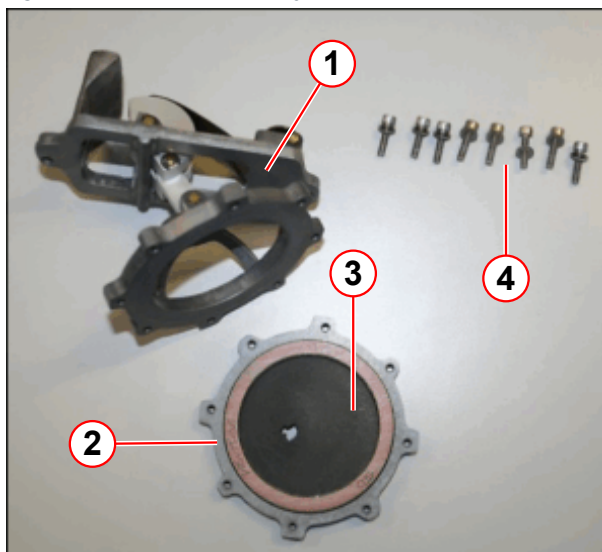


The spring exerts significant force on these screws.



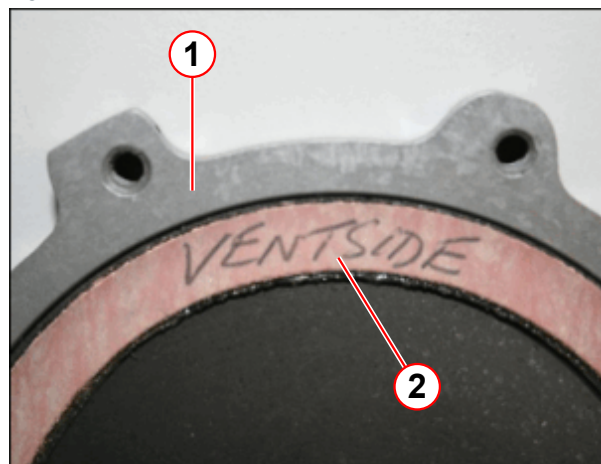
There is no requirement to remove the spring plate on the Mk1 quench valve.

Fig. 61: Burst disk assembly



- (1) Spring plate assembly
- (2) Disk housing
- (3) Broken disk
- (4) M4 screws (8 off)

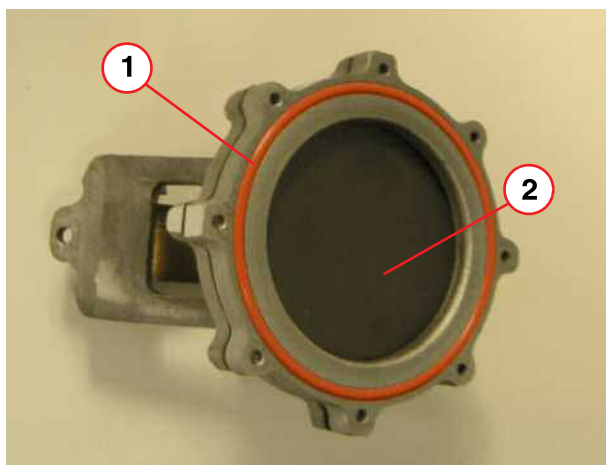
Fig. 62: Burst disk fitment



- (1) Burst disk housing
- (2) Burst disk (Vent side uppermost)

Burst disk Inspection

Fig. 63: Quench Valve Bursting Disk Assembly



- (1) O-ring
- (2) Burst disk

- **Inspect** the burst disk.
 - » If the **burst disk** has ruptured, or is cracked, it must be replaced **immediately**.
- **Inspect** the **burst disk 'O-ring**. Replace if damaged.
- **Clean** the O-ring groove and O-ring before re-fitting.
- **Ensure** the O-ring is **seated evenly** all around the groove.



The O-ring must be checked each time the quench valve is removed, and changed if damaged.

- Re-fit the **burst disk assembly** into the **quench valve body**.
- **Check** the quench valve manifold **O-ring** is in good condition. (→ 2/ Fig. 60 Page 67) **Replace if damaged**.
- **De-ice** the turret flange.

- **Remove** the paper towelling (or NW40 blank if used) from the turret port. Ensure the surfaces are clean and dry.
- **Immediately** re-fit the **quench valve assembly** to the **turret flange**, using **NEW** seals.
 - Ensure correct orientation



DO NOT refit old seals. NEW seals must always be fitted.

- **CLOSE** the magnet **bypass valve**
- **Leak check** the burst disk.
- **If no leaks are detected**, re-fit the **quench manifold** to the **quench valve**.
 - **Mk1**- using the 12 off M8 cap screws (→ 5/ Fig. 55 Page 65)
 - **Mk2** - using the V-clamp (→ 4/ Fig. 56 Page 65)
- **Tighten** the support **bracket** underneath the quench valve **manifold**. (→ Fig. 57 Page 66), (→ Fig. 58 Page 66)
- **Mk1** - Re-fit
 - the **16 psi (A) valve** assembly.
 - **auxiliary vent** burst disk.
 - the **OVC burst disk**.
- **Mk2** - Re-fit
 - **auxiliary vent** burst disk.



If the auxiliary vent burst disk was broken, do not attempt to refit it until a replacement is on site.

OVC Burst disk

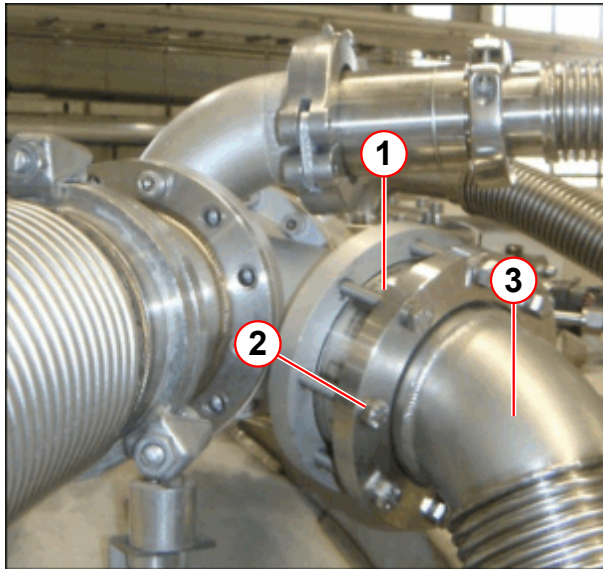


The OVC (DOP) port burst disk and vent line has been removed from the Mk2 magnet design. See Table for system serial number change.

Tab. 5 Venting configurations

Venting	OR105 Avanto serial #	OR122 Espree serial #
Mk1	Before System MR25575	Before System MR30125
Mk2	From System MR25575	From System MR30125
Mk 2 (No DOP)	From System MR57232	From System MR31427

Fig. 64: OR105 Mk1 OVC burst disk assembly



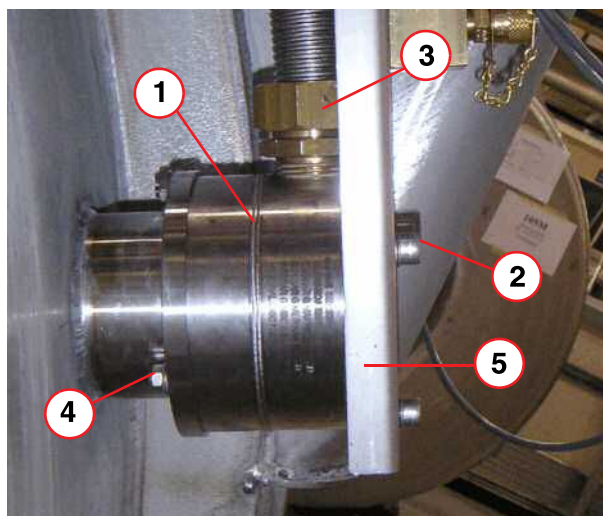
- (1) OVC burst disk
- (2) Screws (8 off)
- (3) OVC vent tube

Mk 1 OVC burst disk

A quench will not cause the OVC metal burst disk to rupture. The OVC burst disk requires checking:

- During routine safety checks and ice inspection of the venting.

Fig. 65: Mk2 OVC Burst Disk Assy



- (1) OVC burst disk
- (2) Cap head screw (3 off)
- (3) OVC vent sealing nut
- (4) Nut (3 off)
- (5) Pressure gauge plate

Mk 2 OVC burst disk

A quench will not cause the OVC metal burst disk to rupture. The OVC burst disk requires checking:

- During routine safety checks and ice inspection of the venting.

Final Steps

- **Re-activate** the pressure control (→ Pressure heater control / Page 57).
- **Re-fill the magnet** with liquid helium.
- Perform a **system leak check**.

- Ramp the magnet to field.



Allow a minimum of 2 hours after filling before ramping the magnet.

- Fill out the **Quench Report report** . (→ Product Information MR, <https://bps-healthcare.siemens.com/productinformation/ProductInformation.aspx?uuid=7abf6bb6-5daf-440c-9920-fab2d4640084>) - Links - MR - Quality Database (and Quench Report) and send to Magnet HSC.

2.1.7 What to do after a Quench (Avanto Dot Ex-factory, Espree with MSUP K2306)

2.1.7.1 General

What to do after a Quench (Avanto Dot Ex-factory, Espree with MSUP K2306)

Should the magnet be quenched deliberately (via the ERDU) or accidentally, (i.e. persistent mode quench), a large volume of helium gas will be generated within the helium reservoir. The quench valve will open, and stay open until the flow subsides. The valve will then re-seat, re-sealing the venting and thus preventing any air (caused by cryo-pumping) from being sucked back into the helium vessel. There is a possibility that the graphite burst disks might have ruptured.

The MSUP should automatically maintain the system pressure above atmospheric. **If this is not the case, contact the Magnet HSC immediately.** (→ Magnet HSC, <mailto:magnet.wscshsc.healthcare@siemens.com>)



When a quench is detected, Magnet HSC must be notified.

Fill out the **Quench Report database** (→ Product Information MR, <https://bps-healthcare.siemens.com/productinformation/ProductInformation.aspx?uuid=7abf6bb6-5daf-440c-9920-fab2d4640084>) - Links - MR - Quality Database (and Quench Report) and send to Magnet HSC.



CAUTION

If the graphite burst disk breaks.

Failure to observe the following will result in ICE FORMING within the magnet.

- » Replace the graphite or auxiliary vent bursting disk IMMEDIATELY.
- » DO NOT RAMP THE MAGNET.
- » Inform Magnet HSC. (→ Magnet HSC, <mailto:magnet.wscshsc.healthcare@siemens.com>)

NOTICE

The magnet helium level can be recovered if the magnet temperature is between LN2 and He temperature.

The CCR (CARBON RES 1) reading must be in the range of between 1200 and 3400 ohms. Compare to the values on the Magnet Acceptance Document, or with HSC.

The magnet must be re-filled immediately with helium.

To re-fill the magnet from zero helium, or warm carbon resistor temperatures, perform a BOTTOM FILL procedure.

If proceeding with a helium fill, immediately place an order for 2000 liters of liquid helium.

» Contact Magnet HSC IMMEDIATELY before proceeding.

**CAUTION**

IF the magnet is allowed to warm up above 77K, a full nitrogen pre-cool will be required.

Helium filling will not cool the magnet sufficiently to collect liquid helium.

A full nitrogen pre-cool takes about 10 days.

» Contact Magnet HSC IMMEDIATELY before proceeding.

Fig. 66: OR105 Magnet Acceptance Document

SIEMENS		Magnet Acceptance Documentation Installation data MR Magnet Technology Quality Regulations	
Cryostat number	21326022	SMT reference	21330107 12.11
System description	1.494T Actively Shielded Superconducting Magnet System OR105		
Measured magnet centre (Offset)	+1 (P) mm		
If magnetic offset is outside the spec (± 5mm), the magnet will be marked (± 5mm).			
 This system requires <u>0</u> Z2 rings and <u>2.92 kg of iron</u> to shim. This is a theoretical calculation that may change due to site conditions.			
Operating parameters			
Description	Specification value	Measured value	
Operating current @ 1.494T	> 500 Amps	510.01 Amps	
Field/Current Ratio	N/A	12.48 kHz/0.1 Amp	
Magnet type:	OR105-01		
TDC calibration values		Carbon resistor values	
Room temperature (23 °C)	1009 Ohms		
Liquid nitrogen temperature (77K)	1309 Ohms		
Liquid helium temperature (4.2K)	2994 Ohms		
Position	Room temperature	Helium temperature	
TDC d - e	1034 Ohms	2837 Ohms	
Top a - e	980 Ohms	3014 Ohms	
Middle b - e	1110 Ohms	2681 Ohms	
Bottom c - e	1006 Ohms	2930 Ohms	
Helium level probe resistance (when using a 250mA supply)			
	Probe 1	Probe 2	
@ 0 resistance	85.36 Ohms	85.36 Ohms	
0% calibration	774	774	
100% resistance	0.22 Ohms	0.22 Ohms	
100% calibration	2	2	
Diode table no.	-1		
Pressure heater resistance	20.5 Ohms		
Final temperature sensor recon values			
Shield bore	Shield Link	Coldhead	Recon margin
47K	47K	42K	658mW
Checked by	Signed <u><i>ANOTHER</i></u>		
V Turner	Date 23/12/2011		
Page 1 of 1		25-AD EPS 000 01	



The magnet acceptance document is an example. Actual values will vary.

Safety



Only Siemens CSEs trained in ramping magnets are allowed to perform this work in the event of a quench.



Switch off and isolate the magnet power supply from the mains before further action.



WARNING

Risk of asphyxiation.

Failure to observe the following safety warning may result in dizziness and loss of consciousness in personnel working with cryogenics.

- » Do not vent helium gas into the magnet room during helium transfer. As an exception, a small quantity of helium gas can be vented to cool the syphon.
- » Ensure the room is ventilated.
- » Ensure that the turret venting kit is connected to the quench pipe.
- » O₂ monitors must be used in the imaging suite.



CAUTION

RISK OF ICING!!

Air ingress will cause INTERNAL ICING OF THE TURRET.

- » Replace broken or damaged burst disk/O-rings IMMEDIATELY.
- » Check that there is a free flow of gas through the magnet bypass valve. If there is no gas flow, there may be an ice blockage.
- » Report the icing to Magnet HSC IMMEDIATELY.
- » Do not fill or ramp the magnet until advised to do so by Magnet HSC.
- » DO NOT OPEN OR REMOVE THE TURRET TOP PLATE OR VENTING.

Required tools

Heat gun

Standard tool kit

Helium leak detector (part no **10614850**)

Laptop and RS232 cable

O₂ monitor
 Eye protection
 Gloves

2.1.7.2 Post quench steps

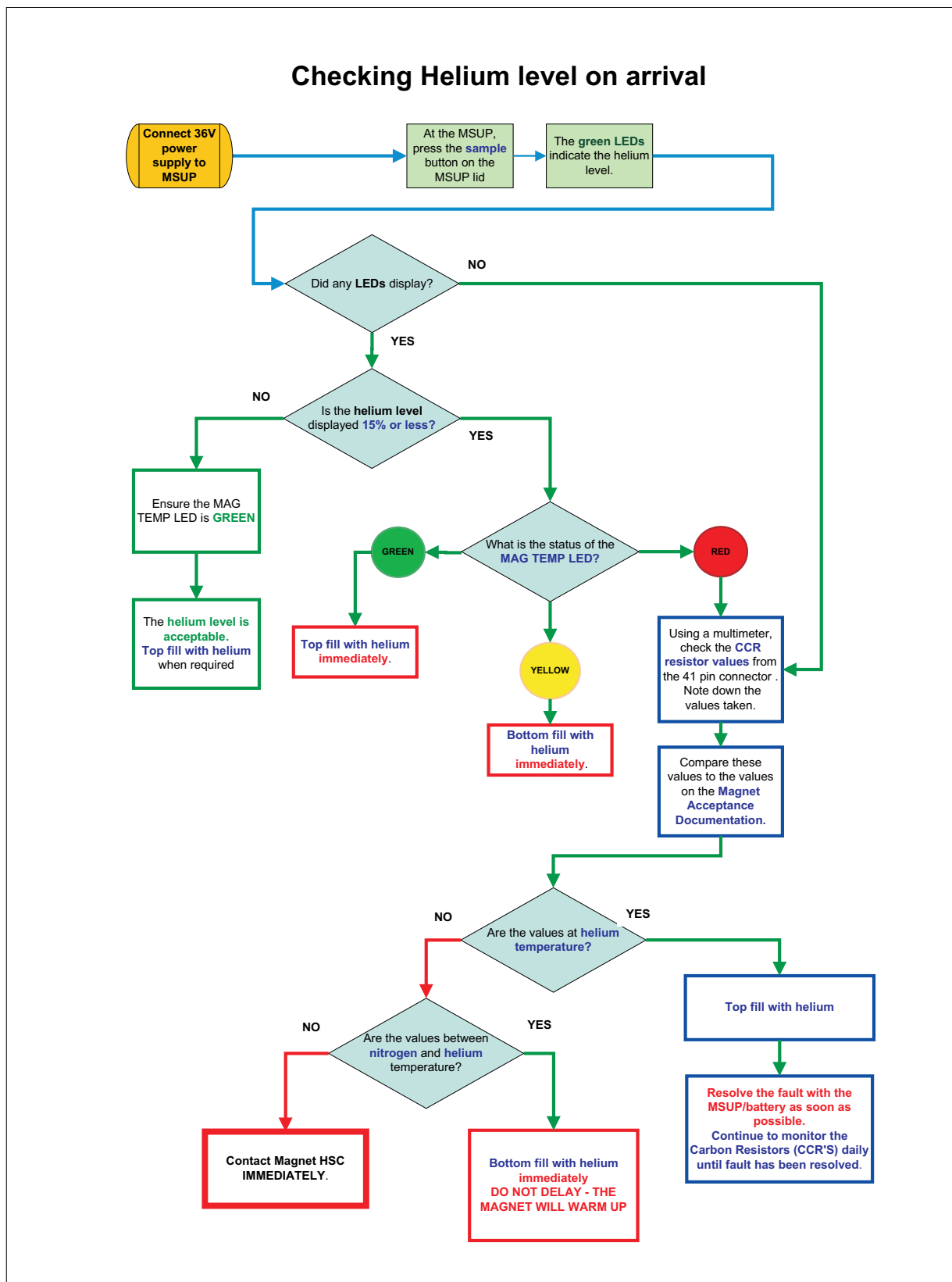
Follow the work steps in the table below for the different types of quench.

	Type of quench		
	Quench during ramping	Quench during He filling	Persistent quench or ERDU button pressed
Step 1	- Isolate the MPSU from the magnet. - Download the 3600 MPSU ramp log files.	Stop filling.	Check that no ERDU buttons are pressed. Re-set ERDU button.
Step 2	Check the liquid helium level using the MSUP-Display Unit.		
Step 3	Order about 1000 liters liquid helium ¹		
Step 4	Warm up and check the burst disk are intact: Auxiliary burst disk:- (→ Auxiliary vent burst disk / Page 84) Quench Valve burst disk:- (→ Burst disk Inspection / Page 88)		
Step 5	After 2 hours : Download the Mag Supervisory log files. Helium fill. Refer to Start up .		
Step 6	Leak check the quench valve and venting. Refer to Turret leak check .		
Step 7	Fill out the Quench report. Refer to Quench report .		
Step 8	Ramp the magnet. Refer to Start up		

	Type of quench		
	Quench during ramping	Quench during He filling	Persistent quench or ERDU button pressed
Step 8	■ Perform the Shim check » If the shim check exceeds the specifications, a re-shim is necessary.		
Step 9	■ Contact Magnet HSC. » Email Mag Power Supply and Mag Supervisory logs to: Magnet HSC with magnet serial no. and all log data. Refer to Startup		

1. The quantity of the order depends on the helium volume already on site.

Fig. 67: Checking Helium level on arrival



2.1.7.3 Procedure

Unattended quench

Unattended quench indicators are:

- If the **Magnet STOP (ERDU)** has been activated:
 - » the yellow **WARNING LED** lights on the alarm box.
 - » the alarm **buzzer** is activated.
- If the **helium level** is low:
 - » the yellow **WARNING LED** lights on the alarm box.
 - » the alarm **buzzer** is activated.
- If the **magnet pressure** is very low, the
 - » **cold head** switches off.
 - » the yellow **WARNING LED** lights on the alarm box.
 - » the alarm **buzzer** is activated.
- Imaging is not be possible if there is no magnetic field.

Press the **AUDIO ALARM OFF** button on the alarm box to silence the audible alarm.

Log file capture

Magnet Supervisory

- The **quench data will be lost**, if:
 - a) The system is **re-booted** within 2 hours of a quench
 - b) The system is **turned off** within 2 hours of a quench
 - c) The data is downloaded within 2 hours of a quench
- Download the MSUP log files.

If the magnet quenches during ramping:-

- Download the **Magnet Power Supply** log file.



The MPSU log files **MUST** be downloaded **BEFORE** the next ramp is performed, otherwise previous data will be overwritten!

- **Switch off and disconnect** the MPSU from the magnet.

If the magnet quenches during helium fill:

- **Stop** the transfer.

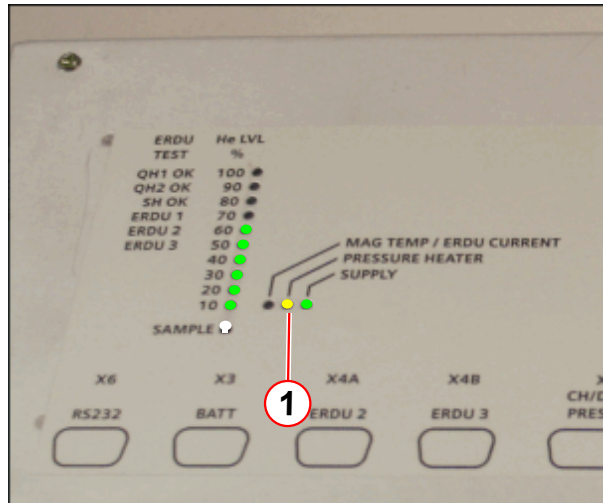


Do not remove or disconnect the syphon as the magnet will be at high pressure.



For all types of quench, allow up to 1 hour for the quench pressure to subside.

Fig. 68: MSUP - Pressure heater LED



(1) Pressure heater LED - yellow

Check the **pressure heater status**

- Check the MSUP **PRESSURE HEATER LED** is lit, **or**
- Access the **Service Software** at the **MRC (MRAWP) Host**:

Magnet&Cooling / Magnet Status / MSUP Control

If the pressure control system is NOT working, the pressure inside the cryostat will fall to atmospheric levels. Ensure the compressor **switches off**.

Check the Ceramic Carbon Resistor (CCR) values

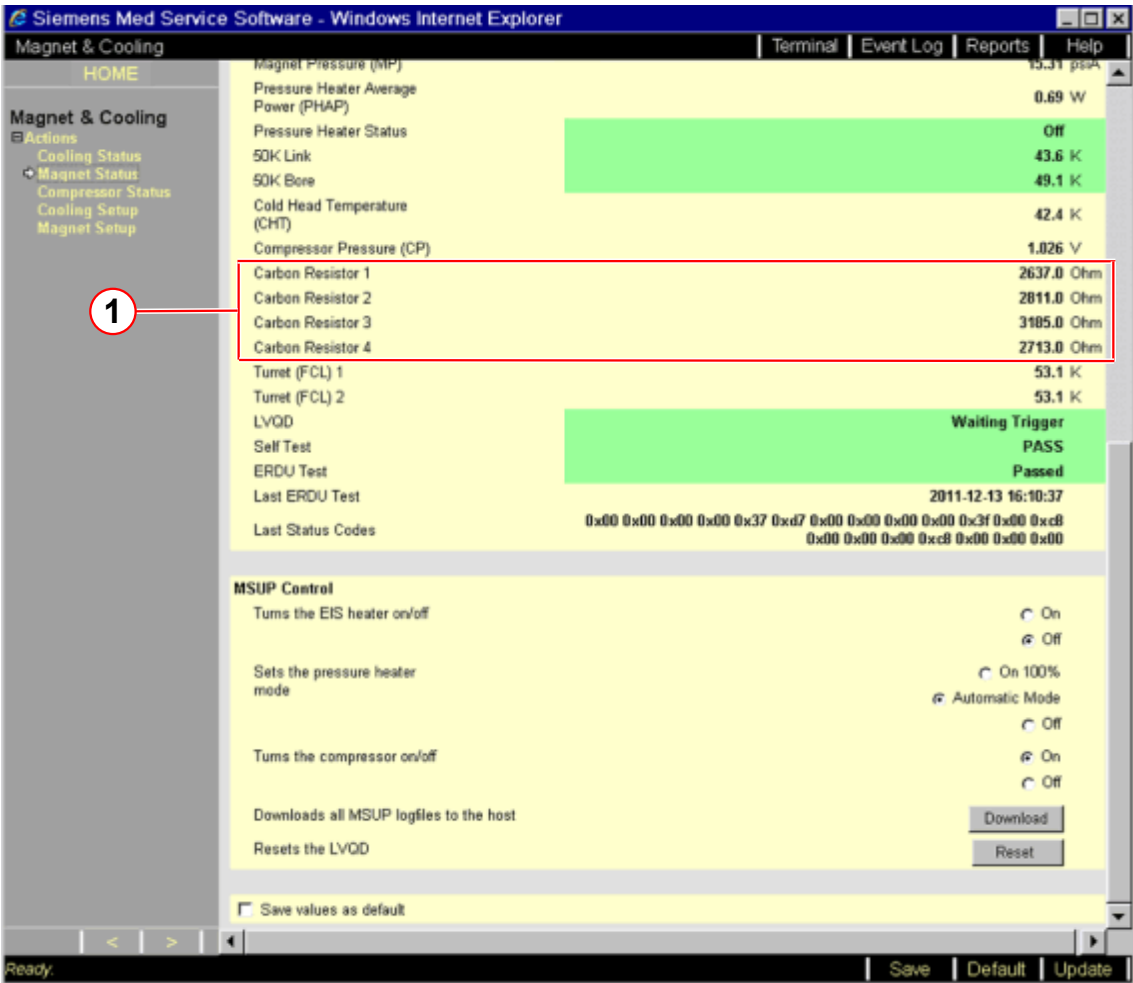


Compare the temperatures with the values on the Magnet Acceptance Document.

- Access the **Service Software** at the **MRC (MRAWP) Host**.

- Select the menu item **Magnet & Cooling - Magnet Status**.

Fig. 69: Checking the Ceramic Carbon Resistors



(1) Resistor values

Tab. 6 Expected CCR resistor values

41 way plug, measured in Ohms	Expected values room temperature	Expected values 77° K temperature	Expected values 4.2 K temperature
All ceramic carbon resistors (CCRs)	880 - 1050	1150 - 1350	2800 - 3380



The above table is a guide only. Always compare the readings with those on the Magnet Acceptance Document.



The magnet helium level can be recovered if the temperature of the CCR is still colder than nitrogen temperature.
The CCR (CARBON RES 1) reading must be in the range of between 1200 and 3400 ohms.



If temperatures are between N2 and helium temperature (77K), a BOTTOM FILL must be performed IMMEDIATELY.



If the host (or laptop), is not available, check CCR values directly from the pins (a-e) on the 41 pin connector on the turret.

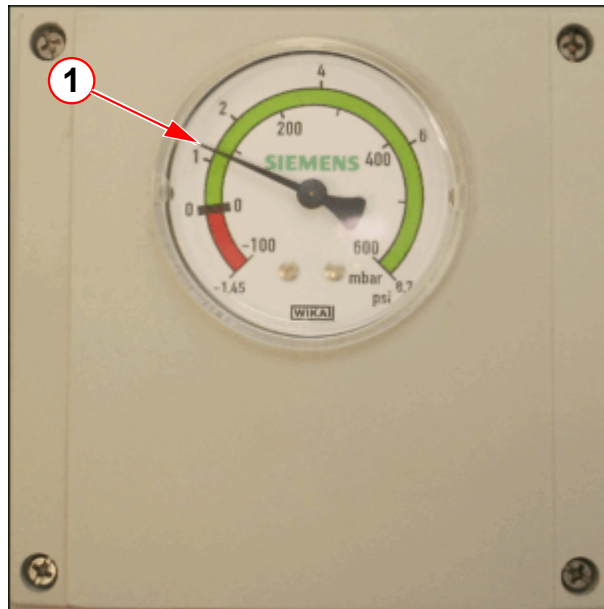
Pressure heater control

After helium filling, the quench valve and venting must be leak checked.

Ensure the **helium vessel** has a **positive** pressure:

- at the **pressure gauge**.
- at the **MRC (MRAWP) Host**. (→ 1/ Fig. 84 Page 90)
- Check for **positive pressure** on the pressure gauge.

Fig. 70: WIKA pressure switch - expected pressure



(1) Expected pressure 1.3 psi (G)

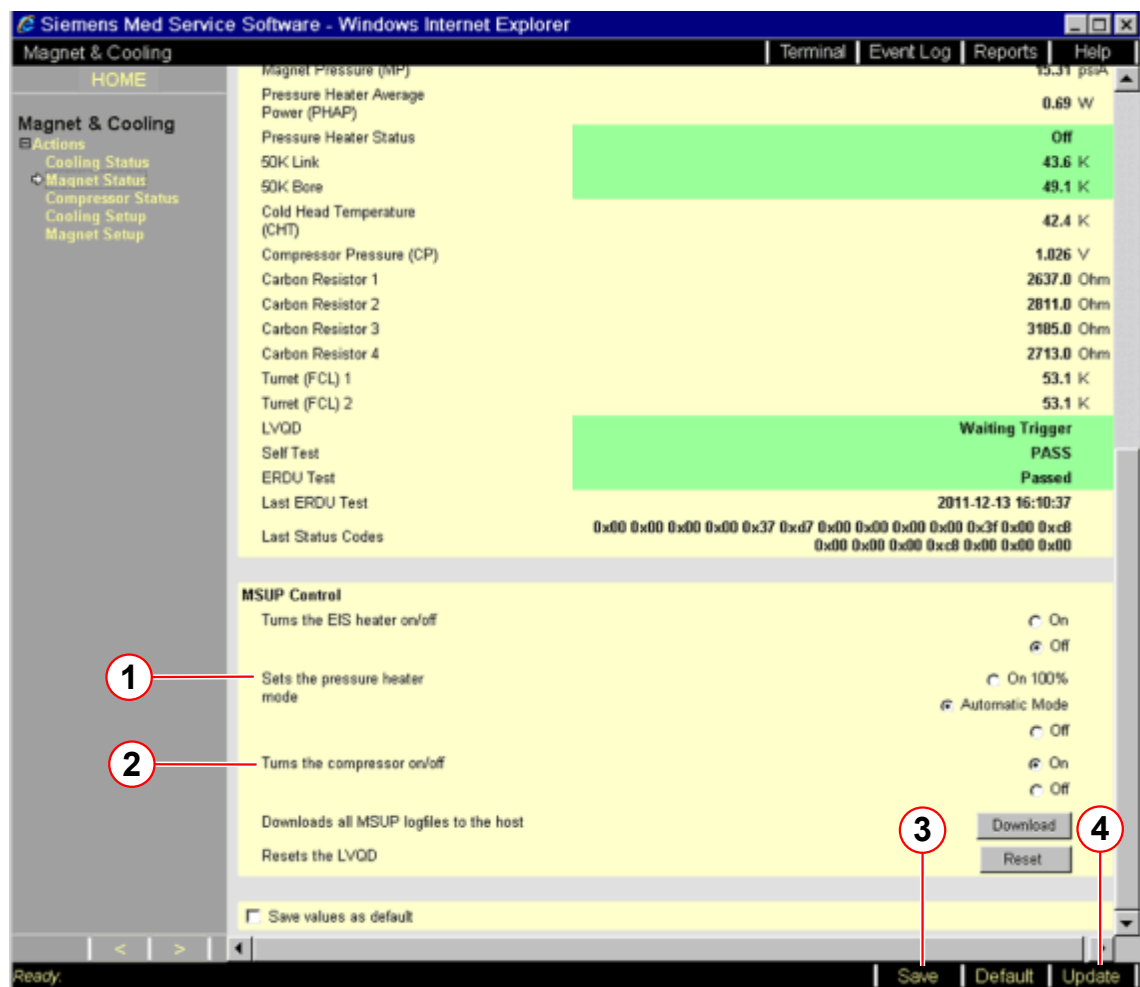
Monitor the helium vessel pressure

- If the **pressure gauge** is within the **RED zone**, the magnet is **sub-atmospheric**.
 - » **DO NOT** open the magnet **bypass valve**, or disturb any high pressure fitting

Deactivate the pressure heater

- Access the **Service Software** at the **MRC (MRAWP) Host**.
- Select the menu item **Magnet & Cooling - Magnet Status**.
- At **MSUP Control - Sets the pressure heater mode** select the **OFF** button..
- Select **SAVE**.

Fig. 71: Service screen Mag&Cooling - Magnet Status - MSUP Control



- (1) Sets the pressure heater mode
- (2) Sets the compressor on/off
- (3) Save
- (4) Update



The pressure heater will reactivate within 60 min.

Depressurize the helium vessel



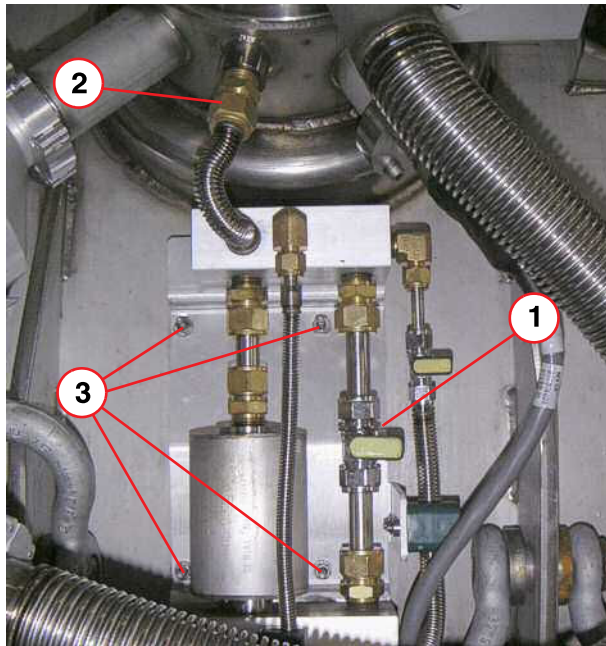
CAUTION

Risk of ice formation.

The magnet bypass valve is only OPENED for depressurization of the helium vessel. The bypass valve is CLOSED during normal operation.

- » To avoid air ingress, CLOSE the magnet bypass valve immediately after the helium vessel is depressurized.

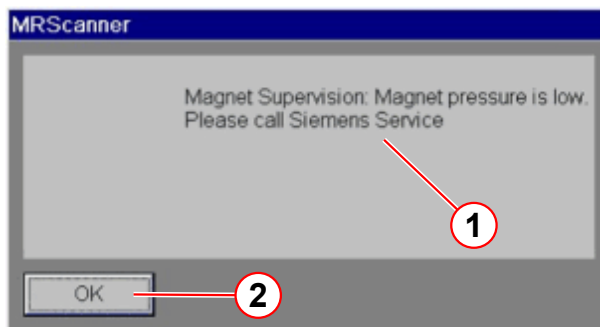
Fig. 72: Mk2 Vent valve assembly



- (1) Bypass valve
- (2) Turret connection
- (3) Mounting screws (4 positions)

- **Fully open the magnet bypass valve to depressurize the helium vessel.**
- Wait until **zero psi** is displayed on the pressure gauge.
- The **pressure switch** will activate.
 - » the **cold head** will turn **off**.
- **Close the magnet bypass valve to prevent air ingress.**
- **Warm up the vent valve assembly.**
 - » **Take care** not to damage any components when using a **heat gun**.

Fig. 73: MSUP - Magnet pressure is low



- (1) Pop-up message
- (2) Click OK to clear

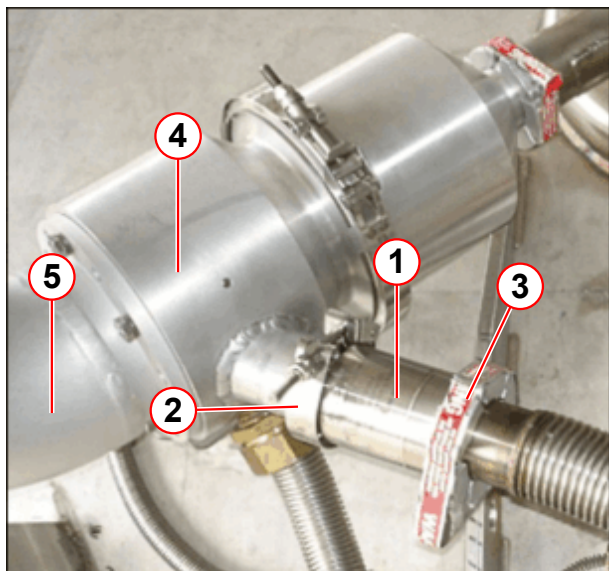
On the MR Scanner:

- A pop-up **alarm message** is displayed.
 - » **Magnet pressure is low.**
 - » Click **OK** to clear the alarm.
 - The **alarm box** will sound when **pressure falls to alarm limits**.
 - » The yellow **WARNING LED** lights on the **alarm box**.
 - » The alarm **buzzer** is **activated**
 - » Press **AUDIO ALARM OFF** to silence the alarm.
- When the magnet has reached **zero pressure**, warm up the **quench valve and venting** to room temperature, using a **heat gun**.

Auxiliary vent burst disk

- **Mk2** - Undo and remove the hinged **V-clamp** from the low pressure side of the auxiliary burst disk

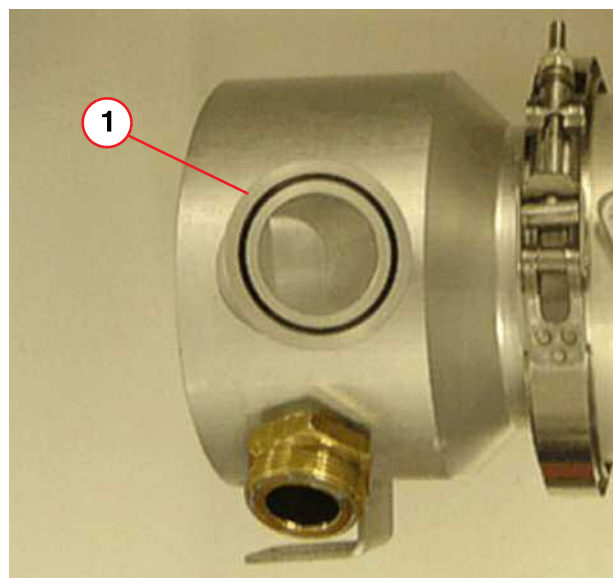
Fig. 74: Mk2 Auxiliary burst disk assembly



- (1) Aux burst disk
- (2) V-clamp (low pressure)
- (3) NW40 clamp (high pressure)
- (4) Manifold
- (5) Quench elbow

- Visually inspect the **internal membrane** of the burst disk for damage.
- Visually inspect the **O-ring** on the low pressure side.

Fig. 75: Mk2 Auxiliary vent interface

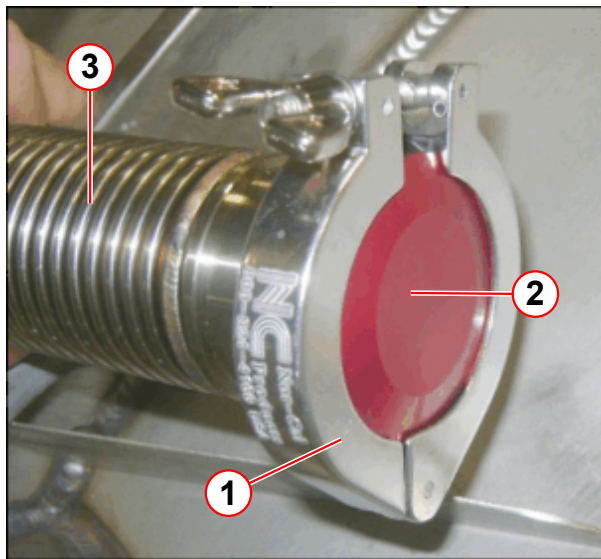


- (1) O-ring seal for aux vent



The membrane is very fragile - DO NOT TOUCH THE SURFACE.

Fig. 76: Aux vent - fitting the NW40 blanking plate



- (1) NW40 high-load clamp
- (2) NW40 red anodized blanking plate
- (3) Aux vent line

If damaged in any way:

- Undo and remove the **NW40 high pressure clamp**
- **Remove** the broken burst disk and metal seal.
- Fit the **red NW40 blanking plate** in place of the burst disk.
- **Refit** the NW40 high pressure clamp.
 - » Contact Magnet HSC **immediately**
 - » order a replacement aux bursting disk
- Place the blanked **aux vent line** to one side and continue with checking the **quench valve**.



WARNING

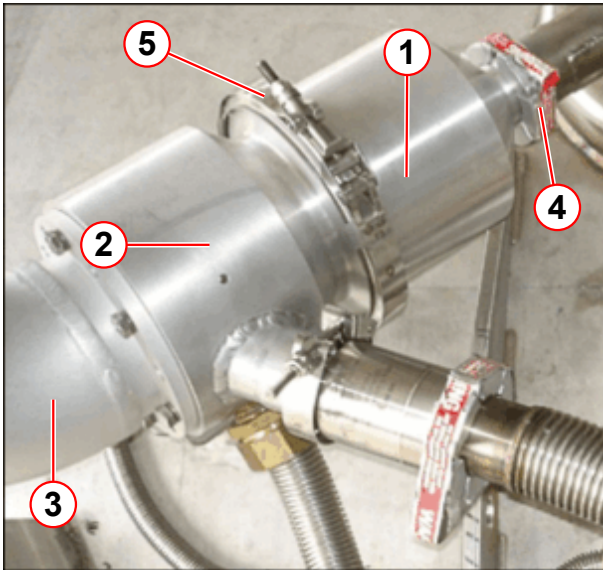
Safety device removed!!!!

The magnet must not be ramped with the auxiliary burst disk removed or broken.

- » **Replace the auxiliary vent burst disk before ramping.**

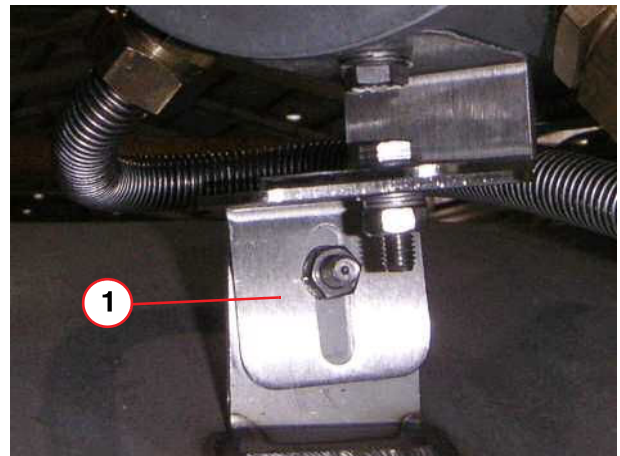
Quench valve burst disk and O-ring seal

Fig. 77: Mk2 Manifold/Quench Valve



- (1) Quench valve
- (2) Manifold
- (3) Quench elbow
- (4) NW40 clamp (high pressure)
- (5) V-clamp

Fig. 78: Mk2 Support bracket



- (1) Support bracket

- **Remove the quench elbow**, or straight coupling if fitted. (→ 3/ Fig. 78 Page 86)
- **Slacken the support bracket** underneath the **quench valve manifold**, (→ Fig. 78 Page 86)
- **Remove the V-clamp** securing the **quench valve manifold** to the **quench valve**. Place the manifold to one side, allowing access to the quench valve. (→ 5/ Fig. 77 Page 86)
- **CLOSE the magnet bypass valve.**
- **Mark the orientation** of the quench valve, then remove the **NW40 (high pressure) clamp and seal**, securing the **quench valve** to the **turret**. (→ 4/ Fig. 77 Page 86)
- **Re-open the magnet bypass valve** (to ensure zero pressure).
- **Lift the quench valve** from the turret flange.

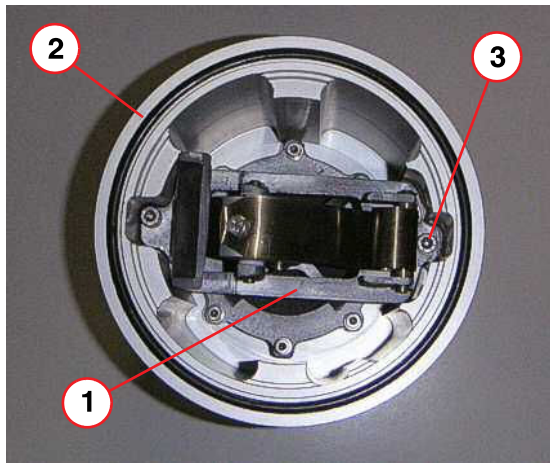


To minimize air ingress, immediately cover the open turret port temporarily, using the NW40 blank.

Use well compacted paper towelling if the blank is not available.

Quench Valve Inspection

Fig. 79: Mk2 Quench valve



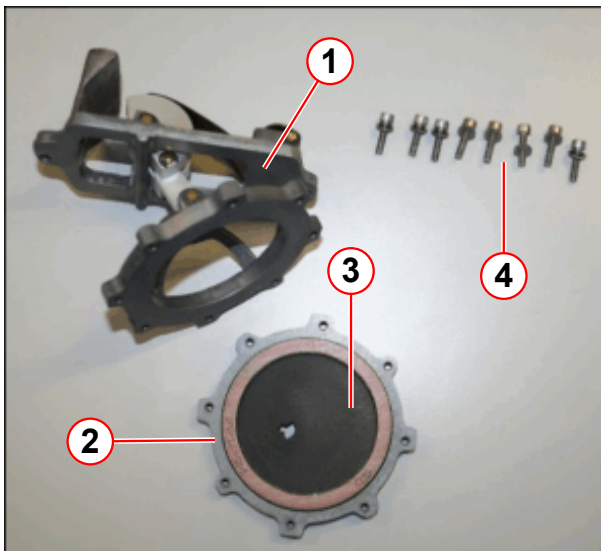
- (1) Burst disk assembly
- (2) O-ring
- (3) M5 Cap head screws (2 off)

- **Evenly** unscrew and remove the (2 off) M5 cap head screws and lift out the **burst disk assembly** from the quench valve body.



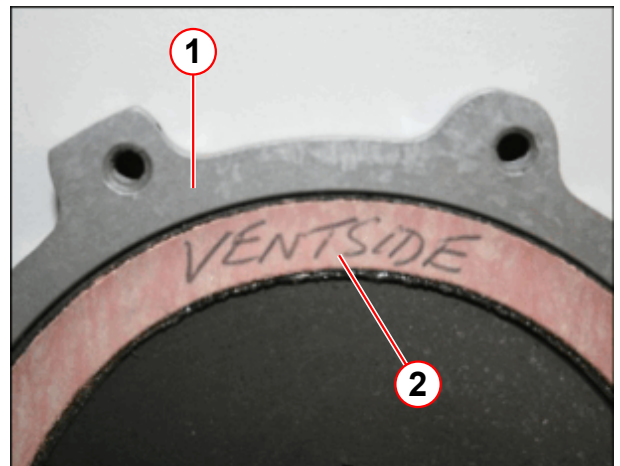
The spring exerts significant force on these screws.

Fig. 80: Burst disk assembly



- (1) Spring plate assembly
- (2) Disk housing
- (3) Broken disk
- (4) M4 screws (8 off)

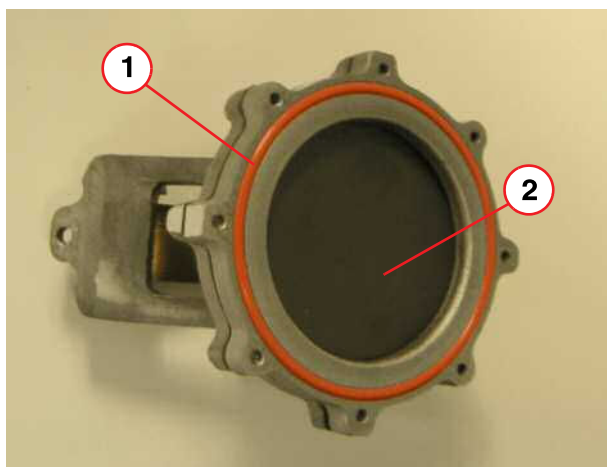
Fig. 81: Burst disk fitment



- (1) Burst disk housing
- (2) Burst disk (Vent side uppermost)

Burst disk Inspection

Fig. 82: Quench Valve Bursting Disk Assembly



- (1) O-ring
- (2) Burst disk

- **Inspect** the burst disk.
 - » If the **burst disk** has ruptured, or is cracked, it must be replaced **immediately**.
- Inspect the **burst disk 'O-ring**. Replace if damaged.
- **Clean** the O-ring groove and O-ring before re-fitting.
- **Ensure** the O-ring is **seated evenly** all around the groove.



The O-ring must be checked each time the quench valve is removed, and changed if damaged.

- Re-fit the **burst disk assembly** into the **quench valve body**.
- **Check** the quench valve manifold **O-ring** is in good condition. (→ 2/Fig. 79 Page 87) **Replace if damaged**.
- **De-ice** the turret flange.
- **Remove** the paper towelling (or NW40 blank if used) from the turret port. Ensure the surfaces are clean and dry.
- **Immediately** re-fit the **quench valve assembly** to the **turret flange**, using **NEW** seals.
 - Ensure correct **orientation**.



DO NOT refit old seals. NEW seals must always be fitted.

- **CLOSE** the magnet **bypass valve**.
- **Leak check** the burst disk.
- **If no leaks are detected**, re-fit the **quench manifold** to the **quench valve**.
 - **Mk2** - using the V-clamp. (→ 4/Fig. 77 Page 86)
- **Tighten** the support **bracket** underneath the quench valve **manifold**. (→ 1/Fig. 78 Page 86)

- **Mk2** - Re-fit the **auxiliary** burst disk.



If the auxiliary vent burst disk was broken, do not attempt to refit it until a replacement is on site.

OVC Burst disk

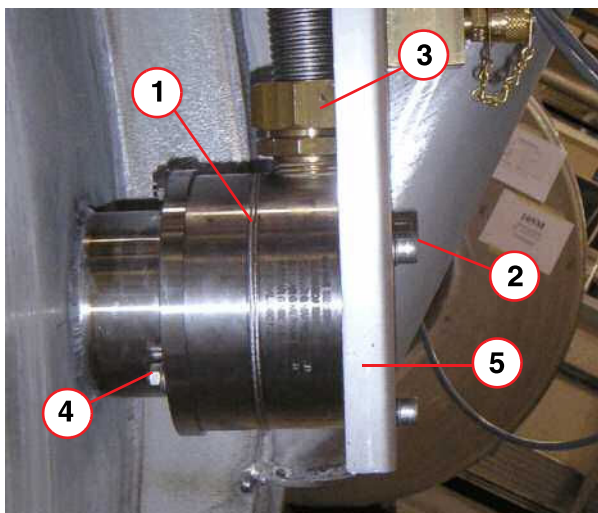


The OVC (DOP) port burst disk and vent line has been removed from the Mk2 magnet design. See Table for system serial number change.

Tab. 7 Venting configurations

Venting	OR105 Avanto serial #	OR122 Espree serial #
Mk2	From System MR25575	From System MR30125
Mk 2 (No DOP)	From System MR57232	From System MR31427

Fig. 83: Mk2 OVC Burst Disk Assy



- (1) OVC burst disk
- (2) Cap head screw (3 off)
- (3) OVC vent sealing nut
- (4) Nut (3 off)
- (5) Pressure gauge plate

Mk 2 OVC burst disk

A quench will not cause the OVC metal burst disk to rupture. The OVC burst disk requires checking:

- During routine safety checks and ice inspection of the venting.

Reactivate the pressure heater

- Access the **Service Software** at the **MRC (MRAWP) Host**.
- Select the menu item **Magnet & Cooling - Magnet Status**.
- At **MSUP Control - Sets the pressure heater mode** select the **On 100%** button.
- Select **SAVE**.

- **At Real-Time Values - Monitor the magnet pressure.**
 - Select **Update** to refresh data until pressure reaches ≥ 15.3 psi (A).

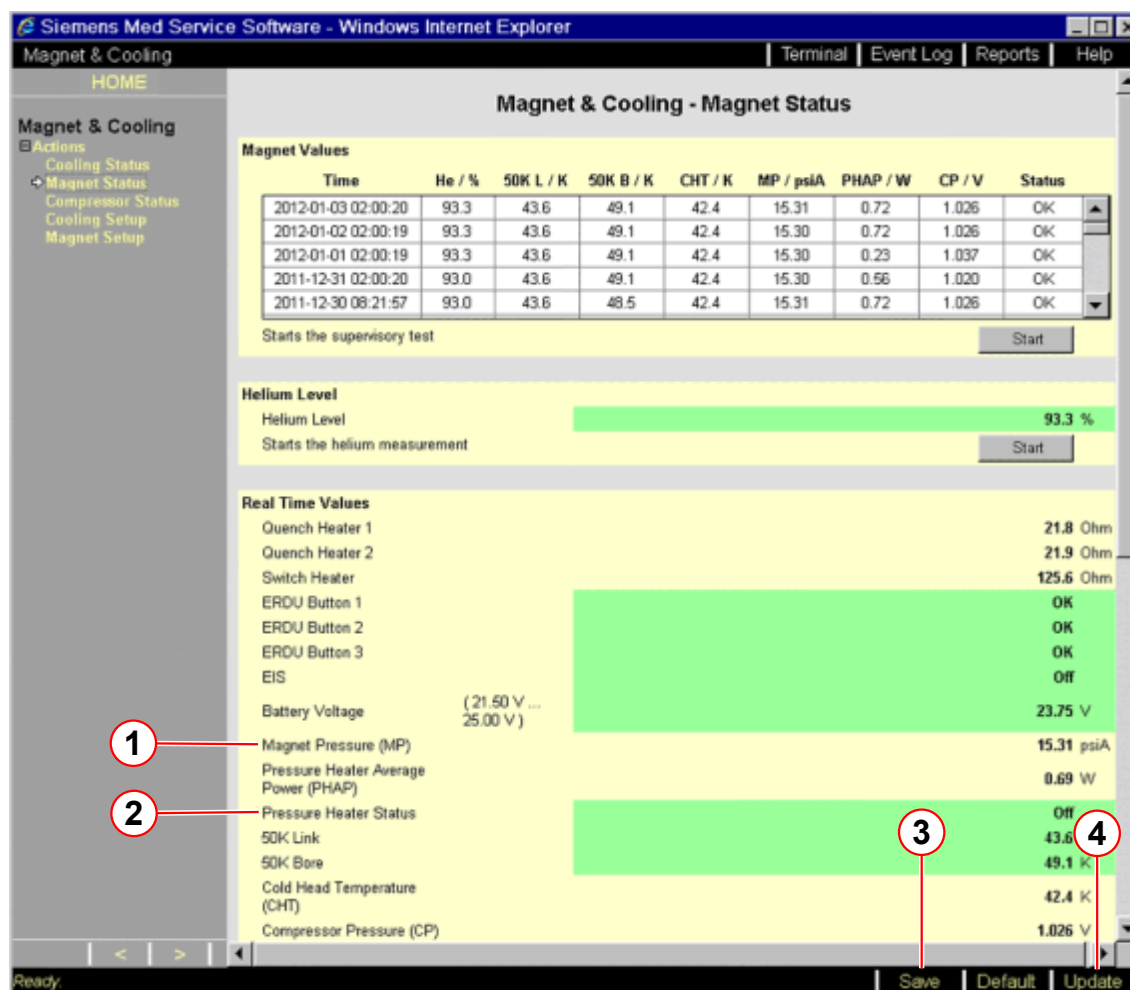


Data is only accurate from the last refresh of the page, or from when the page was opened last.



During pressurization, the cold head will switch on automatically.
If the cold head was turned off at MRC, select.

Fig. 84: Check magnet pressure



- (1) Magnet Pressure (MP)
- (2) Pressure Heater Status
- (3) Save
- (4) Update

- **At MSUP Control - Sets the pressure heater mode select the Automatic mode button.**
- Select **SAVE**.

Final Steps

- **Helium fill.**
- Perform a **system leak check**.
- **Ramp the magnet** to field.



Allow a minimum of 2 hours after filling before ramping the magnet.

- Fill out the **Quench Report report** (→ Product Information MR, <https://bps-healthcare.siemens.com/productinformation/ProductInformation.aspx?uuid=7abf6bb6-5daf-440c-9920-fab2d4640084>) - Links - MR - Quality Database (and Quench Report) and send to Magnet HSC.

2.1.7.4 Troubleshooting post quench

Checking the carbon resistors with a multimeter



CAUTION

Risk of asphyxiation.

Failure to adhere to safety precautions may result in serious bodily injury.

- » **The magnet must be RAMPED DOWN before attempting this procedure.**

The **Ceramic Carbon Resistors** (CCR) are checked using either the **MSUP** or a **multimeter**. The measurement is taken across pins **a-e, b-e, c-e, d-e** on the **41pin turret connector**.

Tooling requirements

- Multimeter and pin leads.
- Laptop/PC with **Terminal Emulation software**.
- RS232 cable.

Procedure

Using a multimeter:

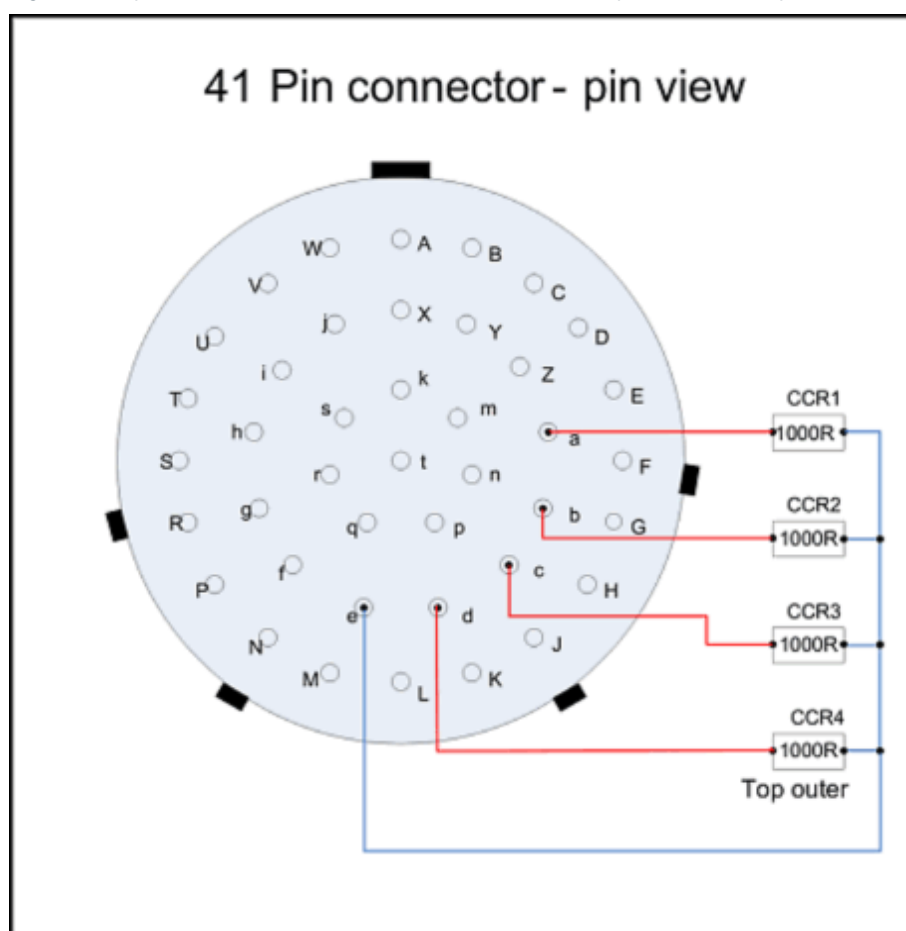
- **Connect the pin leads.**
- Turn the mode selector to **resistor mode Ω** .

Fig. 85: Turret - 41pin connector



- **Disconnect** the 41pin connector.
- **Measure** across pins **a-e**, **b-e**, **c-e**, **d-e**. (→ Fig. 86 Page 92)
- **Compare** the **values** with the Magnet Acceptance Document.
- **Re-connect** the 41pin connector.

Fig. 86: 41pin turret connector - ceramic carbon resistor pins (room temperature values)



2.1.8 De-icing the Syphon Port

2.1.8.1 Prerequisites

General Information



This procedure must only be performed with the magnet RAMPED DOWN.

The service syphon is designed to be fully portable and compatible with all Siemens **4K** magnet products. By using a combination of the service syphon extension legs supplied, a helium top fill, bottom fill, or de-ice of the syphon port can be performed.

A de-ice may be required if:-

- A resistance is felt when inserting the syphon leg.
 - » This suggests the syphon cone is iced-up.
- Poor transfer efficiency.
- No movement on the helium level meter after 20 minutes.

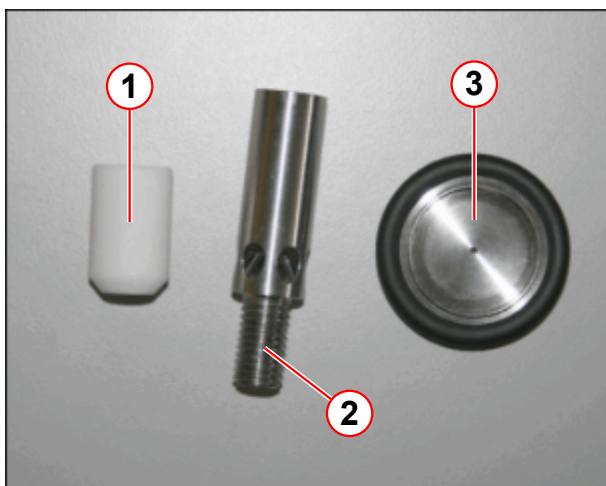
The de-ice tool is designed to blow helium gas (at room temperature), in defined directions within the syphon port dispersing any ice which has formed.

Requirements

Tab. 8 Service syphon kit and upgrades

Material number	Revision	Comment
08396363	00	Service syphon kit
10126691	00	Upgrade kit (200 mm extension leg added)
10129438	00	Upgrade kit (de-ice tooling)
10126651	01	Upgrade Service syphon kit (includes upgrade kits)

Fig. 87: Upgrade kit 10129438 - De-ice tooling



- (1) PTFE tip
- (2) De-ice helium fill diffuser
- (3) NW25 flow restrictor

Tooling

- 99.995% (minimum) helium gas (50 litre) cylinder and high pressure regulator.
- Pressure regulator 200 bar/ 0 - 1.5 bar.
- Gas bottle wrench.
- Protective clothing: gloves, protective goggles or face protection shield, protective clothing, protective shoes (non-magnetic).
- Heat gun.
- Paper toweling.
- 30 mm non-magnetic syphon wrench (MK2 Avanto and Espree turret syphon ports). (→ Fig. 92 Page 98)(→ Fig. 91 Page 98)
- Replacement copper seal for syphon port on the turret (material no **03108094**).
- NW25 hose connector (material number **08396298**)
- O₂ monitor.
- Leak check equipment capable of checking individual leaks down to 1×10^{-4} mbar l/s.
 - Use the Edwards hand-held leak detector (material number **10614850**).

Safety



CAUTION

Strong magnetic field

Failure to adhere to safety measures may result in serious bodily injury.

- » The magnet must be **RAMPED DOWN** before attempting this procedure.

**WARNING**

Pressurized helium vessel.

Cold helium gas.

Failure to observe the following Safety Information may result in physical injury.

- » Do not disconnect any of the fittings without depressurization of the helium vessel.
- » Only trained and accredited CSEs must perform this procedure.

**DANGER**

Risk of asphyxiation.

Failure to observe the following Safety Information may result in dizziness and loss of consciousness.

- » Do not vent large volumes of helium gas directly into the magnet room.
- » Personal Oxygen monitors must be worn if a fixed room monitor has not been installed
- » In a storage facility, ensure that there is sufficient ventilation.

**WARNING**

Risk of cold burns

Failure to observe the following may result in injury to personnel.

- » Do not touch cold surfaces with bare hands.
- » Personal safety equipment must be worn.

2.1.8.2 De-ice Procedure

The de-ice diffuser tip is designed to direct the warm gas downwards past the syphon cone to clear ice from the lower regions of the turret.

When the syphon port is clear of ice the diffuser tip is able to fit into the syphon cone.

**WARNING**

Risk of quench

Failure to observe the following may result in the magnet quenching and loss of helium.

- » Ramp the magnet to zero field before deicing the syphon port.

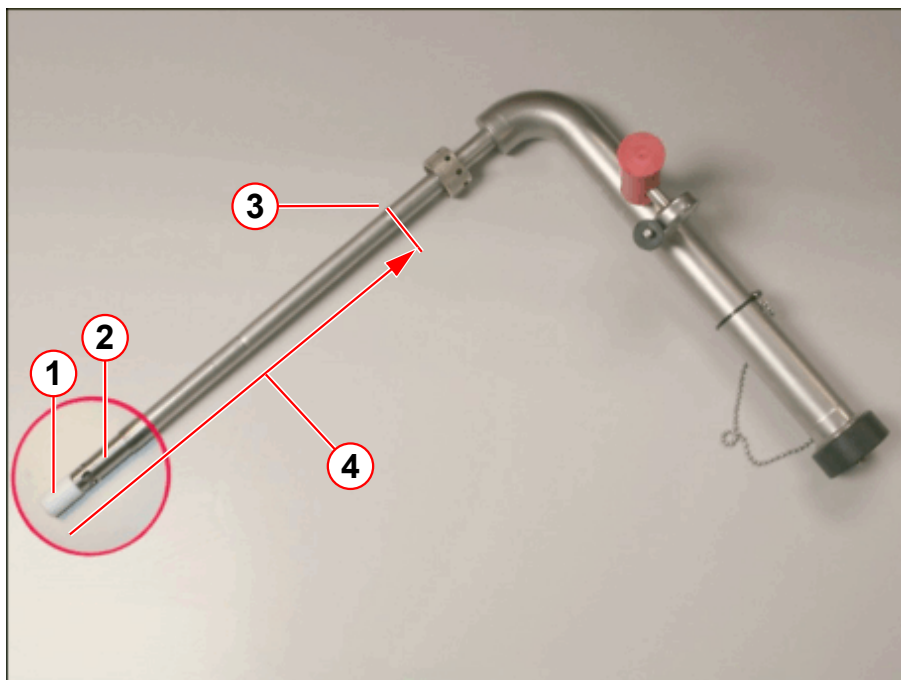
- Ramp the magnet to zero field.
- Deactivate the pressure heater.

- **Open** the magnet bypass valve to depressurize the helium vessel.

Assemble the Syphon

- Fit the required extension legs.
- Fit the de-ice diffuser tip to the syphon leg.
 - Check that the PTFE extension tip is screwed tightly onto the de-ice diffuser.

Fig. 88: Syphon assembly for de-icing the syphon port



- (1) PTFE tip
- (2) 6 hole de-ice diffuser
- (3) Syphon depth guideline
- (4) Depth to cone measurement

Tab. 9 Extension legs and de-ice diffuser requirements

Magnet Type	140 mm Leg	200 mm Leg	60 mm De-ice diffuser and PTFE tip	Depth to cone
Avanto (OR105)	2	1	1	485 mm
Esprea (OR122)	2	1	1	485 mm
Verio (OR97)	2	1	1	535 mm
MAGNETOM Es-senza (OR60)	2	1	1	505 mm
Aera (OR98)	2	1	1	554 mm

Magnet Type	140 mm Leg	200 mm Leg	60 mm De-ice diffuser and PTFE tip	Depth to cone
Skyra (OR99/103)	2	1	1	554 mm
Prisma (OR65)	2	1	1	550 mm

Refer to table above to determine the depth to cone measurement for required magnet type.

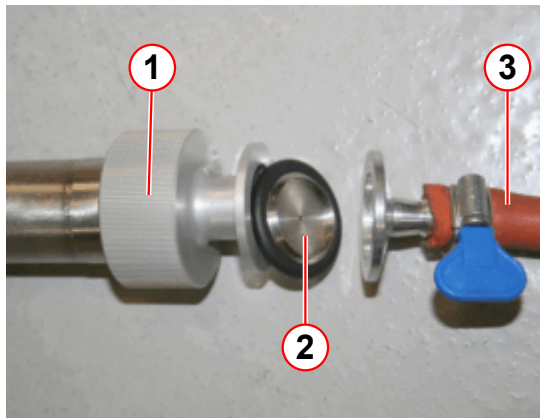
- Select the correct combination of extension legs.



The depth measurement from the top of the ring nut to the syphon cone is a guide. Add an additional 10 mm to ensure that the extension leg reaches the cone.

- Mark the syphon leg with a marker pen.
 - » Example: OR105: Mark a line at **495 mm** (485 mm + 10 mm).
- Connect the high pressure regulator to the helium gas bottle, (99.995% (minimum) helium gas).
- DO NOT turn on the helium gas supply.

Fig. 89: NW25 adaptor and flow restrictor



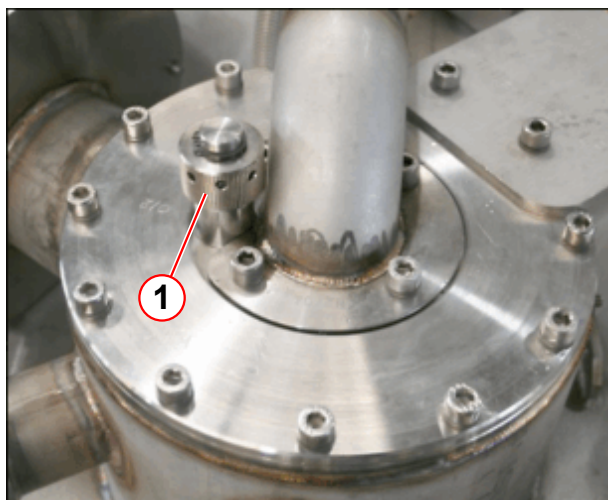
- (1) NW25 fill adaptor
- (2) NW25 flow restrictor
- (3) Helium supply line

- Remove the syphon end cap.
- Fit the NW25 hose adaptor to the coupling end of the syphon .
- Fit the NW25 flow restrictor, between the adaptor and helium supply line.
 - » The NW25 restrictor allows a calibrated flow of gas to pass through it for de-icing.

- Confirm that the magnet is depressurized completely.
- Warm and dry the syphon blanking plug using a heat gun.
- Remove the syphon blanking plug using the syphon 'C' wrench.

OR60/OR65/OR97/OR98/OR99/103/OR105/OR122
turret

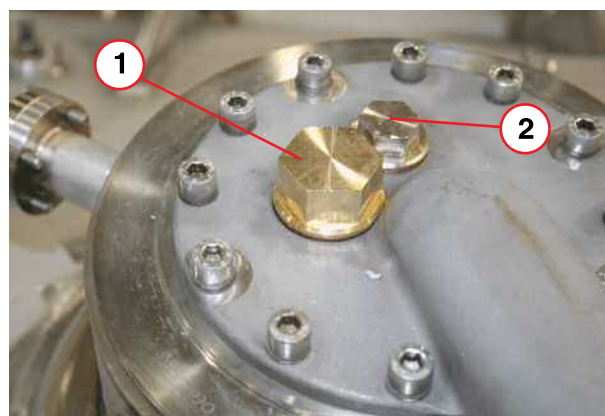
Fig. 90: Syphon entry port



(1) Syphon blanking plug

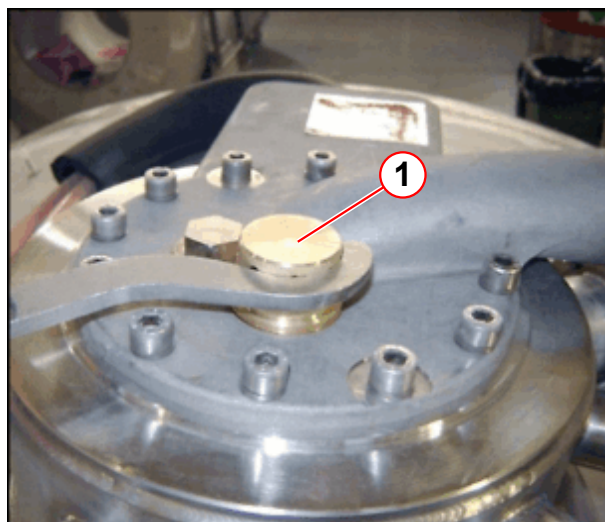
OR105/122/ turret

Fig. 91: Mk2 Syphon port entry



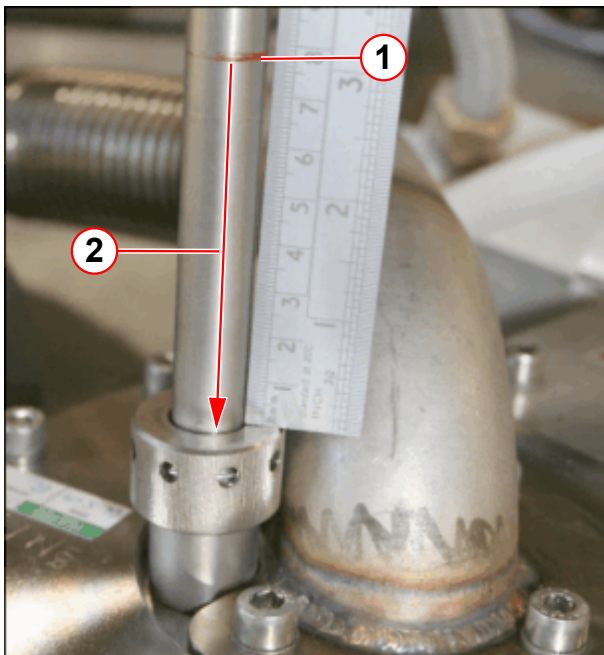
(1) Syphon blanking plug
(2) Sight glass

Fig. 92: Mk2 Syphon port



(1) Syphon blanking plug

Fig. 93: Measure the of depth of ice



- (1) Depth marker
- (2) Depth of ice in the turret

- Turn on the helium regulator and set the outlet pressure to **2 bar** on the gauge.
- Allow the helium gas to purge the syphon for 10 s.
- Insert the service syphon into the syphon port.
- Slowly lower the syphon until a resistance is felt.
 - » The extension tip is now touching the ice.
- The depth of ice in the syphon port is shown in (position 2).

- Wait **1 minute**:-
 - » Unscrew the syphon ring nut enough to allow the syphon to be pushed gently down 10 mm to 15 mm as the ice melts.
- Wait **1 minute**:-
 - » Repeat the process - push the syphon down a further 10 mm to 15 mm.

The internal magnet pressure gradually rises during the de-ice process.

- Continue until the syphon has engaged in the cone either:-
 - » by feeling the PTFE tip hitting the syphon cone.
 - » by reference to the mark made on the syphon extension leg.
(→ 1/ Fig. 93 Page 99)
- Continue the de-icing for a further 10 minutes. Turn off the helium gas pressure at the cylinder regulator.

It is possible to melt 10 mm to 15 mm of ice per minute. The process uses approximately one full (50 litre) cylinder of helium, taking 30 minutes to empty.



The internal magnet pressure quickly rise when there is a clear path through the ice into the helium vessel.

- Wait for the magnet pressure to return to atmosphere (zero).
- Remove the regulator line from the syphon end adaptor.
- Remove the NW25 flow constrictor and adaptor from the end of the syphon.
- Fit the syphon end cap.

- **CLOSE** the magnet bypass valve.
- Remove the syphon and fit the blanking plug or continue with helium filling.
 - Use standard filling procedures.

2.2 MSUP 2306 download instructions



Ensure a safe distance from the magnet for troubleshooting and service actions which require the use of a laptop with extended cabling.

Log files can be downloaded via either:

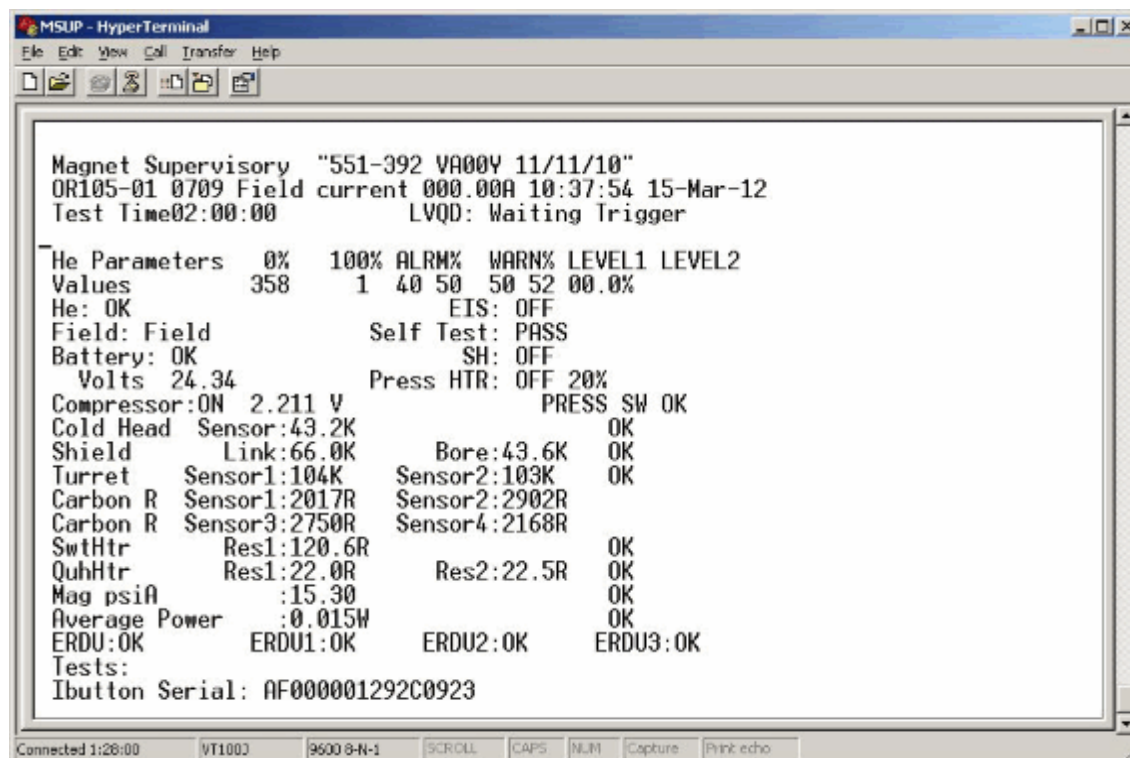
- **Direct connection** to the MSUP via RS232 (**X6**) connection.
- Service software at the **MRC (MRAWP) Host computer**.

2.2.1 Downloading via the RS232 connection

1. Turn on the **laptop** and open the **MSUP Terminal Emulation Software**.
 - » Set bits per second to 9600
 - » Set data bits to 8
 - » Set parity to 1
 - » Set flow control to none
 - » Set Emulation to VT100 (File: Properties:Settings)
2. **Connect** the laptop to connector **X6** on the MSUP **via RS232 cable**.
3. **Press ESC**.
4. Type **L0 6932** <enter>.
5. Type **R** <enter>

6. Press **ESC** to display the **MSUP Process display screen**.

Fig. 94: MSUP Process display screen (Example)



7. **Capture the screenshot.** (→ Capturing a screenshot / Page 106)
8. **Press ESC** to display the **MSUP Command input screen**.
9. Click **Transfer** on the **MSUP Terminal Emulation Screen** tool bar.
10. Click **Capture text**.
11. Click **Browse** and select a directory and a name for the file to be stored. The **Name** should contain the **magnet serial number**.
 - e.g. C:\temp\msn 12345678log.txt
12. Click **Start**.
13. Type the following:
 - **VA** <enter>
 - **VQ** <enter>
 - **VM** <enter>
 - **VP** <enter>
 - **VX** <enter>
14. Wait until data has been captured.
15. Click **Transfer, capture text, Stop**.
16. Type **R** <enter>.
17. **Email** the saved files to (→ Magnet HSC, <mailto:magnet.wscshsc.healthcare@siemens.com>)



After the download is complete, the MSUP must be reset.

This can be done via

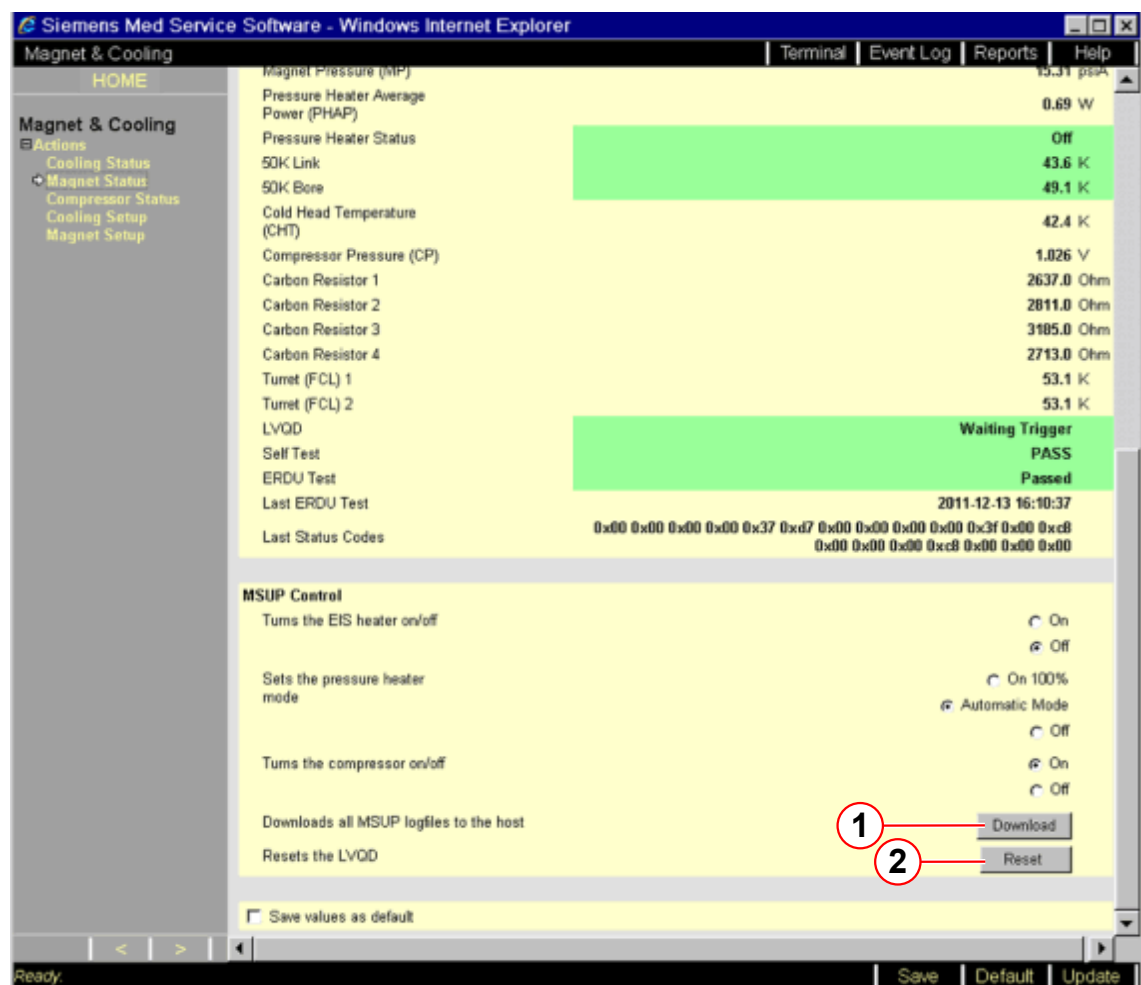
- rebooting the MRC (MRAWP) Host.
- cycling the power to the MSUP.
- allow the MSUP to complete an overnight sample.

2.2.2 Downloading the MSUP 2306 log file data to the MRC via SESO

Access the **Service software** at the **MRC (MRAWP) Host**

- Select menu item **"Magnet & Cooling."**
- Select **"Magnet Status"** in the **"Action"** menu.
- Click **"Download"** to download all the **MSUP logfiles** to the **MRC**.
 - » logfiles are **saved** to the MRC 'C' Drive, under **C:\MEDCOM.LOG**
(→ Fig. 96 Page 104)

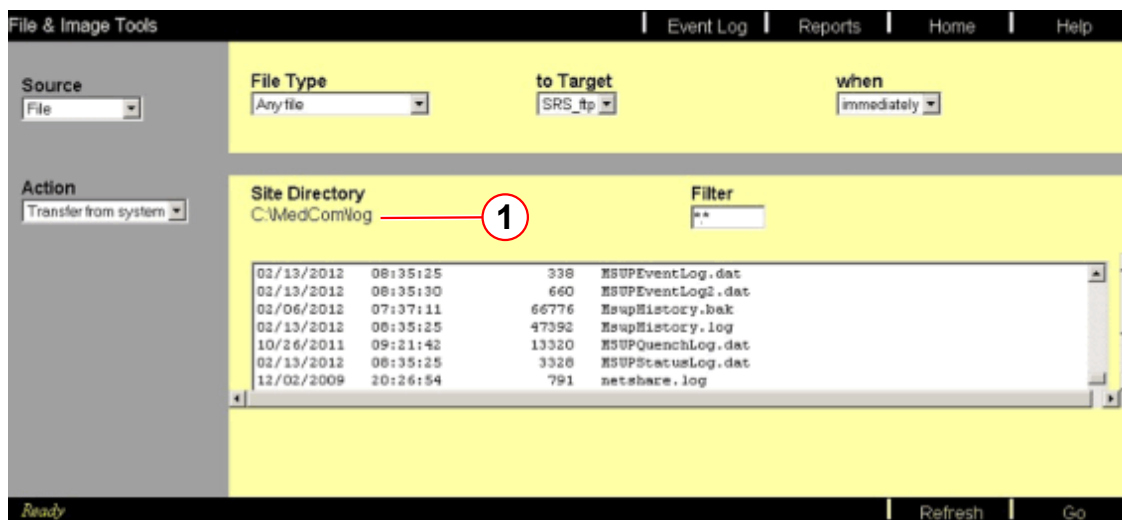
Fig. 95: Downloading MSUP logfiles to the Host



- (1) Download button
- (2) Reset button

- Click **"Reset"** (pos 2).

Fig. 96: Host MRC MSUP logfile location



(1) Site Directory

- **Email** the downloaded log files to (→ Magnet HSC, <mailto:magnet.wscshsc.healthcare@siemens.com>).

2.2.3 Capturing the iButton menu on the MSUP using a laptop

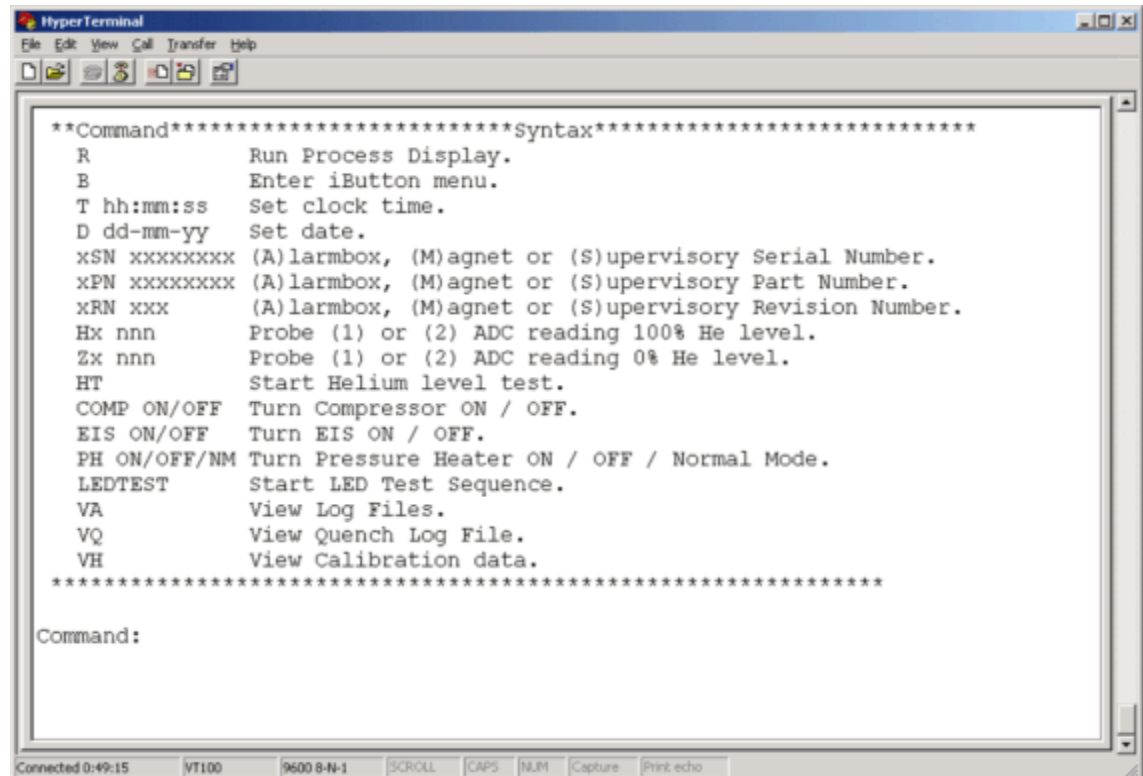


Ensure a safe distance from the magnet for troubleshooting and service actions which require the use of a laptop with extended cabling.

1. Turn on the laptop and open the **MSUP Terminal Emulation Software**.
 - » Set bits per second to 9600
 - » Set data bits to 8
 - » Set parity to 1
 - » Set flow control to none
 - » Set Emulation to VT100 (File:Properties:Settings)
2. **Connect** the laptop to connector **X6** on the MSUP **via RS232 cable**.
3. Press **ESC**.
4. Type **L0 6932** <enter>.
5. Type **R** <enter>.

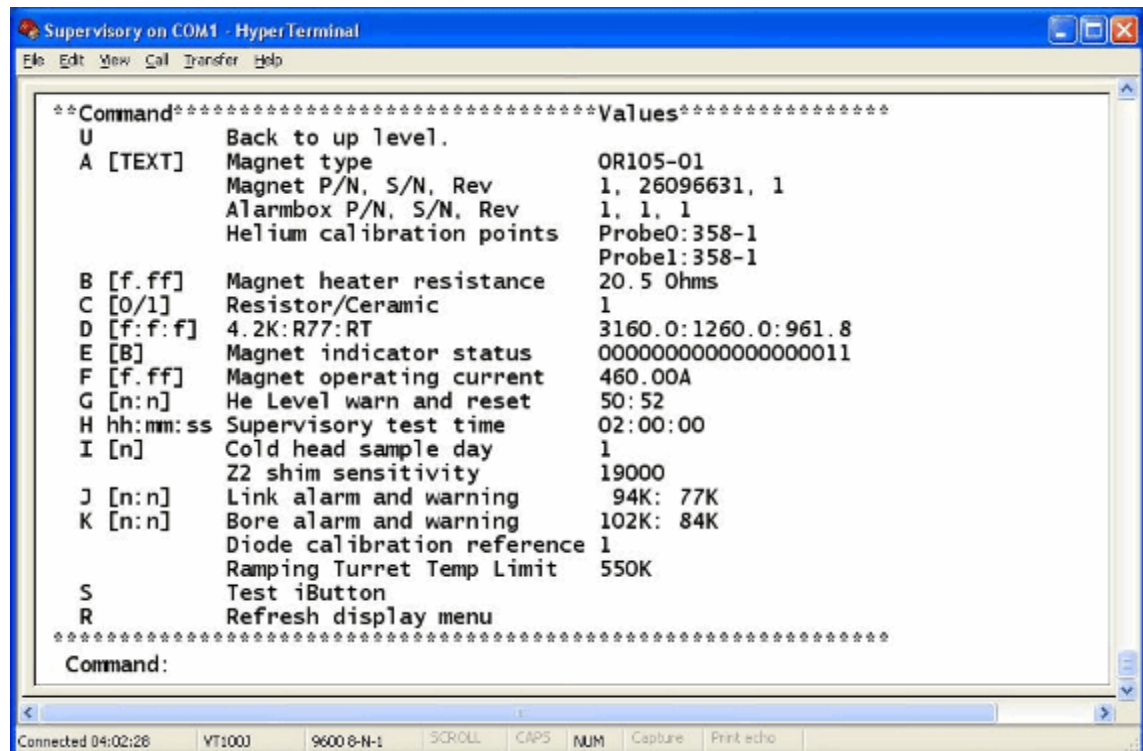
6. Press **ESC** to display the **MSUP Command input screen**.

Fig. 97: MSUP Command Input screen (example)



7. Type **B<enter>** this displays the **iButton menu**.

Fig. 98: MSUP 2306 iButton menu (example)



8. **Capture the screenshot.**(→ Capturing a screenshot / Page 106)

9. Type **U** <enter>.
10. Type **R** <enter>.
11. **Email** the saved files to (→ Magnet HSC, <mailto:magnet.wscshsc.healthcare@siemens.com>).

2.2.3.1 Capturing a screenshot

- Capture a **screenshot** of the **MSUP Terminal Emulation Screen**:
 - a) Press **Ctrl:Alt:Prt Sc**.
 - b) Go to: **Start:Programs:Accessories**:and open Paint.
 - c) Press **Ctrl:V** to paste screenshot.
 - d) **Name** and save as a **.jpg** or **.bmp**.
Name the file DD (**Day**), MM (**Month**), YYYY (**Year**), TTTT (**Time:24hour format**).
i.e. iButtonDDMMYYYYTTTT: MSUPDDMMYYYYTTTT.
- E-mail the saved file to (→ Magnet HSC, <mailto:magnet.wscshsc.healthcare@siemens.com>).

2.2.4 Restoring system calibration parameters

2.2.4.1 Downloading MSUP 2306 data to the MRC

The **MRC (MRAWP) Host** must be updated with the **system parameters**.

Check the **serial/revision/part numbers** are correct for the **Alarm box, MSUP 2306, and Magnet**.

Access the Service software at the MRC (MRAWP) Host.

- Select menu item **"Magnet & Cooling."**

- Select “Magnet Setup”.

Fig. 99: MSUP 2306 - Entering system calibration parameters

Siemens Med Service Software - Windows Internet Explorer

Magnet & Cooling | Terminal | Event Log | Reports | Help

HOME

Magnet & Cooling

- Actions
- Cooling Status
- Magnet Status
- Compressor Status
- Cooling Setup
- ➔ Magnet Setup

Magnet & Cooling - Magnet Setup

Magnet

Helium Level Probe 1	Min. (0%)	85.5	Ohm
	Max. (100%)	0.2	Ohm
Helium Level Probe 2	Min. (0%)	85.5	Ohm
	Max. (100%)	0.2	Ohm
Helium Level Probe Selection		☒ Probe 1 ☐ Probe 2 ☐ Both Probes	
Helium Level Warning	Min.	50	%
Shield Temperature Warning 50K Link	Max.	77.0	K
Shield Temperature Warning 50K Bore	Max.	84.0	K
CCR Calibration	Room Temperature	980.0	Ohm
	Nitrogen Temperature	1259.0	Ohm
	Helium Temperature	2772.0	Ohm
Pressure Heater Resistance		20.5	Ohm
Time Setting	Daily at	02:00	O'Clock
MSUP	Serial Number	1599	
	Revision Number	7	
	Part Number	10523587	
Alarmbox	Serial Number	1508	
	Revision Number	3	
	Part Number	10523563	
Magnet	Magnet Type	OR105-01	
	Serial Number	21259456	
	Revision Number	43	
	Part Number	7758167	
Magnet Current		509.87 A	

MSUP Data

Downloads the magnet setup/calibration data to the host (1)

Uploads the magnet setup/calibration data from the host to the MSUP

☐ Save values as default

Ready. (2) (3)

- (1) Download button (MSUP data stored on MRC)
- (2) Save button - (transfer calibration values to MSUP)
- (3) Update button (refreshes on-screen data)

- Compare the values with the **Magnet Acceptance Document**. (→ Fig. 100 Page 109)
- **Correct** MSUP parameters as necessary. Compare **acceptance data** to the **serial data** captured **from the MSUP**.

- Click **"UPDATE"**
 - » This **refreshes** the on-screen values.(→ 3/Fig. 99 Page 107)
- Update the **serial/revision/part** numbers (if required).
- Click **"SAVE"** to transfer calibration values **to the MSUP**. (→ 2/Fig. 99 Page 107)
- Click **"DOWNLOAD"**
 - » This takes the **MSUP data** and stores it in a file on the **MRC**.
(→ 1/Fig. 99 Page 107)

Fig. 100: OR105 Magnet Acceptance Document

SIEMENS		Magnet Acceptance Documentation Installation data MR Magnet Technology Quality Regulations	
Cryostat number	21326022	SMT reference	21330107 12.11
System description	1.494T Actively Shielded Superconducting Magnet System OR105		
Measured magnet centre (Offset)	+1 (P) mm		
If magnetic offset is outside the spec (± 5mm), the magnet will be marked (± 5mm).			
 This system requires <u>0</u> Z2 rings and <u>2.92 kg of iron</u> to shim. This is a theoretical calculation that may change due to site conditions.			
Operating parameters			
Description	Specification value	Measured value	
Operating current @ 1.494T	> 500 Amps	510.01 Amps	
Field/Current Ratio	N/A	12.48 kHz/0.1 Amp	
Magnet type:	OR105-01		
TDC calibration values		Carbon resistor values	
Room temperature (23 °C)	1009 Ohms		
Liquid nitrogen temperature (77K)	1309 Ohms		
Liquid helium temperature (4.2K)	2994 Ohms		
Position	Room temperature	Helium temperature	
TDC d - e	1034 Ohms	2837 Ohms	
Top a - e	980 Ohms	3014 Ohms	
Middle b - e	1110 Ohms	2681 Ohms	
Bottom c - e	1006 Ohms	2930 Ohms	
Helium level probe resistance (when using a 250mA supply)			
	Probe 1	Probe 2	
@ 0 resistance	85.36 Ohms	85.36 Ohms	
0% calibration	774	774	
100% resistance	0.22 Ohms	0.22 Ohms	
100% calibration	2	2	
Diode table no.	-1		
Pressure heater resistance	20.5 Ohms		
Final temperature sensor recon values			
Shield bore	Shield Link	Coldhead	Recon margin
47K	47K	42K	658mW
Checked by	Signed <u><i>ANOTHER</i></u>		
V Turner	Date 23/12/2011		
Page 1 of 1		25-AD EPS 000 01	

2.3 Magnet Supervision (MSUP)

The MSUP is a Field Replaceable Unit. The fuses are the only elements which can be replaced if required.

This procedure can be used to fault find the MPSU. An understanding of Tim systems is required to use the following information.

2.3.1 MSUP fuses

For location of these fuses, refer to (→ Fig. 101 Page 112).

Fuse	Rating	Title
F1	125 mA F/250 mA F*	EIS control
F2	4 A T	+36 V DC input supply
F3	500 mA T	He probe
F4	5 A T	Battery supply
F5	3 A F	Quench heater 1
F6	3 A F	Quench heater 2
F7	3 A F	Pressure heater control
F8	375 mA T/1A F*	Switch heater

*Higher value fitted after Rev 14.



Fuse F7 will blow if pressure heater is on for 150 minutes continuously.

2.3.2 MSUP test points

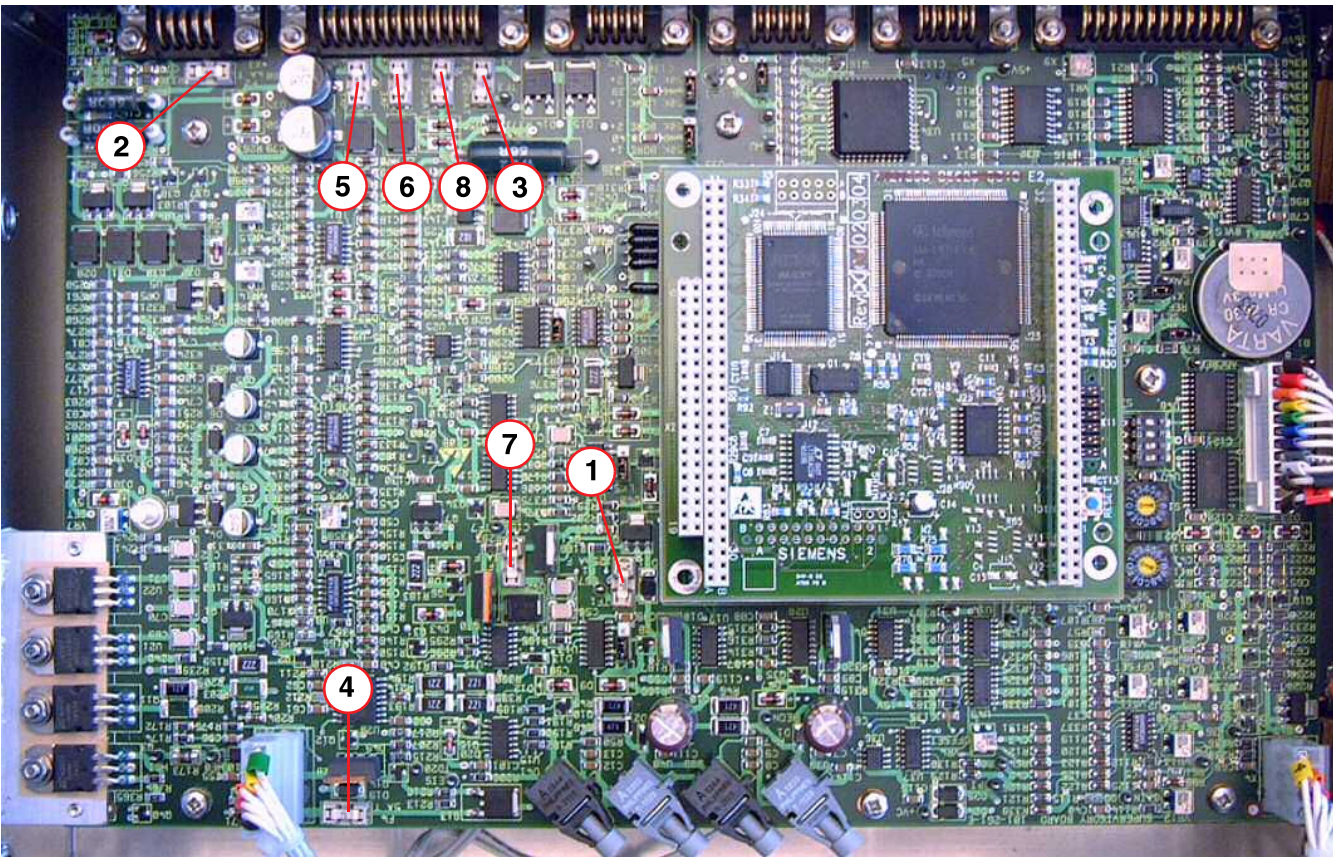
For location of these items, refer to PCB diagram (→ Fig. 102 Page 113).

Ref	Function
+36	Incoming +36V supply (ref 0 test point)
P	Magnet pressure signal to ADC
P0	Pressure monitor 0V reference test point
+VC	+VCONT supply to +15V regulator (ref 0 test point)
PI	Magnet pressure transducer signal (ref P0 test point)
+15V	15 V supply (ref 0 test point)
EIS	EIS output current signal (ref 0 test point)

Ref	Function
HE0	Helium probe 0V reference test point
HEI	Helium probe current signal (ref HE0 test point)
+5VREF	5V reference supply (ref P0 test point)
0	0V star point test point
PS	Pressure transducer supply (ref P0 test point)
+5V	5V supply (ref 0 test point)
+B	Battery voltage (ref 0 test point)
0V(S)	Sensor 0V test point
+5VREF2	5V reference supply #2
+20V	20V sensor supply (ref 0V (S) test point)

2.3.2.1 Fuses

Fig. 101: MSUP Fuses



- (1) Fuse F1, 125 mA, EIS Control (250 mA for MSUP with Rev. $\geq$ 14)
- (2) Fuse F2, 4 AT, +36 V MSUP supply
- (3) Fuse F3, 500 mA, LHe level probe
- (4) Fuse F4, 5 AT, Battery Supply
- (5) Fuse F5, 3 AF, Quench Heater 1
- (6) Fuse F5, 3 AF, Quench Heater 2
- (7) Fuse F7, 3 AF, Pressure Heater Control
- (8) Fuse F8, 375 mA, Switch Heater (1 AF for MSUP with Rev. $\geq$ 14)

2.3.3 MSUP Jumper links

For location of these items, refer to PCB diagram (→ Fig. 102 Page 113).

Link	Normal Operating Position	Description
LK1	50K LINK 1+ Normally linked pin1/pin2	Link Temp Sensor 1 or 2 Fit a link pin1/pin2 to enable Link Temp Sensor 1+ Fit a link pin2/pin3 to enable Link Temp Sensor 2+
LK2	50L BORE 1+ Normally linked pin1/pin2	Bore Temp Sensor 1 or 2 Fit a link pin1/pin2 to enable Bore Temp Sensor 1+ Fit a link pin2/pin3 to enable Bore Temp Sensor 2+
LK3	Normally linked pin2/pin3	Test or Normal operate delay EIS Crowbar. Fit a link pin1/pin2 to Test EIS Crowbar. Fit a link pin2/pin3 to enable Normal EIS Crowbar operation.
LK4	Normally linked pin1/pin2	Enable or Disable Hardware Pressure Control. Fit a link pin1/pin2 to Test Pressure Heater Crowbar. Fit a link pin2/pin3 to Disable Hardware Pressure Control.
LK5	Normaly linked pin2/pin3	Test or Normal operation of Pressure Heater Crowbar. Fit a link pin1/pin2 to Test Pressure Heater Crowbar. Fit a link pin2/pin3 to enable Normal Pressure Heater Crowbar operation.
LK6	Normally linked pin1/pin2	Internal or External Magnet Field Sensor. Fit a link pin1/pin2 to select an Internal Magnet Field Sensor. Fit a link pin2/pin3 to select an External Magnet Field Sensor.
X10	Normally linked	For Development purposes only. U3 U43 Chip Select Line. Disables data logging to EPROM.
X11	Normally linked	For development purposes only. Disables the ADC reference voltage.

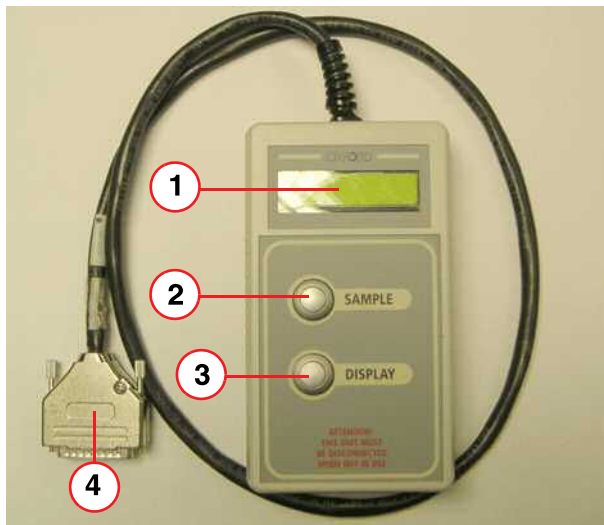
2.3.4 MSUP-Display Unit

The only operations that can be performed independently of the controlling host computer are through the **MSUP-display unit**.

Required tools

MSUP-display unit

Fig. 103: MSUP- Display Unit



- (1) LCD Display
- (2) Sample button
- (3) Display scroll button
- (4) 25 way 'D' connector

Performing the procedure

- Connect the display unit to **X9** on the MSUP via the 25 way 'D' connector (4)
- Ensure that power has been applied to the MSUP.
- Switch on the hand held unit by simultaneously pressing the **Sample** button (2) and **Display** button (3).
- Press **Display** (3).
- The unit will **automatically** carry out a helium level test. The **helium fill level** is shown in the display (1).

- Scroll through the parameters by pressing the **DISPLAY** button, one parameter for each press.

The parameters in order of presentation are:

1. He level (press sample to recheck level)
2. Battery volts
3. 50K link, bore (K, K)
4. Carbon resistor 1, 2 (ohm, ohm)
5. Carbon resistor 3, 4 (ohm, ohm)
6. Turret 1, 2 (K, K)
7. Switch heater (Ohm, status)
8. Quench heaters 1, 2 (Ohm)
9. Cold head (K)
10. ERDU buttons (status)
11. Magnet pressure (psiA)
12. Pressure heater (Off/On, %) (press Sample to switch on, 50 min timeout)
13. Pressure heater (average power over last hour of previous measurements)
14. EIS (Off, Sample to switch On, 60 min timeout)
15. Self-test



Disconnect the display unit from the MSUP after test is finished.

2.3.5 MSUP download instructions



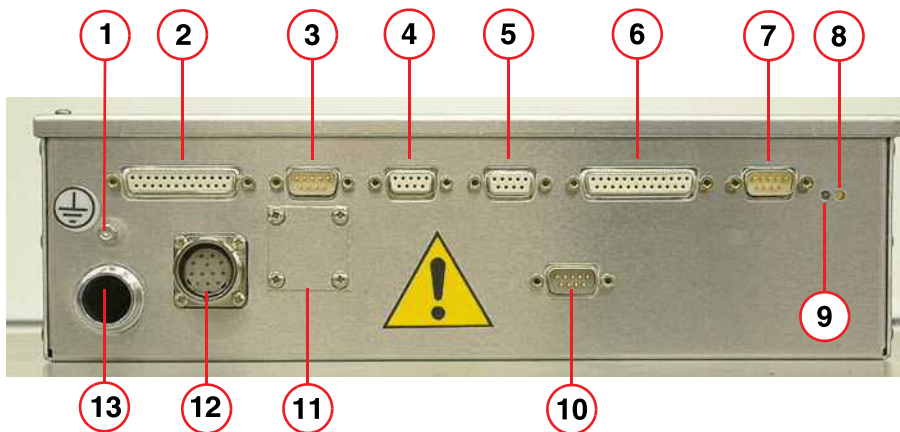
Ensure a safe distance from the magnet for troubleshooting and service actions which require the use of a laptop with extended cabling.

Instruction on how to download the **log file** of the MSUP

1. Turn on the laptop and open the hyper terminal application. Set to the following settings. Select the appropriate Com Port.
 - » Set bits per second to 9600
 - Set data bits to 8
 - Set parity to none
 - Set stop bits to 1
 - Set flow control to none

In the file, properties, settings tab, set Emulation to VT100.
2. Connect the laptop via an **RS232 cable** to connector **X5** on the MSUP (pos 3).

Fig. 104: MSUP 2657 Indicators and input/output connectors



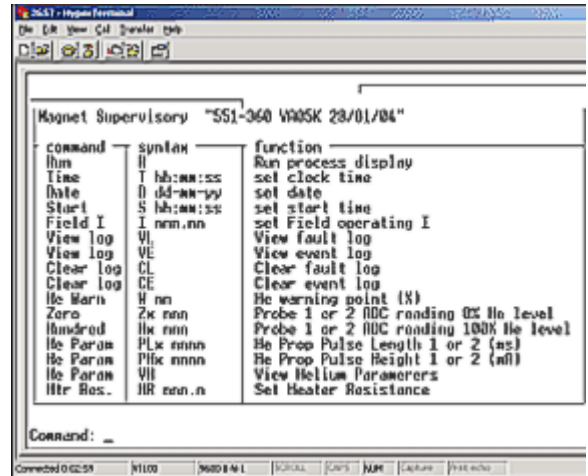
- (1) Earth stud (6mm)
- (2) X9 Display
- (3) X5 RS232
- (4) X6 ERDU/Field
- (5) X2 CH/DIAG/PRESS
- (6) X1 Turret
- (7) X3 Power/SL/UPS
- (8) Pressure heater LED (yellow)
- (9) Power ON LED (green)
- (10) X12 Battery
- (11) ERDU switch cover
- (12) X7 Magnet ID
- (13) CAN Fiber optics wave guide

3. Press **Esc** and then type **R.** (→ Fig. 106 Page 117) should be displayed.

Fig. 105: Download



Fig. 106: Download



4. Press **Esc.** (→ Fig. 105 Page 117) should be displayed.
5. Press **Esc** again.
6. Click "**Transfer**"
7. Click "**Capture text**"
8. Click "**Browse**" and select a directory and a name for the file to be stored. Name should contain the magnet serial number.
 e.g. C:\Temp\MSN 12345678log.txt
9. Click "**Start**"
10. Type the following
 - » **VL** <enter>
 - VE** <enter>
 - VC** <enter>
 - VH** <enter>
 - VQ** <enter>
 - VD** <enter>
 - VP** <enter>
 - VX** <enter>
11. Click "**Transfer**", "**capture text**", "**Stop**" after downloaded data has been transferred.
12. Capture a screenshot of the **MSUP terminal emulation** screen:
 - a) Press **Ctrl:Alt:Prt Sc**
 - b) Go to: **Start:Programs:Accessories:**and open Paint
 - c) Press **Ctrl:V** to paste screenshot
 - d) Name and save as a .jpg or .bmp.

13. E-mail the saved file and screenshot to (→ Magnet HSC, magnet.hsc.med@siemens.com)

2.4 Magnet Supervision (MSUP 2657B)

The MSUP is a **Field Replaceable Unit**. The fuses are the only elements which can be replaced if required.

This procedure can be used to fault find the MSUP. An understanding of Tim systems is required to use the following information.

2.4.1 MSUP fuses

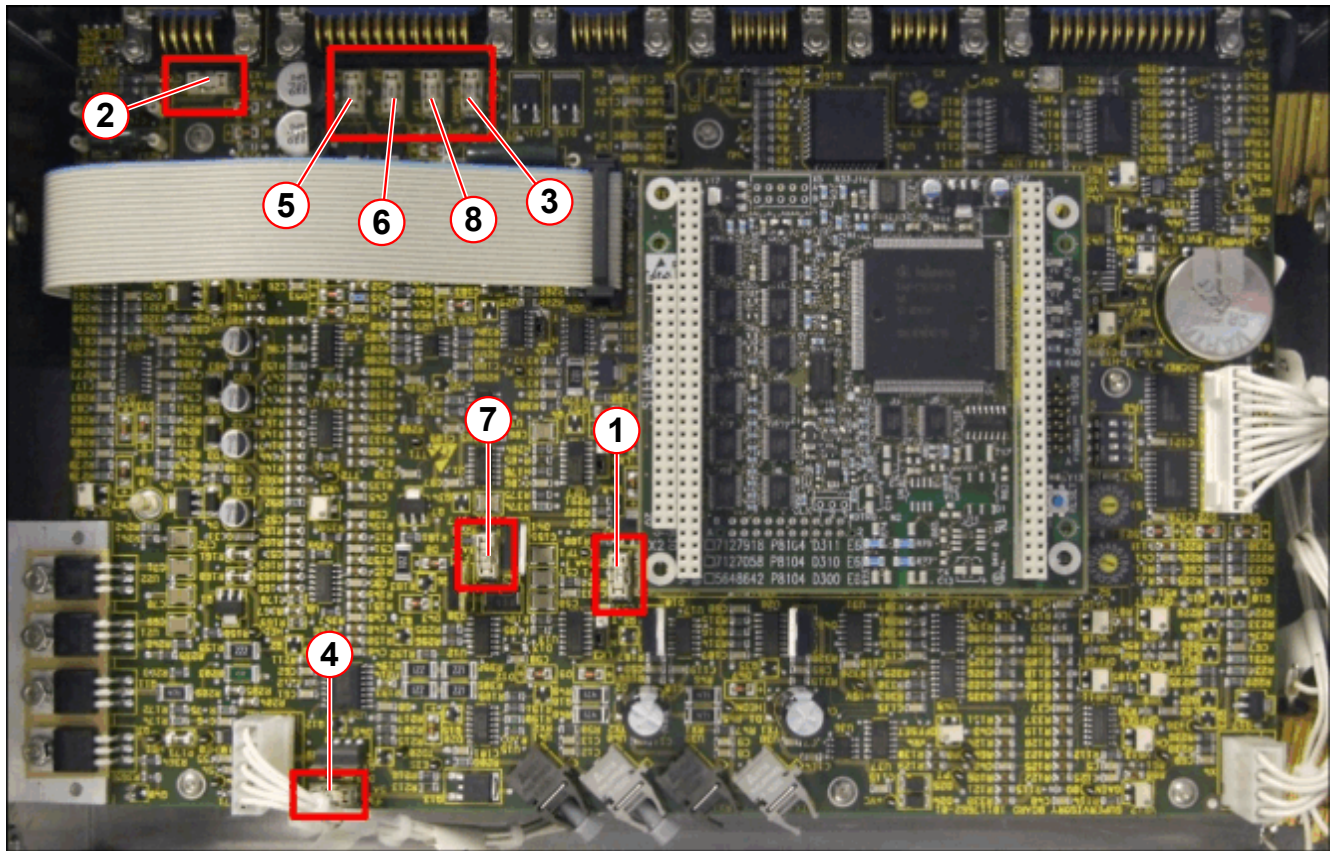
For location of these fuses, refer to (→ Fig. 107 Page 120).

Fuse	Rating	Title
F1	125 mA F/250 mA F*	EIS control
F2	4 A T	+36 V DC input supply
F3	500 mA T	He probe
F4	5 A T	Battery supply
F5	3 A F	Quench heater 1
F6	3 A F	Quench heater 2
F7	3 A F	Pressure heater control
F8	375 mA T/1A F*	Switch heater

*Higher value fitted after Rev 14.

2.4.2 Fuses

Fig. 107: MSUP 2657B - Fuses



- (1) Fuse F1, 125 mA, EIS Control (250 mAF for MSUP with Rev. = 14)
- (2) Fuse F2, 4 AT, +36 V MSUP supply
- (3) Fuse F3, 500 mA, LHe level probe
- (4) Fuse F4, 5 AT, Battery Supply
- (5) Fuse F5, 3 AF, Quench Heater 1
- (6) Fuse F5, 3 AF, Quench Heater 2
- (7) Fuse F7, 3 AF, Pressure Heater Control
- (8) Fuse F8, 375 mAF, Switch Heater (1 AF for MSUP with Rev. = 14)

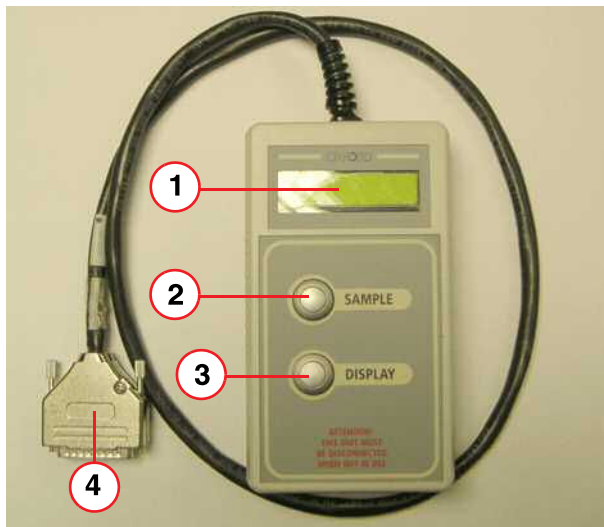
2.4.3 Hand-held MSUP Display-Unit

The only operations that can be performed independently of the controlling host computer are through the **hand-held MSUP Display-Unit**.

Required tools

Hand-held MSUP Display-Unit.

Fig. 108: MSUP- Display Unit



- (1) LCD Display
- (2) Sample button
- (3) Display scroll button
- (4) 25 way 'D' connector

Performing the procedure

- Connect the display unit to **X9** on the MSUP via the 25 way 'D' connector (4)
- Ensure that power has been applied to the MSUP.
- Press **SAMPLE** button (2).
 - The unit will carry out a helium level and battery test.
 - The helium fill level is shown in the display (1).
- Press **DISPLAY** (3), and **SAMPLE** (2) simultaneously to unlock the scroll function.

- Scroll through the parameters by pressing the **DISPLAY** button, one parameter for each press.

The parameters in order of presentation are:

1. He level (press sample to recheck level)
2. Battery volts
3. 50K link, bore (K, K)
4. Carbon resistor 1, 2 (ohm, ohm)
5. Carbon resistor 3, 4¹ (ohm, ohm)
6. Turret 1, 2 (K, K)
7. Switch heater (Ohm, status)
8. Quench heaters 1, 2 (Ohm)
9. Cold head (K)
10. ERDU buttons (status)
11. Magnet pressure (psiA)
12. Pressure heater (Off/On, %) (press Sample to switch on, 50 min time-out)
13. Pressure heater (average power over last hour of previous measurements)
14. EIS (Off, Sample to switch On, 130 min time-out)
15. Self-test
16. Compressor status (OFF/ON,P,T)
17. LVQD (Waiting trigger/quench location) Time & Date
 - Time and date

1. Carbon resistor 4 is shown in Kelvin for CCR type magnets.

- Magnet type
- Part no and software version



Disconnect the display unit from the MSUP after test is finished.

2.4.4 MSUP download instructions



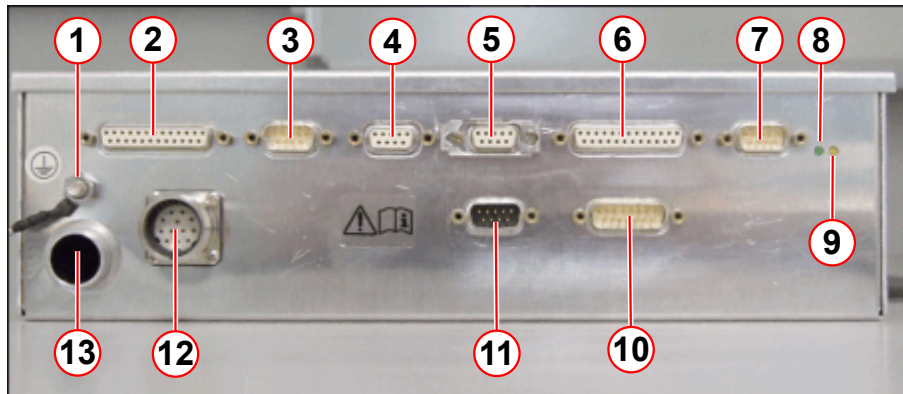
Ensure a safe distance from the magnet for troubleshooting and service actions which require the use of a laptop with extended cabling.

Instruction on how to download the **log file** of the MSUP

1. Turn on the laptop and open the hyper terminal application. Set to the following settings. Select the appropriate Com Port.
 - » Set bits per second to 9600
 - Set data bits to 8
 - Set parity to none
 - Set stop bits to 1
 - Set flow control to none
 - In the file, properties, settings tab, set Emulation to VT100.

2. Connect the laptop via an **RS232 cable** to connector **X5** on the MSUP (pos 3).

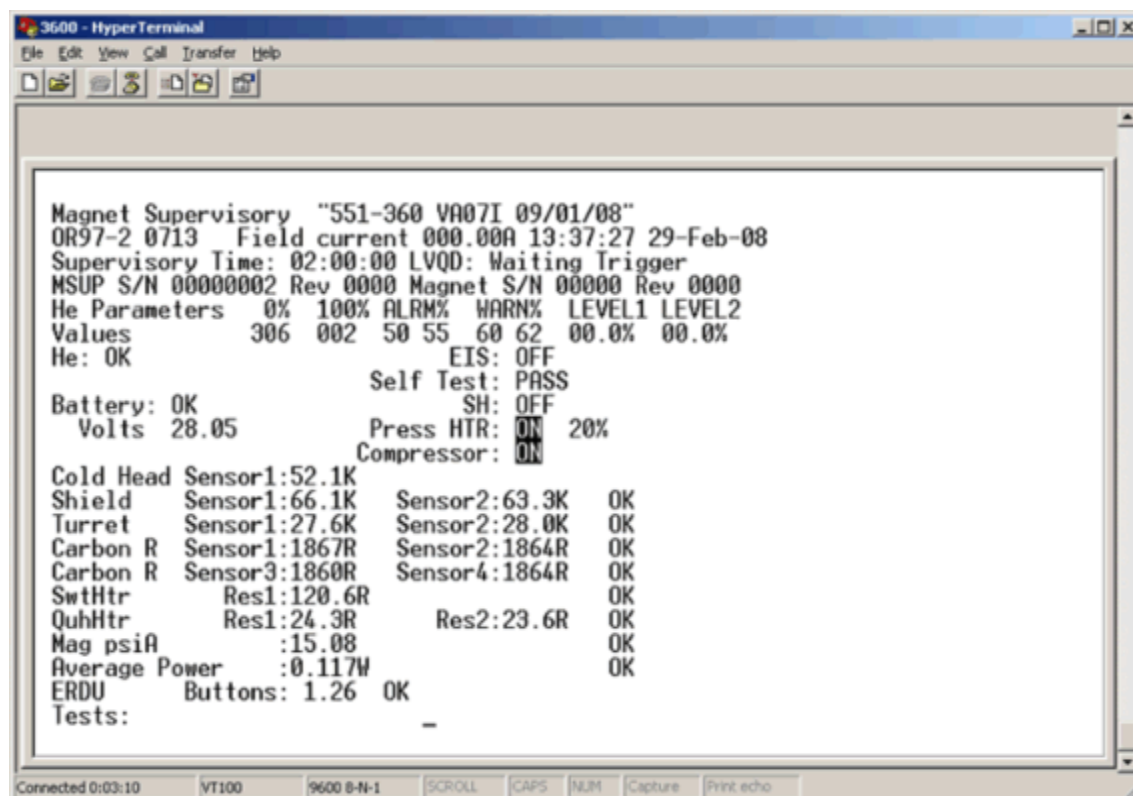
Fig. 109: MSUP 2657B - Indicators and input/output connections



- (1) Earth stud (6mm)
- (2) X9 DISPLAY
- (3) X5 RS232
- (4) X6 ERDU
- (5) X2 CH/DIAG/PRESS
- (6) X1 TURRET
- (7) X3 POWER/SL/UPS
- (8) Power ON LED (green)
- (9) Pressure heater LED (yellow)
- (10) X14 LVQD
- (11) X12 BATTERY
- (12) X7 Magnet ID
- (13) CAN Fiber optics wave guide

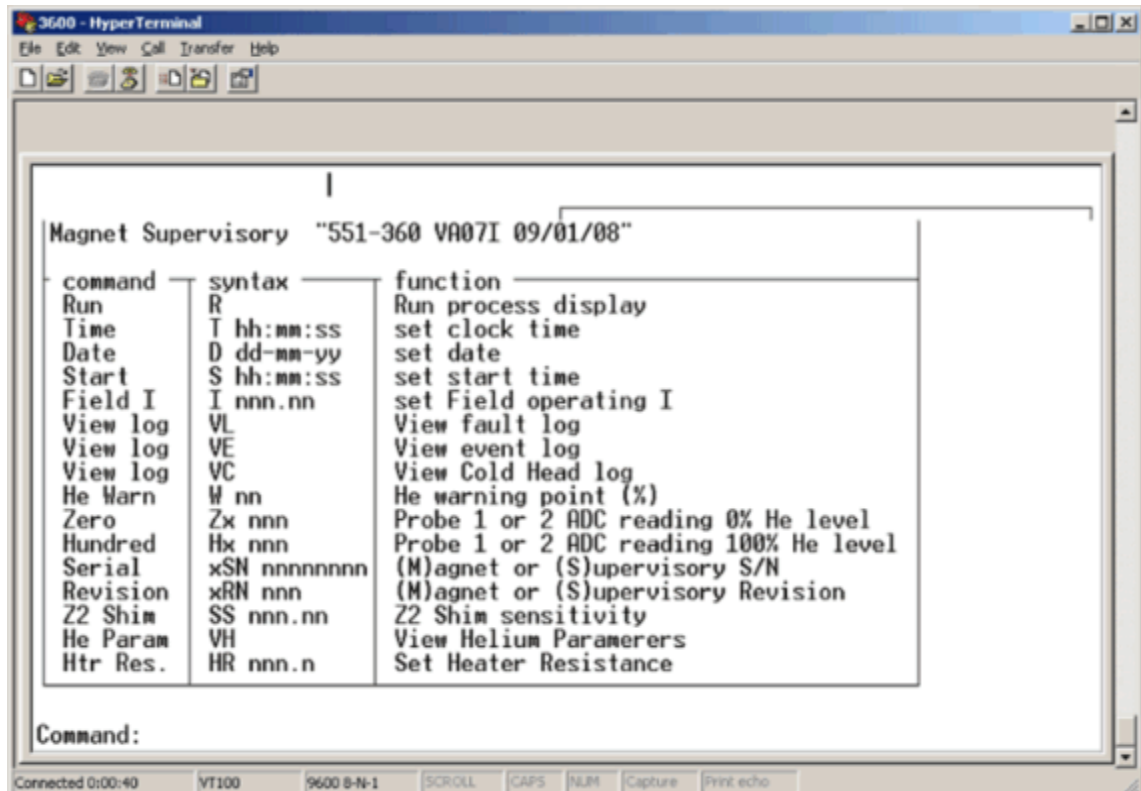
3. The **MSUP Process Display** is displayed.

Fig. 110: MSUP 2657B - Process Display screen (typical)



4. Press **Esc.** to display the **MSUP Command Input** screen.

Fig. 111: MSUP 2657B - Command Input screen (typical)



5. Click **"Transfer"**
6. Click **"Capture text"**
7. Click **"Browse"** and select a directory and a name for the file to be stored. Name should contain the **magnet serial number**.
e.g. C:\Temp\MSN 12345678log.txt
8. Click **"Start"**
9. Type the following
 - » VL <enter>
 - VE <enter>
 - VC <enter>
 - VH <enter>
 - VQ <enter>
 - VD <enter>
 - VP <enter>
 - VX <enter>
10. Click **"Transfer"**, **"capture text"**, **"Stop"** after downloaded data has been transferred.
11. Capture a **screenshot** of the **MSUP terminal emulation** screen:
 - a) Press **Ctrl:Alt:Prt Sc**
 - b) Go to: **Start:Programs:Accessories:**and open Paint

- c) Press **Ctrl:V** to paste screenshot
 - d) Name and save as a .jpg or .bmp.
12. E-mail the saved file and screenshot to (→ Magnet HSC, <mailto:magnet.wscshsc.healthcare@siemens.com>)

Tab. 10 MSUP 2657B Input Codes

Description	Command	Operation/Example
Time	T hh:mm:ss<R>	Set time in real time clock 1 hour 30mins 30secs - enter as T 13:30:30<R>
Date	D dd-mm-yy<R>	Set date in real time clock 1st Jan 2008 - enter as D 01-01-08<R>
Start test time	S hh:mm:ss<R>	Set supervisory test time 2 hour 45mins 45secs - enter as T 2:45:45<R>
Magnet operating current	I nnn.nn<R>	Set Mag Operating Current Operating current 685.02 - enter as I 685.02<R>
Helium warn point	W nn<R>	Set Helium Warning point 50% enter as W 50<R>
0% Calibration (probe 1)	Z1 nnnn<R>	Set He Probe 1 0% Probe1: 88.8 ohms -enter as Z1 805<R>
0% Calibration (probe 2)	Z2 nnnn<R>	Set He Probe 2 0% Probe2: 88.8 ohms - enter as Z2 805<R>
100% Calibration (probe 1)	H1 nnnn<R>	He Probe 1 100% Probe1: 1.8 ohms - enter as H1 16<R>
100% Calibration (probe 2)	H2 nnnn<R>	He Probe 2 100% Probe2: 1.8 ohms - enter as H2 16<R>
Magnet Serial No.	MSN nnnnnnnn<R>	Adjust Magnet Serial No Enter as MSN12345678<R>. Password required
Magnet Revision No.	MRN nnn<R>	Adjust Mag Rev No Magnet Rev 4 - enter as MRN4<R> Password required
Supervisory Serial No.	SSN nnnnnnnn<R>	Adjust Supervisory Serial No. Supervisory Serial No S0014 - enter as SSN S0014<R>
Supervisory Revision No.	SRN nnn<R>	Adjust Supervisory Revision No Supervisory Revision No 14 - enter as SRN14<R> Password required
Pressure heater resistance	HR nnn.n<R>	Adjust Pressure Heater Res 19.4 ohms - enter as HR 19.4<R> Password required
Shield bore warning	L4 2947<R> SBW nnn<R>	Shield Bore Warning Shield Bore warning of 77K - enter as SBW 77<R>

Description	Command	Operation/Example
CCR Calibration	SR <R> XXX.X <R> YYYY.Y <R> ZZZZ.Z <R>	Adjust CCR calibration to RT (XXX.X), 77K (YYYY.Y), 4.2K (ZZZZ.Z) Password required Refer to Replacement of Parts for Calibration of CCRs .
Run process image	R<R>	Refresh the screen
View status log	VL<R>	View status log
View error log	VE<R>	View error log
View month log	VM<R>	View month log
View cold head log	VC<R>	View cold head log
View quench log	VQ<R>	View quench log
View LVQD log	VD<R>	View LVQD log
View helium log	VH<R>	View the helium calibrations
View helium pulse log	VP<R>	View the helium pulse log
Notes: ■ <R> = 'Enter' or 'Return'. ■ Where a password is required, type L 9912<R> before the command is input. ■ CCR Calibration - The MSUP will prompt the user to input values after the command SR has been used. - Refer to the Magnet Acceptance Document for TDC CCR4 Calibration Values.		

2.5 Refrigerator

2.5.1 Helium gas charging CSW-71



Make sure the refrigerator is powered off and compressor and cold head temperatures are almost at room temperature before adjusting the helium gas pressure.

If the compressor has been running, switch off and leave for a minimum of 30 minutes before checking static pressure.



CAUTION

High pressure helium gas.

Failure to observe the following safety warning may result in personal injury.

- » Protective clothing, gloves and eye protection must be worn.

Required tools

Charge and vent adapter (gas charging tool)(part number **08396660**)

Bleed adapter assembly (part number **10125874**)

High pressure regulator for helium gas bottle.

30bar Helium pressure hose (charging line).

Helium gas bottle, purity of 99.999% .



Use only helium of purity 99.999% or better.

3/4" (19mm) Open-end wrench/spanner.

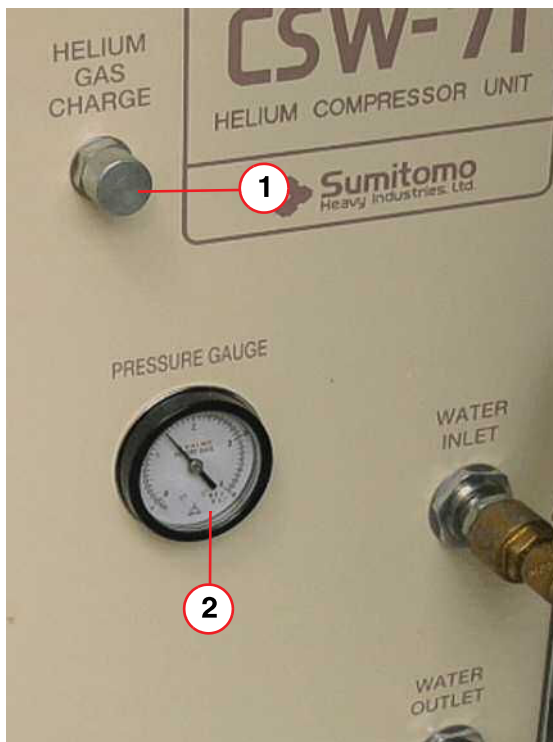
5/8" (16mm) Open-end wrench/spanner.

Performing the procedure

- Turn **off** the compressor at the **main power switch**.
 - » The Alarm Box will display a red **COMPRESSOR** LED, and an audible alarm will sound.
 - » Press the **ACKNOWLEDGE** button to cancel the audible alarm

- Wait for at least **30 minutes** to allow the static pressure to stabilize.

Fig. 112: Compressor He Gas charge & pressure gauge



- Remove the **protective cap (1)** from the **helium gas charge coupling** on the compressor unit.
- Check the **rubber aeroquip gasket** for dirt or dust and ensure it is correctly fitted. Replace if required (part no **03861494**)

NOTE: Incorrect fitment may cause a helium gas leak.

- (1) Helium gas charge
(2) Helium pressure gauge

- Connect a **high pressure regulator** and **charging line** to the helium **gas bottle** (purity 99.999%).

Fig. 113: He compressor - charge and vent adaptor (gas charging tool)



- (1) Charging line (30bar)
(2) Charge and vent adaptor valve
(3) Aeroquip coupling (female)
(4) Bleed adaptor assembly

- **Open** the valve on the **helium gas bottle**.
 - Ensure the **regulator valve** is closed.
 - Ensure that **30bar** of helium gas is available.
- Screw the **charging line** (→ 1/ Fig. 113 Page 130) into the **gas charging tool**.
- Screw the **bleed adaptor assembly** (→ 4/ Fig. 113 Page 130) into the **gas charging tool**.
- **Purge the charging line** of any air contaminants by setting the high pressure regulator to 10psi for 30 seconds and **opening** the valve on the **gas charging tool**.
- Unscrew and remove the **bleed adaptor assembly**.
- **Close** the **gas charging tool valve** (→ 2/ Fig. 113 Page 130)
- Set the **gas pressure regulator** to:
 - » **16.5 - 17 bar or 239 - 246 psig**
- Using two wrenches, connect the **gas charging tool** to the **helium gas charge coupling** on the compressor unit.
- Slowly **open** the gas charging tool valve.

Fig. 114: Helium Compressor



(1) Gas charging tool with charging line

- Wait 60 seconds for gas pressure to **stabilize** on the compressor gauge.
- If the compressor pressure still indicates **lower** than the required:-
 - Adjust the gas pressure regulator until the compressor gauge reads **16.0 - 16.5 bar**.
 - Wait 60 seconds for gas pressure to **stabilize** on the compressor gauge.
- When pressure is achieved, **close** the **gas charge tool valve**.



It is possible to overpressurize the compressor. To adjust pressure, refer to (→ Adjusting the equalization pressure / Page 132).

- **Disconnect** the **gas charging tool** from the **compressor**.
 - Take care not to dislodge the aeroquip gasket seal.
- Replace the **protective cap** on the **helium gas charge coupling**.
- **Close** the valve on the **helium gas bottle**.
- Screw the **bleed adaptor assembly** into the end of the **charging line**.
- Slowly open the **gas charging tool valve** to release pressure from the **charging line**.
 - » The regulator gauge should read **zero**.
- **Unscrew** and remove the **bleed adaptor assembly**.
- **Disconnect** the high pressure **regulator** from the **gas bottle**.
- Store the **charge line** to keep it clean.
- Re-start the compressor.

2.5.1.1 Adjusting the equalization pressure

If the compressor has been **overpressurized**, the excess helium gas must be vented to atmosphere.



The compressor should not be operated above the specified pressure. Overpressure will cause the compressor to run in continuous bypass mode.

- Turn **off** the compressor at the **main power switch**.
 - » The Alarm Box will display a red **COMPRESSOR** LED, and an audible alarm will sound.
 - » Press the **ACKNOWLEDGE** button to cancel the audible alarm.
- Wait for at least **30 minutes** to allow the static pressure to stabilize.
- **Remove** the **protective cap** from the **helium gas charge coupling** on the compressor unit (→ 1/Fig. 112 Page 130).
- Make sure the **gas charge tool valve** is **closed** (→ 2/Fig. 113 Page 130).
- Using two wrenches, connect the **gas charging tool** to the **helium gas charge coupling** on the compressor unit.
- **Slowly open** the **gas charging tool valve**.
- **Vent** the **helium gas** until the required **equalization pressure** is achieved.
- **Close** the **gas charging tool valve**.
- **Disconnect** the **gas charging tool** from the **compressor**.
 - Take care not to dislodge the aeroquip gasket seal.
- Replace the **protective cap** on the **helium gas charge coupling**.
- Re-start the compressor.

2.5.2 Helium gas charging F70



Make sure the refrigerator is powered off and compressor and cold head temperatures are almost at room temperature before adjusting the helium gas pressure.



CAUTION

High-pressure helium gas

Failure to observe the following safety warning may result in personal injury.

- » Protective clothing, gloves, and eye protection must be worn.

Required tools

- **Charge & Vent tool Kit** (part number **10125858**)
 - Adaptor valve (gas charging tool) (part number **10125873**)
 - Bleed adapter assembly (part number **10125874**)
- High pressure regulator for helium gas bottle
- 30bar Helium pressure hose (charging line)
- Helium gas bottle, purity of 99.999%



Use only helium of purity 99.999% or better.

3/4" (19 mm) Open-end wrench/spanner.

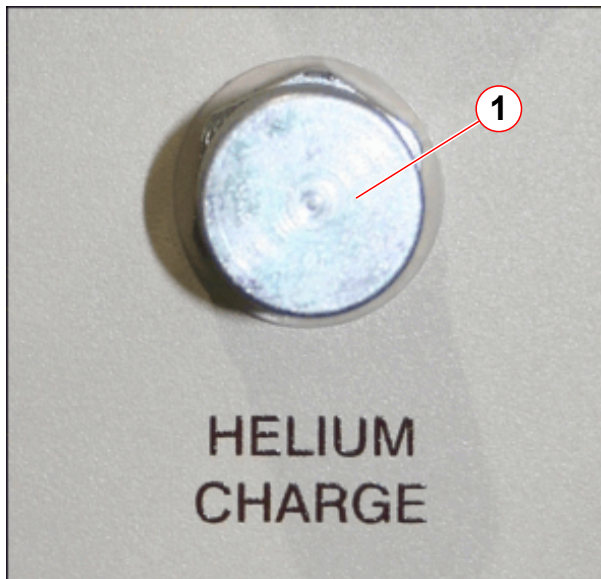
5/8" (16 mm) Open-end wrench/spanner.

Performing the procedure

- Turn **off** the compressor at the main power switch.
 - » The Alarm Box will display a warning LED, and an **audible** alarm will sound.
 - » Press the **ACKNOWLEDGE** button to cancel the audible alarm

- Wait for at least **30 minutes** to allow the static pressure to stabilize.

Fig. 115: F70 Compressor - Helium charge port



(1) Protective cap

- Remove the **protective cap (1)** from the **helium gas charge coupling** on the compressor unit.

Fig. 116: F70 Compressor - Helium Charge fill port gasket seal



(1) Aeroquip gasket seal

- Check the **rubber aeroquip gasket** for dirt or dust and ensure it is correctly fitted. Replace if required (part number **03861494**).

NOTE: Incorrect fitment may cause a helium gas leak.

- Connect a **high pressure regulator** and **charging line** to the helium **gas bottle** (purity 99.999%).

Fig. 117: He compressor - charge and vent adaptor (gas charging tool)



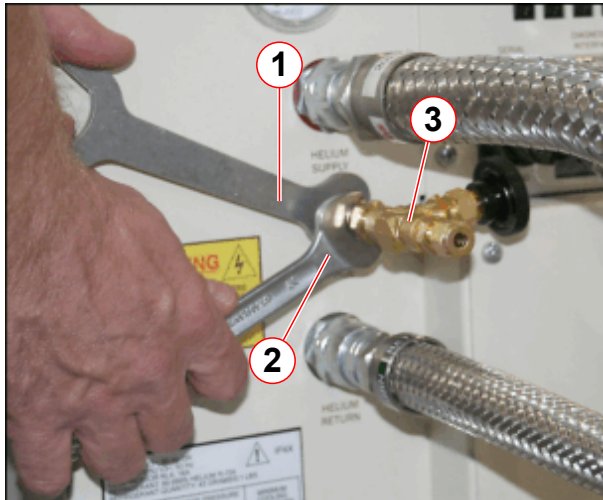
- (1) Charging line (30bar)
- (2) Charge and vent adaptor valve
- (3) Aeroquip coupling (female)
- (4) Bleed adaptor assembly

- **Open** the valve on the **helium gas bottle**.
 - Ensure the **regulator valve** is closed.
 - Ensure that **30bar** of helium gas is available.
- Screw the **charging line** (→ 1/Fig. 117 Page 135) into the **adaptor valve**.
- Screw the **bleed adaptor assembly** (→ 4/Fig. 117 Page 135) into the **adaptor valve**.
- **Purge the charging line** of any air contaminants by setting the **high pressure regulator** to 10 psi for 30 seconds and **opening** the valve on the **adaptor valve**.
- Unscrew and remove the **bleed adapter assembly**.
- **Close** the **adaptor valve**.
- Set the **gas pressure regulator** as required - see table below:-

Tab. 11 Helium gas pressures

Cold head	Static pressure	Dynamic pressure
4 K - 50/60 Hz	13.5 - 14.0 bar (13.6 nominal) 196 - 203 psi (G) 1350 - 1400 kPa	21 - 22 bar 304 - 319 psi (G) 2100 - 2200 kPa

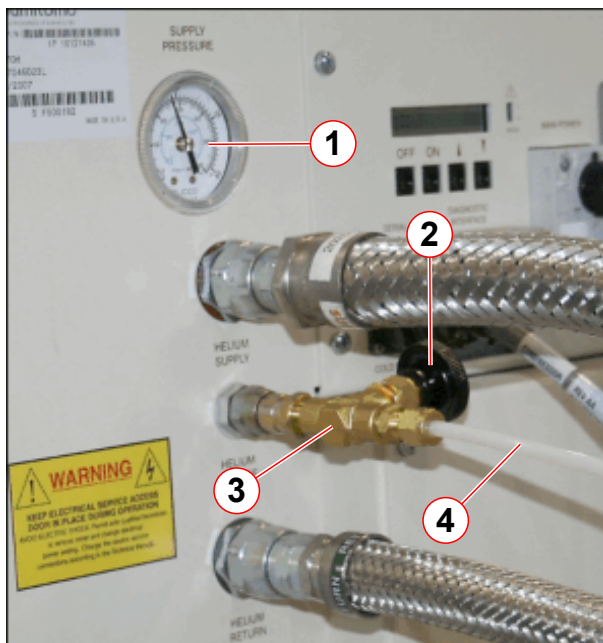
Fig. 118: F70 Compressor - Fit/remove charge & vent adaptor (gas charging tool)



- (1) Turn 5/8" (16mm) spanner/wrench
- (2) Hold 3/4" (19mm) spanner/wrench
- (3) Charge & vent adaptor (gas charging tool)

- Using two wrenches, connect the **adaptor valve** to the **helium gas charge coupling** on the compressor unit.

Fig. 119: F70 - Charge & vent adaptor (gas charging tool)



- (1) Pressure gauge
- (2) Valve
- (3) Charge and vent adaptor (gas charging tool)
- (4) Charging line

- **Slowly open** the adaptor valve 2.

- Wait **60 seconds** for gas pressure to **stabilize** on the compressor gauge.
- If the compressor pressure still indicates **lower** than the required:-
 - Slowly open the **high pressure regulator** until the required pressure has been achieved.
 - Wait 60 seconds for gas pressure to stabilize on the compressor gauge.

- When pressure is achieved, **close** the **adaptor valve**. (→ 2/Fig. 119 Page 136)



It is possible to overpressurize the compressor. To adjust pressure, refer to Adjusting the equalization pressure.

- **Disconnect** the **adaptor valve** from the **compressor**.
 - Take care not to dislodge the aeroquip gasket seal.
- Replace the **protective cap** on the **helium gas charge coupling**.
- **Close** the valve on the **helium gas bottle**.
- Screw the **bleed adaptor assembly** into the end of the **charging line**.
- **Slowly** open the **adaptor valve** to release pressure from the **charging line**.
 - » The regulator gauge should read **zero**.
- **Unscrew** and remove the **bleed adaptor assembly**.
- **Disconnect** the high pressure **regulator** from the **gas bottle**.
- Store the **charge line** to keep it clean.
- Restart the compressor.

2.5.2.1 Adjusting the equalization pressure

If the compressor has been **over pressurized**, the excess helium gas must be vented to atmosphere.

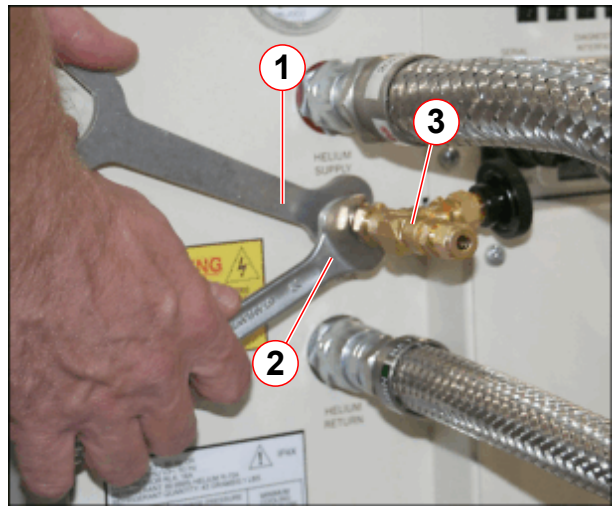


The compressor should not be operated above the specified pressure. Overpressure will cause the compressor to run in continuous bypass mode.

- Turn **off** the compressor at the **main power switch**.
 - » The Alarm Box will display a red warning LED, and an audible alarm will sound.
 - » Press the **ACKNOWLEDGE** button to cancel the audible alarm.
- Wait for at least **30 minutes** to allow the static pressure to stabilize.
- **Remove** the **protective cap** from the **helium gas charge coupling** on the compressor unit.

- Make sure the **gas charge tool valve** is closed.

Fig. 120: F70 Compressor - Fit/remove charge & vent adaptor (gas charging tool)



- (1) Turn 5/8" (16mm) spanner/wrench
- (2) Hold 3/4" (19mm) spanner/wrench
- (3) Charge & vent adaptor (gas charging tool)

- Using two wrenches, connect the **adaptor valve** to the **helium gas charge coupling** on the compressor unit.

- **Slowly open the adaptor valve.**
- **Vent the helium gas** until the required **equalization pressure** is achieved.
- **Close the adaptor valve.**
- **Disconnect the adaptor valve** from the **compressor**.
 - Take care not to dislodge the aeroquip gasket seal.
- Replace the **protective cap** on the **helium gas charge coupling**.
- **Restart** the compressor.

2.5.3 Error corrections

An error condition will cause either a system alarm or a shutdown. The following table lists the alarm and the shutdown errors that are monitored. Overload trip of the Mains Power switch is not monitored or reported.

Tab. 12 F70 Error Conditions

LCD Display First Line	Type	Error	Indication
Helium Temp - ERR	Shutdown	High helium discharge temperature	Temperature > 93°C (200°F)
Water Temp - ERR	Alarm	High water supply temperature	Temperature > 35°C (95°F)
Water Flow - ERR	Alarm	Low water flow	Temperature > 46°C (115°F)

LCD Display First Line	Type	Error	Indication
Motor Temp - ERR	Shutdown	High compressor motor winding temp (internal protector open)	Return Pressure >140psig while compressor is running
Phase Seq - ERR	Shutdown	Phase sequence or open fuse	Monitored by microprocessor
Return Pressure- ERR	Shutdown	Loss of gas charge	Return pressure < 15psig
Oil Level - ERR	Shutdown	Loss of circulating oil	Oil level switch opens
DB-25 Config 1 - ERR (or Config 2)	Shutdown	Remote cable does not match switch setting.	Monitored by microprocessor

If the compressor has been shut down by one of these interlocks, do not restart until the problem has been found and corrected. Refer to the Troubleshooting Guide to identify the problem.

In the event of a shutdown error, the compressor motor and cold head valve motor are turned off. The microprocessor will annunciate the error condition via signals at the System Diagnostics Connector and via the LCD Display.

The Mains Power switch trips when steady-state current exceeds 1.2 x the front panel set point. When tripped, the switch knob locates halfway between the on and off positions.

2.5.3.1 Clearing Error Conditions



Errors can only be cleared locally when configuration mode is set for Mode 1. When configuration mode is set for Mode 2, errors must be cleared using the DB25 diagnostic interface.

When an error is corrected, the message(s) can be cleared from the display and the system diagnostics connector by any of the following procedures:

- Press the **DISPLAY UP** and **DISPLAY DOWN** (arrow) buttons simultaneously.
- Turn the **MAIN POWER SWITCH** to **OFF**, pause briefly and switch back **ON**.

The MSUP will automatically apply a reset signal via the DB25 diagnostic interface.

2.5.3.2 Restarting after an Error Condition

When the shutdown is caused by helium discharge high-temperature, the compressor will be able to restart only after it has cooled enough for the thermistor to read a temperature below the cut-off point. After waiting for the compressor to cool, clear the error message then push the ON button to restart. Should the compressor fail to start, allow more cooling time. Repeat the restart procedure. Check cooling water temperature and flow. Compare with Specifications.

If the shutdown is due to a motor high temperature, the motor winding high temperature relay will automatically reset in approximately 30 minutes. To restart the compressor, clear the error message then push the ON button. If the compressor fails to start, allow more cooling time. Repeat the restart procedure.

If the motor circuit protector opens, the knob moves part way toward OFF. Reset it by turning the knob completely to OFF, then clockwise to ON. The compressor and the cold head should start automatically.

2.5.4 F70 Compressor download instructions

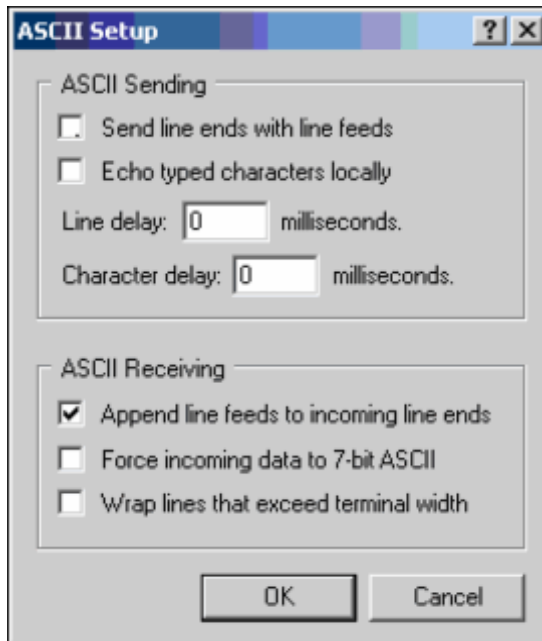
2.5.4.1 Tooling required

- RS232 Cable
- Laptop

2.5.4.2 Download Instructions

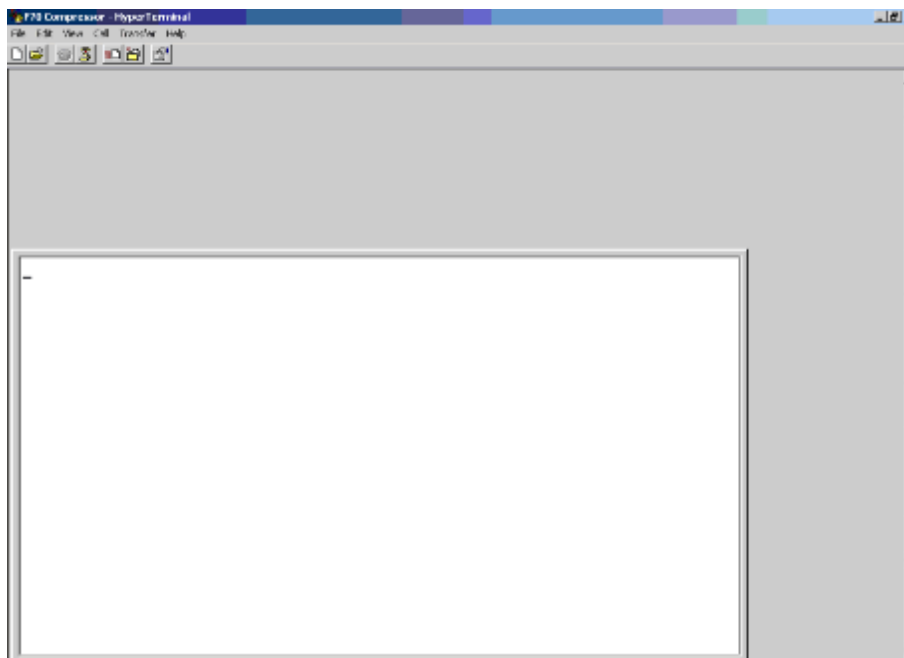
1. **Turn off** the compressor **main breaker**.
2. Turn on laptop and open the Terminal emulation software (e.g HyperTerminal) application. Set to the following settings. Select the appropriate **Com Port**.
 - » Set bits per second to 9600
 - » Set data bits to 8
 - » Set parity to none
 - » Set stop bits to 1
 - » Set flow control to none
 - » Set Emulation to VT100 (File: Properties:Settings)
 - » Click OK
 - » Set ASCII setup (File: Properties:Settings)
 - » Click **OK**

Fig. 121: ASCII Setup



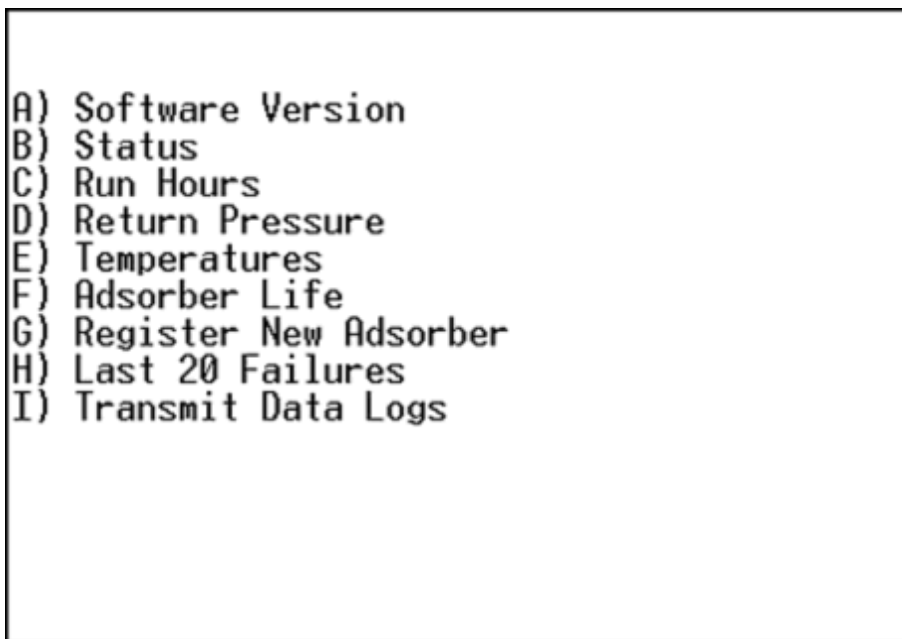
3. Connect the **laptop** via an **RS232 cable** to **serial port** on front of F70 compressor.
4. Turn **on** compressor **main breaker**.
5. Open **terminal emulation program**, the screen below is displayed.

Fig. 122: F70 Terminal emulation program



6. Press letter '**M**' <Enter> to open menu. See below:.

Fig. 123: Select from the menu



7. Click '**Transfer**'.
8. Click '**Capture text**'.
9. Click '**Browse**'. Select a directory and a name for the file to be stored. Name should contain **Compressor serial number** e.g. **C:\Temp\Comp 12345678log.tx**.
10. Click '**Start**'.
11. Type the following:-



Allow 10 seconds between each letter to ensure complete download of data.

- A <Enter>
 - B <Enter>
 - C <Enter>
 - D <Enter>
 - E <Enter>
 - F <Enter>
 - G <Enter>
 - H <Enter>
 - I <Enter>
12. Click '**Transfer**' '**Capture text**' '**Stop**' after completing transfer.
 13. Turn **off** the compressor.
 14. **Disconnect** the RS232 cable from the serial port of the F70.
 15. Turn **on** the compressor **main breaker**.

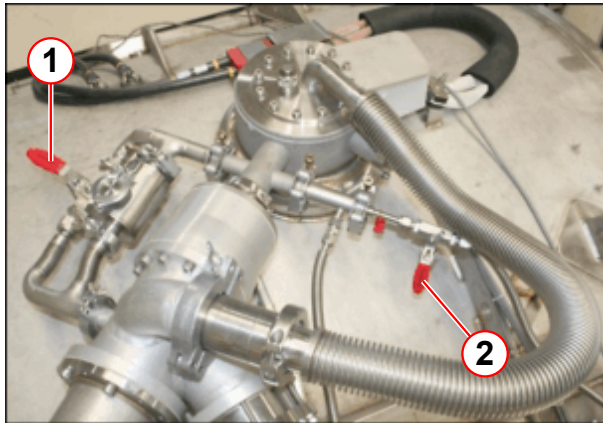
16. **Email** captured file to
(→ , (Magnet HSC, <mailto:magnet.wscshsc.healthcare@siemens.com>)).

2.5.5 Refrigeration Failures with Magnet at Field

Failure of the refrigeration system when the magnet is at field, will see increased magnet pressure with the loss of liquid helium.

To reduce helium loss and risk of the magnet quenching the cold head bypass must be opened after **30 hours**, (not before).

Fig. 124: Avanto - Espree Mk1 vent valves



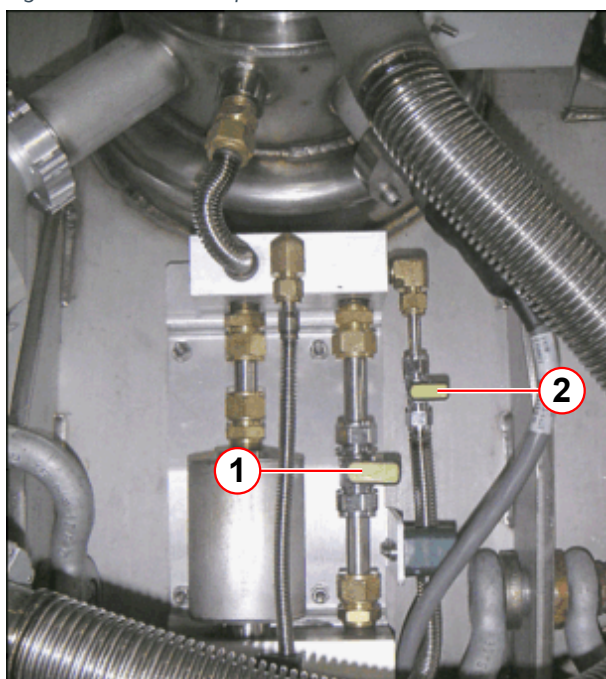
- (1) Magnet bypass valve (closed)
(2) Cold head bypass valve (closed)

MK1 Venting

For long periods of refrigeration down time, greater than **30 hours**:

- Open the cold head valve to reduce helium loss.
- Close cold head valve as soon as the refrigeration is running correctly.

Fig. 125: Avanto - Espree Mk2 vent valves



- (1) Magnet bypass valve (closed)
 (2) Cold head bypass valve (closed)

MK2 Venting

For long periods of refrigeration down time, greater than **30 hours**:

- Open the cold head valve to reduce helium loss.
- Close cold head valve as soon as the refrigeration is running correctly.

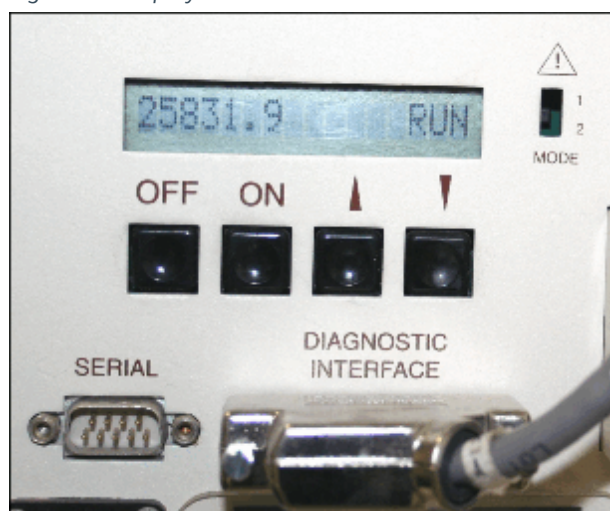
2.5.6 F70 Hours run clock display

2.5.6.1 Counters / Timers

Compressors running software version 1.7 have 3 counters. Each one can help with service operations.

- **Display Run Time.**

Fig. 126: Display Run Time



This timer is the run hours that are displayed on the liquid crystal display. The run hours can also be viewed on a compressor download.

The Display Run Time is a counter that displays the total running time (in hours) since the compressor has last visited a Sumitomo service center

- **Master Run Time.**

This timer can only be viewed on a compressor download.

The master run time is a timer that displays (in hours) the total run time of the compressor since it was first manufactured. An example of this will read '000844.8 Master Run Time'.

- **Adsorber Life.**

This timer can only normally be viewed on a compressor download. It is a count down of when the next Adsorber on the compressor should be changed.

After an Adsorber is changed in the compressor the time must be reset, this takes the timer to 30,000 hours. The timer will then count down while the compressor is running and will even become negative once zero hours is passed. At the next adsorber change the timer is reset back to 30,000 hours. An example of this will read '29155 HRS Adsorber Life'.

The Adsorber Life timer can be reset through the **front panel control buttons**:-

1. With the Main Power switch turned **OFF**, press and hold **all four control buttons** below the LCD window.
2. Keep these buttons held and turn the Main Power Switch **ON** to enter "**Service Level**" mode. Release the buttons.
3. Scroll using the arrow control buttons until the display shows "**NEW ADS? ON=Yes.**"
4. Press the **ON** control button. Display will show "**SAVE? ON = Yes OFF = NO.**"
5. Press the **ON** control button again to complete. (Press the **OFF** control button to back out of the function). When **ON** is pressed, display will show: "**Saving New ADS**". When function is complete, display will again show "**NEW ADS? ON =YES.**"
6. Turn the **main power** switch to OFF to leave "**Service Level**".
7. Switch **ON** main power breaker.
8. Scroll **UP/DOWN** arrows, ADS Life = **30,000Hrs.**
9. The new adsorber is now **registered**.

2.5.7 Fuse Location F70



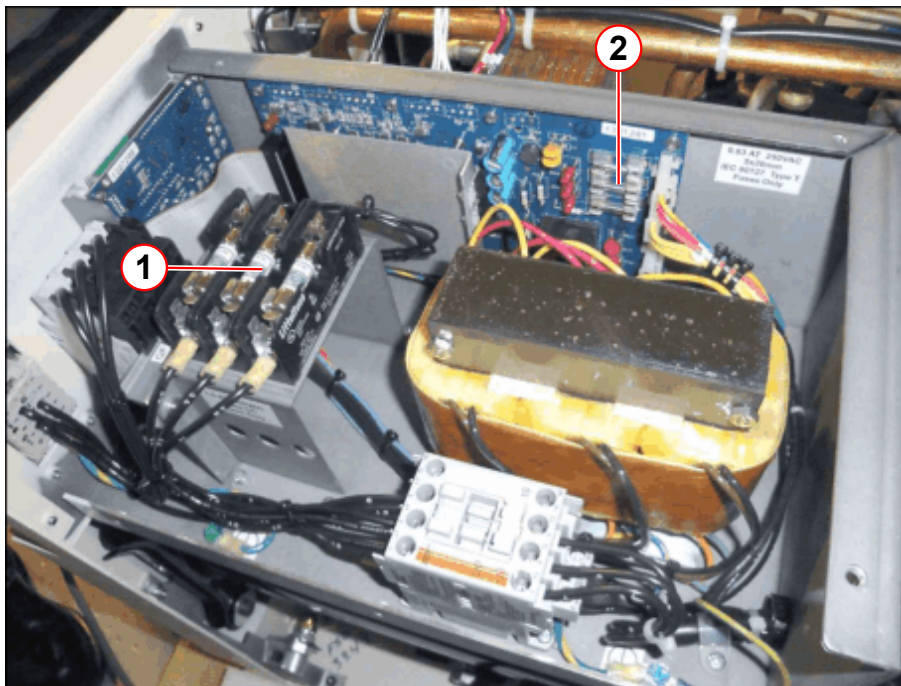
WARNING

AVOID ELECTRIC SHOCK

Failure to observe this warning can result in serious injury.

- » Disconnect the power to the compressor before troubleshooting the electrical components.
- » Permit only qualified electrical technicians to open electrical enclosures.

Fig. 127: F70 - Electrical Chassis



- (1) Fuses, FU-1, FU-2, FU-3
- (2) PCB 1 with Fuses, FU-4, FU-5, FU-6

2.5.8 Static Compressor Pressure Test

2.5.8.1 General



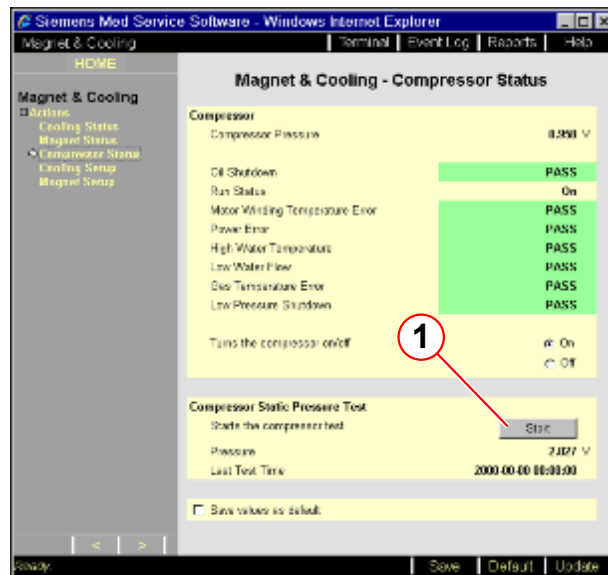
Static Compressor Pressure Test process can only be completed by DOT / Fit systems

On the MRC host there is an option to complete a Compressor Static Pressure Test. This can only be completed on site or with remote full access to the system. The voltage read on the system is equal to the static pressure reading on the compressor.

2.5.8.2 Procedure

- Log on to the MRC host (gaining full access if required).
- Navigate to 'Magnet and Cooling', then on to the Compressor Status.

Fig. 128: Magnet & Cooling - Compressor Status

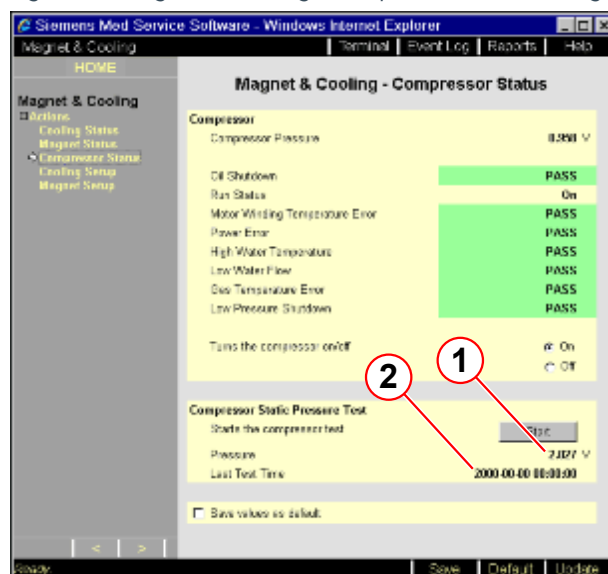


(1) Start compressor test button

- Click the start button. (under Compressor Static Pressure Test)
This will initialize a compressor test.

The first step of the automatic test is the compressor is turned off, this in turn turns off the cold head.

Fig. 129: Magnet & Cooling - Compressor Status Voltage



(1) Voltage reading
(2) Logged time and date

- After approximately one hour the voltage reading is taken and then displayed with the time and date of when the reading was taken.

Once the system has finished taking and recording the readings and logged them the cold head will restart automatically.

2.5.8.3 Converting the voltage reading to a compressor static pressure.

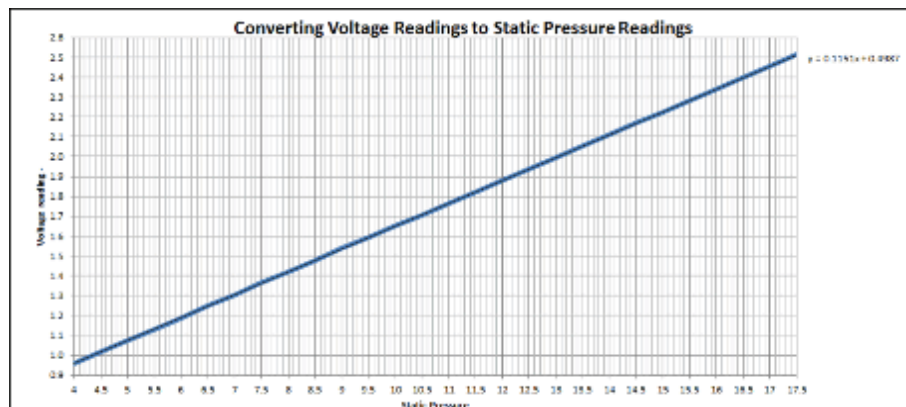
Once the system has produced a voltage reading a pressure can be obtained.



The voltage reading obtained using the method is OK to convert to a static pressure reading.

DO NOT USE any other voltage reading to convert to a pressure as the reading will be incorrect.

Fig. 130: Converting voltage readings to static pressure readings



The graph above can be used to convert this voltage to a Static pressure. Or this equation can be used.

Static Pressure (in Bar) = (Voltage reading – 0.4987) / 0.1151

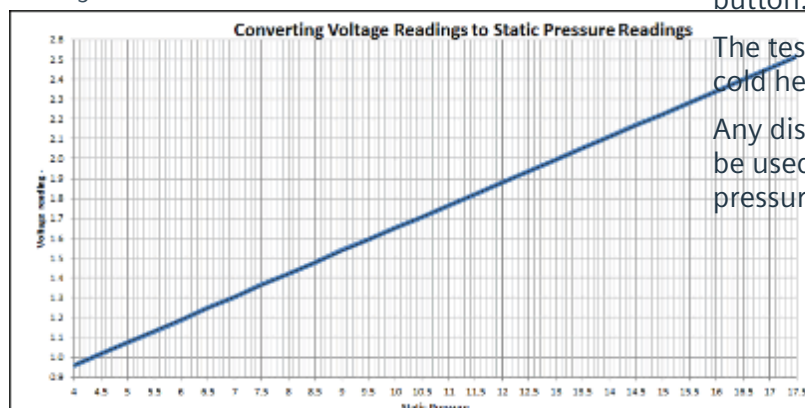
The calculated pressure has an approximate tolerance +/- 0.3 Bar

2.5.8.4 Aborting the test.

The test runs and completes automatically.

Fig. 131: Converting voltage readings to static pressure readings

- To abort the test click the **Abort** button.



The test will be stopped and the cold head will restart.

Any displayed voltage should not be used to work out compressor pressures.

- » The common print number (M6-020.840.01) from the Troubleshooting Guide is divided into separate Component print numbers.

Version 02

- Chapter Magnet System
 - Post-quench procedure reused from SMT document
 - Repairing leaks reused from SMT document
 - Alarm box reused from SMT document
 - Strategy information updated.
- Chapter MSUP
 - Strategy and procedure reused from SMT document
- Chapter Refrigerator
 - Strategy and procedure reused from SMT document
- Chapter Magnet Power Supply
 - Strategy information updated.

Version 03

- Procedure
 - Service platform added.
 - MSUP 2657 added.
 - MSUP download instruction updated.

Version 04

- Strategy
 - Link to "Magnet Error messages" has been added.

Version 05

- Procedure
 - Post Quench procedure updated.
 - CSW71: Corrected static pressure to 16.0 - 16.5 bar.

Version 06

- Procedure
 - Update of the Service platform installation.
 - New section: De-icing the syphon port.

Version 07

- Procedure
 - Update of the MSUP download instruction.

Version 08

- Procedure
 - Update of "What to do after a Quench".
 - F70 download instruction

Version 09

- Strategy
 - Compressor re-start time added.
 - Fuse location added.
 - MSUP 2306 added
 - Alarmbox added
- Procedure
 - Update of "De-icing of the Syphon Port".
 - F70 Hours run clock display added.
 - MSUP 2306 added

Version 10

- Procedure
 - Fuse Location F70 added. (→ Fuse Location F70 / Page 145)
 - Static Compressor Pressure Test added.(→ Static Compressor Pressure Test / Page 146)

Version 11

- Strategy
 - Link correction.
- Procedure
 - Updated: What to do after a Quench (→ What to do after a Quench (Avanto, Avanto Dot Upgrade, Espree with MSUP K2256) / Page 49) MR Charm_00453494.
 - Updated: What to do after a Quench (→ What to do after a Quench (Avanto Dot Ex-factory, Espree with MSUP K2306) / Page 71) MR Charm_00453494.

Version 12

- Procedure
 - Refrigerator (→ Refrigeration Failures with Magnet at Field / Page 143) MR Charm_00472208, added.

There are no Hazard IDs in this document.

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